

The Normal Distributions Indistinguishability Spectrum and its Application to Privacy-Preserving Machine Learning

Abstract—The output of several prominent randomized machine learning (ML) algorithms follows a (multivariate) Gaussian distribution. Consequently, such algorithms achieve some inherent level of privacy. However, beyond the Gaussian mechanism, where Gaussian noise is explicitly added to an algorithm, little is known about the privacy guarantees of Gaussian-output algorithms. Motivated by the above, the core question addressed in this work is:

What is the DP of any algorithm with multivariate Gaussian output?

We answer the above question with a new general lemma which we call *Normal Distributions Indistinguishability Spectrum* (NDIS), a closed-form analytic computation of the hockey-stick divergence δ between an arbitrary pair of multivariate Gaussians, parameterized by privacy parameter ϵ . We demonstrate the importance of this lemma in theory and in practice by developing a *toolbox* of corollaries that can be used to derive a conceptual simpler and tighter privacy proofs for Gaussian-output algorithms. Concrete applications include a tighter parametrization of the privacy of *random projection* (RP), which leads to a more noise-frugal DP mechanism that leverages the randomness of RP to reduce the added noise whenever possible.

Beyond random projection, we show how NDIS can lift any Gaussian-output algorithm satisfying our sensitivity condition into a Gaussian-output DP mechanism. The resulting wrapper calibrates the amount of additional Gaussian randomness needed to control the worst-case indistinguishability over neighboring datasets, while still exploiting the algorithm's existing Gaussian randomness. Lastly, as a consequence that is of independent interest, NDIS entails an accurate white-box auditing approach. A mechanism's output distributions on neighboring inputs can be turned into a provably accurate lower bound on its DP parameters via generalized χ^2 CDF evaluations.

Index Terms—differential privacy, normal distribution, random projection, privacy auditing

1. Introduction

Differentially privacy (DP) [1] is the most widely accepted privacy definition for the release of (queries on) sensitive data. Informally, a mechanism (randomized algorithm) is DP if slightly changing its input data(base)—adding or removing a single record/database-row—yields only somewhat *indistinguishable* effect on the mechanism's

output. The databases that are the result of this small change are referred to as *neighboring*.

More concretely, an algorithm $\mathcal{M}$ is (ϵ, δ) -DP if for all neighboring databases $D \sim D'$, and all events $\mathcal{S}$ over the mechanism's output space:

$$\Pr[\mathcal{M}(D) \in \mathcal{S}] \leq e^\epsilon \Pr[\mathcal{M}(D') \in \mathcal{S}] + \delta$$

We are interested in the optimal (i.e., smallest) (ϵ, δ) parameters (often called the mechanism's *privacy profile* [2]). It is well-known that one can trade off a worse/larger parameter ϵ for better/smaller δ and vice-versa, thus we will consider the optimal δ , as a function of ϵ , written as the following:

$$\delta(\epsilon) = \sup_{D \sim D'} \sup_{\mathcal{S} \subseteq \mathcal{R}^T} \left(\Pr[\mathcal{M}(D) \in \mathcal{S}] - e^\epsilon \Pr[\mathcal{M}(D') \in \mathcal{S}] \right)_+,$$

where we write $(x)_+ \stackrel{\text{def}}{=} \max\{x, 0\}$, meaning $\delta(\epsilon) \geq 0$

Throughout the paper, we study mechanisms $\mathcal{M}$ with Gaussian output, i.e., for every input D , $X = \mathcal{M}(D)$ is a (multivariate) Gaussian random variable (r.v.). For brevity, instead of $\mathcal{M}(D)$ and $\mathcal{M}(D')$, we consider Gaussian-distributed r.v.'s X, Y that are the output of $\mathcal{M}$ on neighboring databases. In addition, our main result (NDIS) specifically describes $\delta(\epsilon)$ for any fixed pair of Gaussians X, Y (later applied to describe (ϵ, δ) -DP). We define:

$$\delta_{X,Y}(\epsilon) \stackrel{\text{def}}{=} \sup_{\mathcal{S} \subseteq \mathcal{R}^T} \left(\Pr[X \in \mathcal{S}] - e^\epsilon \Pr[Y \in \mathcal{S}] \right)_+.$$

We will call $\delta_{X,Y}(\epsilon)$ the *indistinguishability* (or IS for short) between X, Y . When fixing an $\epsilon \geq 0$, this is also called the *hockey-stick divergence* [3].

The investigation of the IS for a pair (X, Y) of *Gaussian outputs* is motivated by the pervasive role of Gaussian distributions throughout private machine learning and statistics, well beyond the explicit addition of zero-mean Gaussian noise (known as the Gaussian mechanism [4] in DP literature). For instance, Gaussianity arises in sketching and random projection (RP) routines (which have applications such as randomized PCA and SVM), randomized numerical linear algebra, and classic asymptotic normality phenomena [5], [6], [7], [8], [9]. In such settings, the algorithm itself produces a Gaussian (or nearly Gaussian) output. However, a complication in these algorithms is that the shape of its output distribution may depend on the input dataset, which makes analyzing DP—the worst-case IS over *all* neighboring databases—a difficult task!

1. Where both $D, D' \in U^*$ for some *universe* U that defines the set of possible values for each database row.

In the simplest case of the standard Gaussian mechanism, Gaussian noise is added to the result of a deterministic query on the input database. The mechanism’s outputs on neighboring inputs would then differ only by a *mean shift* (whose magnitude is bounded by the ‘sensitivity’ of the query, if this sensitivity exists), but otherwise share the same covariance. As such, the IS $\delta_{X,Y}(\varepsilon)$ admits a clean closed form (which, given the sensitivity bound, gives a nice form for (ε, δ) -DP). In contrast, many practically important Gaussian-output algorithms induce *data-dependent covariances*, so neighboring databases may yield Gaussians with *different* covariance matrices. This is precisely where standard privacy analyses may fall back on loose upper bounds, indirect divergences, or worst-case sensitivity arguments, obscuring what truly governs privacy and potentially leading to unnecessary noise. One concrete example is privacy-preserving Gaussian random projection [10]²: the induced pair of outputs can have the same mean but different covariances (that depend on the input), a setting not covered by standard closed-form analyses. The same phenomenon also in common second-order and Bayesian procedures, such as Bayesian Logistic Regression [9] and Gaussian Process Regression [17]. The above challenges lead to the following core question:

Q1. *Is there a general way to exactly describe the indistinguishability between any two Gaussian distributions, and how can we use it to analyze the DP of (data-dependent) Gaussian-output algorithms?*

Answering the above is not only of theoretical interest, but has several practical implications which we explore in this work (discussed in Contributions subsection). We prove a general lemma (which we call NDIS, for **n**ormal **d**istributions **i**ndistinguishability **s**pectrum) for the indistinguishability (IS) between any two Gaussians. Then, we present a general toolbox for proving the privacy of Gaussian-output algorithms, demonstrating it for random projection and expose the *actual* quantities that govern its privacy loss.

Beyond analyzing specific mechanisms, a major task in DP literature is creating new DP mechanisms. This raises a natural design question, which we address in this work:

Q2. *Can a tight characterization of IS be used to design new DP mechanisms?*

1.1. Our Contributions.

In this paper, we answer both research questions in the affirmative.

NDIS lemma and toolbox NDIS exactly describes the IS between two Gaussian r.v.’s X, Y . Specifically, it is the expectation of a quadratic expression where the constants

depend on the means/covariances of X, Y , and the variable is a standard normal r.v..

To aid the reader in applying NDIS, we construct a toolbox which helps simplify the IS in various cases. The first tool is a connection between the IS and the CDF of the generalized- χ^2 distribution, which can be easier to work with. Since there exist widely-used algorithms [18], [19], [20] (and actively going research [21], [22] on how) to numerically compute the CDF, another nice implication is then the easy numerical computation of IS. One application of this is white-box auditing, discussed later in this subsection.

The second tool simplifies $\delta_{X,Y}(\varepsilon)$ into *clean analytic forms* (exact expressions in special cases and explicit upper bounds in more general regimes). These analytic characterizations make it possible to prove monotonicity and to identify the worst-case neighboring pair for a mechanism (as demonstrated in our analyses of random projection and of the general NDIS-calibrated wrapper that lifts Gaussian-output algorithms to (ε, δ) -DP mechanisms).

In the easiest case, we give an equivalent but simpler IS expression for X, Y when they have the same covariance matrices, and possibly different means. This captures the well-known Gaussian mechanism (and its linear query variant [23]). In the case when X, Y share the same mean but have differing covariances, with covariances comparable by Loewner order (‘Loewner-comparable’), we provide simpler upper-bound expressions for the IS of X, Y . We show and use a property of IS that when the covariance of X is ‘larger’ than Y (by Loewner order), the IS of (X, Y) is always larger than the IS of (Y, X) . For the most complex case of any two Gaussians X, Y , we decompose this case into two pairs of simpler distributions, where one pair has the same mean and the other has the same covariance, then combine the IS expressions of these two pairs based on the triangle inequality property of IS.

We apply the first tool (connections between NDIS with CDF of generalized χ^2) in conjunction with the property of IS when one covariance in the pair of distributions being larger than the other, to exactly characterize the (ε, δ) -DP of random projection, discussed below.

Applications to privacy-preserving random projection (RP). Gaussian RP projects a high-dimensional database D onto a lower-dimensional space using a Gaussian matrix. The NDIS toolbox can be applied to achieve an exact DP characterization of RP on neighboring inputs (and thus tighter than previous analysis from [10], [11], [12], [24]). This in turn creates a more noise-frugal privacy-preserving RP mechanism that achieves any desired DP parameters. Empirically, on the linear-regression benchmarks used by recent random projection based work, the resulting mechanism improves both downstream OLS test error and covariance-release quality over previous work [11], [12], [24].

Informally, for RP (which is well-studied) we know the mean and covariance of its output as a function of the input database. Applying the NDIS lemma, we can determine a ‘worst-case’ neighbor pair that maximizes IS. We then use the NDIS toolbox to obtain, and then simplify the IS expres-

2. First explored in [10], with follow-up works [11], [12], and those extending and applying this to private ML, e.g., private large graph queries [10], [13] and linear regression [14], [15], [16]

sion to show that the privacy of RP is best described by a quantity of the input databases called the *leverage* (denoted by $p_{(i)}$, where higher leverage implies worse privacy). In order to achieve (ϵ, δ) -DP, our mechanism regularizes (via ridge regularization) the input database to lower its leverage down to a target value, that the NDIS lemma identifies as corresponding to the desired (ϵ, δ) . Additionally, we optimized our RP mechanism *subsampling* and *Propose-Test-Release (PTR)* techniques. In the appendix, we discuss under what conditions on the input database can our optimized mechanisms achieve better utility than our baseline RP mechanism.

Lifting any Gaussian-output algorithm to a DP mechanism with Gaussian output. We present a definition of ‘sensitivity’ that generalizes that of the standard Gaussian mechanism. Here, we consider a Gaussian-output algorithm whose output’s mean/covariance may depend on the input. Assuming this sensitivity exists (and we show examples of such sensitivity in useful Gaussian algorithms), it allows us to define a smallest sufficient boost of the algorithm’s randomness, to achieve a desired level of (ϵ, δ) -DP. Inside this more general Gaussian mechanism is the NDIS theorem—by introducing sufficient additional randomness, the mechanism’s privacy can be described as the IS $(\delta_{X,Y}(\epsilon))$ for a known pair of Gaussians X, Y . NDIS can then be applied to analyze this IS to compute the privacy parameters (ϵ, δ) -DP of the mechanism. We provide proof-of-concept experiments for the utility of our wrapper on Böhning-style Bayesian logistic regression and Gaussian process regression.

White-box auditing. One neat consequence of NDIS is a simple tool to white-box audit a Gaussian-output mechanism. *Auditing* refers to attempting to find violations of privacy in a mechanism claiming to be (ϵ, δ) -DP, and ‘white-box’ refers to having certain knowledge about the algorithm (such as its code). One way to find such violations is to compute/estimate the IS of the mechanism outputs for a specific pair of neighboring databases, and treating this IS as a *lower-bound* of the true privacy of the mechanism. When the lower-bound is higher than the claimed privacy, we have found a violation. Computing IS for a specific pair of Gaussian distributions is exactly in our wheelhouse, and using the connection between IS and the CDF of the generalized- χ^2 distribution (which can be estimated efficiently), we obtain an efficient audit/lower-bound.

2. Related Work

The distance between pairs of multivariate Gaussian is a fundamental problem, with widespread applications in ML, e.g., [25], [26], and in security and privacy, e.g., [27], [28]. For *statistical distance* (different from the divergence used in DP), Devroye et al. [29] established a lower and an upper bound on the distance between two multivariate Gaussian with the same mean; more recently, Arbas et al. [30] generalized this result and proved a tight bound on the distance between two arbitrary multivariate Gaussians. In contrast, our work centers on (closed-form expressions

for the) *DP distance* between any two Gaussians, instead of statistical distance. Moreover, while [31] also considers the DP between two Gaussians, their work constructs an empirical verifier (with some error) verifying the DP-distance of a specific pair of one-dimensional Gaussians, which does not readily extend to proofs of DP guarantees.

The Gaussian Mechanism is a standard DP mechanism introduced by [32], [33] and further analyzed in e.g., [34], which adds a zero-mean Gaussian vector to the output of a (deterministic) query. Building on this concept, Chanyaswad et al. [35] proposed the Matrix-Variate Gaussian (MVG) mechanism, which incorporated matrix noise from a matrix-variate Gaussian distribution. Liu [36] presented a generalized Gaussian mechanism, which adds noise whose scale is computed from a more generalized sensitivity function. In this work, our goal is to solve a strictly more general problem: the indistinguishability between arbitrary multivariate Gaussians, whose mean and covariance can both vary (unlike for Gaussian mechanisms).

Random Projection (RP). RP projects a high-dimensional database D onto a lower-dimensional space, with various applications such as randomized PCA and SVM. The seminal work of Blocki et al. [10] showed that RP achieves DP, if the input database’s least singular value is sufficiently large and each record’s norm is bounded. Sheffet [11], [12] studied an r -fold composition of RP and improved on previous works’ DP analysis. Concurrently, another line of works [37], [38], [39] studied the DP of random projection using random matrices other than standard Gaussian matrices. Beyond this, RP has been explored as a pre-processing step to achieve DP in many applications [13], [40], [15], [16], [14]. Our objective aligns with [10], [11], [12], focusing on Gaussian RP, and our results improve upon the analyses of previous work. The NDIS lemma derives a *tight* privacy statement for Gaussian RP, as a function of the (global) *leverage score*.

To put our leverage-based characterization in context, prior work showed that Gaussian RP can be made DP by calibrating its randomness using a lower bound on the worst-case least singular value (LSV) over input datasets. In the same spirit, our RP mechanism $\mathcal{M}_{RP}$ takes a target privacy budget (ϵ, δ) and computes a *maximum allowable leverage* threshold $p_{(i)}$ and calibrate the noise using this threshold. Since larger leverage corresponds to weaker privacy (by Lemma 4 and proposition 4), $p_{(i)}$ directly captures the privacy-critical regime. Moreover, leverage can be upper bounded from LSV and norm of records. Consequently, LSV-based calibration is inherently more conservative: it controls leverage only indirectly (via an upper bound) and thus can yield strictly looser privacy/utility tradeoffs than working with leverage directly, while the reverse implication does not hold in general. We refer Figure 2 in Section 5.2 for further illustration.

Using an algorithm’s inherent randomness for privacy. At the heart of the concept of ‘lifting’ an already randomized algorithm to a DP mechanism (in our work, we show as examples Gaussian RP, Bayesian Logistic Regression, and Gaussian Process Regression), is the idea of using the inher-

ent randomness in queries or randomization operations for privacy. This concept exists in previous work, e.g., “privacy amplification by subsampling” [41], [42], [43], i.e., running a DP mechanism on a random subsample of a dataset offers stronger privacy guarantees compared to applying it to the entire dataset, used in algorithms such as DP-SGD [44]. In addition, there is a growing body of work investigating the inherent privacy of specific randomized algorithms, such as the Flajolet-Martin Sketch [45], [46], Thompson Sampling [47], and random projection [10], [15], [12], [16]. Most recently, GaussMix revisits Gaussian sketching for private linear regression through Rényi differential privacy and obtains improved privacy–utility bounds [24]. Our work considers more general settings: GaussMix focuses on RP-style linear-regression releases, whereas NDIS gives an exact (ϵ, δ) indistinguishability characterization for arbitrary Gaussian pairs, allowing both the mean and covariance to change. In the RP specialization, NDIS yields an exact leverage-based calibration, which we compare empirically against GaussMix-style releases and find improved downstream and covariance-release utility.

Terminology of indistinguishability (IS). We use the term *indistinguishability* (IS) to denote the optimal δ as a function of ϵ for a *specific pair* of distributions (X, Y) . We do so to distinguish it from the notion of the optimal (ϵ, δ) -DP parameters of a *mechanism*—the worst-case $\delta(\epsilon)$, over *all pairs* of neighboring databases. The optimal DP parameters of a mechanism has been introduced and studied in prior work [2], [48], [49], [50], [51], [52], [53] under names such as privacy profile/privacy curve.

Studies on optimal (ϵ, δ) -DP The study of a optimal (ϵ, δ) -DP parameters via the privacy profile (or, the hockey-stick divergence of the mechanism’s *privacy loss random variable*) has been extensively developed for subsampled Gaussian mechanisms [43], [53] and for the standard Gaussian mechanism (i.e., two normal distributions with the same covariance matrix but different means) [34], [51]. To the best of our knowledge, there has been no prior work analyzing the optimal (ϵ, δ) pairs for arbitrary pairs of Gaussian distributions, particularly in the case where the covariance matrices differ. The NDIS lemma can also be viewed as a more explicit (in fact, quadratic polynomial) form of the privacy loss random variable, which can then be further simplified (using our toolbox) to prove (ϵ, δ) -DP.

DP Variants. The study of the privacy loss r.v. for tighter composition bounds and more tractable privacy analysis led to several variants of DP, such as (Zero-)Concentrated DP (zCDP) [49], [54] and Rényi DP (RDP) [55]. These frameworks enable refined analysis of iterative algorithms like stochastic gradient descent, particularly in the context of subsampled Gaussian mechanisms [56]. Unlike the NDIS lemma, however, zCDP and RDP measure privacy using Rényi divergence rather than the standard DP distance. While conversions from RDP and zCDP to (ϵ, δ) -DP are well established, it remains unclear how to obtain closed-form expressions for the optimal (ϵ, δ) parameters in the DP setting, especially for arbitrary Gaussian distributions, from

the existing results under RDP or zCDP. Lastly, Gaussian (f -)differential privacy (GDP) [57] is a recently proposed variant of differential privacy that emphasizes measuring privacy using a function (specifically, a tradeoff function) rather than a fixed parameter pair. In particular, GDP uses the tradeoff function between two Gaussian distributions with the same (unit) variance and different means as the privacy measure. Although conversions from the GDP tradeoff function to the optimal (ϵ, δ) -DP curve are known, it remains an open question how to derive closed-form expressions for the optimal (ϵ, δ) parameters—especially for arbitrary Gaussian distributions—based on the existing results in GDP.

(White-box) auditing of DP. Auditing a (DP) mechanism refers to attempting to find counterexample(s) to a claim that a given algorithm satisfies (ϵ, δ) -DP. In general, an auditor can be either *black-box* (can only feed inputs to the audited algorithm and observe outputs), or *white-box*/*semi*-*white-box*³, where the auditor knows everything (or some partial information) about the algorithm. While black-box auditing allows auditing without access to the actual code, white-box auditing can be more accurate [62], and in some restricted cases, even *prove* DP through analysing the algorithm’s code (e.g., when the algorithm’s code is differentiable [63]). Nevertheless, we are not aware of auditors (white- or black-box) that are specialized for an arbitrary pair of Gaussian distributions (that can differ in mean and covariance)—[31] verifies algorithms that use one-dimensional Gaussians through computing/estimating their distributions. Thus we believe our tool can be a useful addition to the arsenal of any white-box auditor. While our auditing tool still requires empirically estimating the IS and treating this as the lower-bound on privacy parameters, we remark that its white-box nature gives advantages over black-box counterparts, such as [64] which also produces tight estimation of the IS for a pair of (neighbouring) databases ([65] is also tight, but requires enumerating the mechanism’s output space, making it inapplicable for Gaussians). In contrast to black-box tools that audit with some error probability, we guarantee our estimates are close to the real IS assuming a generalized χ^2 CDF oracle. Lastly, while a general-purpose f -DP (and thus Gaussian DP) auditor exists (e.g., [66]), Gaussian DP considers a pair of Gaussian distributions with the *same* variance, rather than arbitrary Gaussians.

3. Preliminaries

Definition 1 (Generalized- χ^2 distribution). *A real-valued random variable W is said to have a generalized- χ^2 distribution if there exist an integer $r \geq 1$, coefficients $\lambda_1, \dots, \lambda_r \in \mathbb{R} \setminus \{0\}$, degrees of freedom $\nu_1, \dots, \nu_r \in \mathbb{N}_{>0}$, noncentrality parameters $\rho_1, \dots, \rho_r \geq 0$, a Gaussian coefficient $\sigma \geq 0$, and a shift $\kappa \in \mathbb{R}$, such that*

$$W \stackrel{d}{=} \kappa + \sum_{j=1}^r \lambda_j X_j + \sigma G,$$

3. A sample of the many works on white-box auditing: [58], [59], [60], [61]

where $X_1, \dots, X_r$ are independent noncentral chi-square random variables $X_j \sim \chi_{\nu_j}^2(\rho_j)$ and $G \sim \mathcal{N}(0, 1)$ is independent of $\{X_j\}_{j \in [r]}$.

Definition 2 (Gaussian random projection). *Fix an integer $r \geq 1$. Let $G \in \mathbb{R}^{n \times r}$ have i.i.d. $\mathcal{N}(0, 1)$ entries. For a database $D \in \mathbb{R}^{n \times d}$, define the Gaussian random projection (RP) as a $\mathbb{R}^{d \times r}$ random matrix*

$$\text{gRP}_r(D) \stackrel{\text{def}}{=} D^\top G.$$

Equivalently, if we write $\text{gRP}_r(D) = [X^{(1)}, \dots, X^{(r)}]$ by columns, then $X^{(1)}, \dots, X^{(r)}$ are i.i.d., and for each $j \in [r]$,

$$X^{(j)} \sim \mathcal{N}(0, D^\top D).$$

Leverage score. Assume D has full column rank, and let $v^\top \in \mathbb{R}^{1 \times d}$ denote the i -th row of D . The *leverage (score)* of record i with respect to database D is

$$p_{(i)} \stackrel{\text{def}}{=} v^\top (D^\top D)^{-1} v. \quad (1)$$

Equivalently,

$$p_{(i)} = \left\| (D^\top D)^{-1/2} v \right\|_2^2$$

so $p_{(i)}$ is the squared norm of the record v after *whitening* by the covariance matrix $D^\top D$. Intuitively, $p_{(i)}$ is large when the record has a large ℓ_2 norm, or when it is aligned with directions in which $D^\top D$ is poorly conditioned (small eigenvalues). Let $P \stackrel{\text{def}}{=} D(D^\top D)^{-1}D^\top$ denote the projection matrix onto the column space of D . Then $p^{(i)} = e_i^\top P e_i$, and we will use the following standard fact.

Fact 1. *For every $i \in [n]$,*

$$0 \leq p_{(i)} \leq 1, \quad \text{and} \quad \sum_{i=1}^n p_{(i)} = \text{tr}(P) = d.$$

Moreover, $p_{(i)} = 1$ if and only if $P e_i = e_i$ (equivalently, $e_i \in \text{col}(D)$).

4. The NDIS Lemma and Toolbox

The Normal-Distributions Indistinguishability (NDIS) lemma (Lemma 1) provides a closed-form characterization of the indistinguishability between any two multivariate Gaussians X and Y with means μ_1, μ_2 and covariances Σ_1, Σ_2 , respectively. The characterization is an expectation over a *standard* Gaussian $Z \sim \mathcal{N}(0, I_d)$, where the expression's dependence on (μ_1, Σ_1) and (μ_2, Σ_2) is captured by a simple quadratic polynomial $g_\varepsilon(Z)$, thereby making explicit how privacy is governed by the relative geometry of the two Gaussians. Intuitively, $g_\varepsilon(Z)$ can be viewed as the ε -shifted privacy loss (log-likelihood ratio) expressed in a whitened/diagonalized coordinate system: the region $g_\varepsilon(Z) \leq 0$ corresponds exactly to where the likelihood ratio exceeds e^ε , and the hinge form $\mathbb{E} [1 - e^{g_\varepsilon(Z)}]_+$ captures both the probability mass and the severity of this privacy loss. This expression connects δ to the classical theory of

(generalized) quadratic forms in Gaussian random variables, enabling efficient evaluation and further analysis (as we do in the next sections).

Lemma 1 (NDIS Lemma; Proof is in Section A). *Let $X \sim \mathcal{N}(\mu_1, \Sigma_1)$ and $Y \sim \mathcal{N}(\mu_2, \Sigma_2)$ on $\mathbb{R}^d$, where $\Sigma_1, \Sigma_2 \succ 0$. Let $\Delta\mu \stackrel{\text{def}}{=} \mu_1 - \mu_2$ and $M = \Sigma_1^{1/2} \Sigma_2^{-1} \Sigma_1^{1/2}$. Let $U \text{diag}(\tau_1, \dots, \tau_d) U^\top$ be an Eigenvalue Decomposition (EVD) of M , and define*

$$a_i \stackrel{\text{def}}{=} 1 - \tau_i \text{ for } i \in [d], \quad b \stackrel{\text{def}}{=} -U^\top \Sigma_1^{1/2} \Sigma_2^{-1} \Delta\mu$$

$$c(\varepsilon) \stackrel{\text{def}}{=} \varepsilon + \frac{1}{2} \log \frac{\det \Sigma_1}{\det \Sigma_2} - \frac{1}{2} \Delta\mu^\top \Sigma_2^{-1} \Delta\mu.$$

Then for $Z \sim \mathcal{N}(0, I_d)$,

$$\delta_{X,Y}(\varepsilon) = \mathbb{E} \left[\left(1 - \exp(g_\varepsilon(Z)) \right)_+ \right],$$

where $g_\varepsilon(Z) \stackrel{\text{def}}{=} c(\varepsilon) + b^\top Z + \frac{1}{2} \sum_{i=1}^d a_i Z_i^2$.

To prove Lemma 1, we work directly from the standard expression $\sup_{\mathcal{S}} (\Pr[X \in \mathcal{S}] - e^\varepsilon \Pr[Y \in \mathcal{S}])$. At a high level, the goal is to rewrite this set-optimization problem as an expectation under a *standard* Gaussian measure, so that the expression's dependence on (μ_1, Σ_1) and (μ_2, Σ_2) becomes explicit. We first expand $\Pr[X \in \mathcal{S}]$ and $\Pr[Y \in \mathcal{S}]$ using their Gaussian density formulas and combine them inside a single integral over $\mathcal{S}$. Next, we re-express X via the change of variables $x = \mu_1 + \Sigma_1^{1/2} z$, which turns the X -density into the standard Gaussian density $\varphi(z)$ and turns $\mathcal{S}$ into $\mathcal{S}_z = \Sigma_1^{-1/2}(\mathcal{S} - \mu_1)$. After this standardization, the integrand becomes $\varphi(z)(1 - e^{\psi_\varepsilon(z)})$, where $\psi_\varepsilon(z)$ is an explicit quadratic function obtained by expanding the exponent of the Y density and rearranging the expression; the quadratic term is characterized by $M = \Sigma_1^{1/2} \Sigma_2^{-1} \Sigma_1^{1/2}$.

We then diagonalize $M = U \text{diag}(\tau_1, \dots, \tau_d) U^\top$ and rotate coordinates via $v = U^\top z$. This preserves the standard Gaussian measure but separates out the quadratic part, yielding $\psi_\varepsilon(z) = g_\varepsilon(v) = c(\varepsilon) + b^\top v + \frac{1}{2} \sum_{i=1}^d a_i v_i^2$. Finally, the set-optimization: since $\varphi(v) \geq 0$, the integral is maximized by choosing the region where the bracket is positive, i.e., where $1 - e^{g_\varepsilon(v)} > 0$ or equivalently $g_\varepsilon(v) \leq 0$. This gives the claimed representation in Lemma 1.

4.1. Tool 1: Efficient Computation via Generalized- χ^2 CDFs

Building on the NDIS lemma, we next highlight a key *computational* consequence. Theorem 1 shows that, by pushing the NDIS representation one step further, $\delta_{X,Y}(\varepsilon)$ can be written as the difference of *two* tail probabilities of quadratic forms, namely $\Pr[g_\varepsilon(Z) \leq 0]$ and $\Pr[g_\varepsilon(\tilde{Z}) \leq 0]$. Crucially, both $g_\varepsilon(Z)$ and $g_\varepsilon(\tilde{Z})$ are generalized- χ^2 random variables (cf. Definition 1). Thus, computing indistinguishability between arbitrary multivariate Gaussians reduces to a well-studied numerical task:

evaluating generalized- χ^2 CDFs, for which there exist mature and widely used algorithms based on characteristic-function inversion and related techniques (e.g., Imhof-type and Davies-type methods).

Two applications of this tool include the analysis of random projection (Sec. 5) and white-box auditing (Sec. 7).

Theorem 1 (Connecting IS and CDF of generalized- χ^2 r.v.s; Proof is in Section E.1). *Let ε, X, Y and the quantities $g_\varepsilon(\cdot)$, b , and $\tau_1, \dots, \tau_d$ be as in Lemma 1. Let $D = \text{diag}(\tau_1, \dots, \tau_d)$ and $m = D^{-1}b$. Let $Z \sim \mathcal{N}(0, I_d)$ and $\tilde{Z} \sim \mathcal{N}(m, D^{-1})$. Then both $g_\varepsilon(Z)$ and $g_\varepsilon(\tilde{Z})$ are generalized- χ^2 random variables (cf. Definition 1), and*

$$\delta_{X,Y}(\varepsilon) = \Pr[g_\varepsilon(Z) \leq 0] - e^\varepsilon \Pr[g_\varepsilon(\tilde{Z}) \leq 0].$$

Consequently, computing $\delta_{X,Y}(\varepsilon)$ reduces to evaluating two generalized- χ^2 CDFs.

4.2. Tool 2: Simplifying NDIS to Cleaner Analytic Forms, Beyond CDF Evaluation

While Theorem 1 shows that $\delta_{X,Y}(\varepsilon)$ for an arbitrary Gaussian pair can be computed efficiently via generalized- χ^2 CDF evaluation, for several purposes it is also valuable to have *explicit analytic* expressions or bounds. An analytic form is useful when performing further analysis, for example: proving monotonicity properties of this expression, or deriving sufficient conditions for achieving (ε, δ) -DP. Motivated by this, we next develop analytic characterizations of $\delta_{X,Y}(\varepsilon)$ in progressively more general settings (using the NDIS lemma as the starting point of their proofs).

We begin with the ‘mean-shift’ case, where X and Y share the same covariance but have different means (a well-known example is the Gaussian mechanism). We then turn to the ‘covariance-shift’ case, where X and Y share the same mean but different covariances (a well-known example is the random projection-based mechanisms). Finally, we combine these ingredients to obtain a simple analytic upper bound for simultaneous mean-and-covariance changes in the Loewner-comparable covariance regime (Theorem 2).

Equal covariance (mean shift). In the simplest nontrivial regime, X and Y differ only in their means while sharing a common covariance. In this case, Proposition 1 shows that $\delta_{X,Y}(\varepsilon)$ admits a closed form that depends *only* on the Mahalanobis norm of the difference of the mean $\Delta\mu$ under the common covariance, namely $\tau = \sqrt{\Delta\mu^\top \Sigma^{-1} \Delta\mu}$. This leads to the well-known privacy profile of the Gaussian mechanism [34] (and its linear-query variants [23], [67], [68]) as a special case.

Proposition 1 (IS for mean shift; Proof is in Section E.2). *Let $\varepsilon, X \sim \mathcal{N}(\mu_1, \Sigma_1), Y \sim \mathcal{N}(\mu_2, \Sigma_2)$, and $\Delta\mu$ be as in Lemma 1. Define the Mahalanobis distance*

$$\tau \stackrel{\text{def}}{=} \sqrt{\Delta\mu^\top \Sigma^{-1} \Delta\mu}.$$

Assume $\Sigma_2 = \Sigma_1$. Then,

$$\delta_{X,Y}(\varepsilon) = \Phi\left(-\frac{\varepsilon}{\tau} + \frac{\tau}{2}\right) - e^\varepsilon \Phi\left(-\frac{\varepsilon}{\tau} - \frac{\tau}{2}\right),$$

where Φ is the CDF of $\mathcal{N}(0, 1)$.

Equal mean (covariance shift). We next consider the case where X and Y share the same mean but differ in their covariances. In this case, Lemma 2 shows that when the covariances are comparable in the Loewner order (denoted $\Sigma_1 \succeq \Sigma_2$), $\delta_{X,Y}(\varepsilon)$ admits an analytic *upper bound* controlled by the relative eigen-geometry of (Σ_1, Σ_2) through the eigenvalues $\tau_1, \dots, \tau_d$ from Lemma 1. In this direction, the privacy-loss event becomes an upper-tail event for a weighted χ^2 sum. This is the direction needed for worst-case covariance leakage, since Lemma 3 shows that the more-spread-to-less-spread orientation dominates the reverse orientation. Lemma 2 provides a closed-form Markov-type bound that can be optimized over two scalar parameters (s, u) . We note in passing that this covariance-shift setting also covers random-projection-based mechanisms, where neighboring datasets can induce different (but ordered) post-projection covariances.

Lemma 2 (Upper bound IS for covariance shift; Proof is in Section E.2). *Let $\varepsilon \geq 0, X \sim \mathcal{N}(\mu_1, \Sigma_1), Y \sim \mathcal{N}(\mu_2, \Sigma_2)$, and let $\tau_1, \dots, \tau_d$ be as in Lemma 1. Define*

$$f(\varepsilon) \stackrel{\text{def}}{=} \log \frac{\det \Sigma_1}{\det \Sigma_2} + 2\varepsilon, \quad k \stackrel{\text{def}}{=} \left(2 \max_{i \in [d]} (\tau_i - 1)\right)^{-1} \in (0, \infty],$$

with the convention $k = \infty$ if $\max_i (\tau_i - 1) = 0$. Let $\Sigma_1 \succeq \Sigma_2$ and $\mu_1 = \mu_2$. Then,

$$\delta_{X,Y}(\varepsilon) \leq \inf_{\substack{s \in [0, k) \\ u \geq 0}} \left[\exp\left(-sf(\varepsilon) - \frac{1}{2} \sum_{i=1}^d \log(1 - 2s(\tau_i - 1))\right) - e^\varepsilon \left(1 - \exp\left(uf(\varepsilon) - \frac{1}{2} \sum_{i=1}^d \log\left(1 + 2u \frac{\tau_i - 1}{\tau_i}\right)\right)\right) \right].$$

To see how Lemma 2 is proved, we specialize the NDIS lemma (Lemma 1) to the covariance-only setting $\mu_1 = \mu_2$. In this case, the NDIS polynomial simplifies substantially: the linear term disappears ($b = 0$) and $g_\varepsilon(Z)$ becomes a shifted quadratic form. Consequently, the privacy-loss region $\{g_\varepsilon(Z) \leq 0\}$ can be rewritten as a one-sided upper-tail constraint on a weighted χ^2 sum,

$$\sum_i (\tau_i - 1) Z_i^2 \geq f(\varepsilon),$$

where $f(\varepsilon) = \log \frac{\det \Sigma_1}{\det \Sigma_2} + 2\varepsilon$ measures the determinant gap between the two covariances together with the privacy parameter.

We then invoke the two-CDF decomposition from Theorem 1 and control the resulting quadratic-form probabilities using exponential (Markov/Chernoff) bounds. At a high level, we upper bound the first term by applying Markov to a positively tilted version of the weighted χ^2 upper tail. For the second term, after writing $\tilde{Z}_i = \tau_i^{-1/2} Z'_i$, the event becomes $\sum_i \frac{\tau_i - 1}{\tau_i} (Z'_i)^2 \geq f(\varepsilon)$. We lower bound this probability by upper bounding its complementary lower-tail event using a negatively tilted exponential moment. The closed-form χ^2 moment generating functions then produce

the explicit log-product expressions in (s, u) , and optimizing over these (s, u) yields the stated analytic upper bound.

$\delta_{X,Y}(\varepsilon)$ is inherently asymmetric in (X, Y) , which means in general $\delta_{X,Y}(\varepsilon) \neq \delta_{Y,X}(\varepsilon)$. In the covariance-only setting, however, this asymmetry becomes *structured*. Lemma 3 shows that if $\Sigma_1 \succeq \Sigma_2$ (so X is the “more spread” Gaussian), then $\delta_{X,Y}(\varepsilon) \geq \delta_{Y,X}(\varepsilon)$ for all $\varepsilon \geq 0$. Thus, in the comparable-covariance regime there is a “hard” direction.⁴ Consequently, it suffices to upper bound $\delta_{X,Y}(\varepsilon)$ in this direction to upper bound the covariance-induced leakage, which is exactly what Lemma 2 provides.

Lemma 3 ($\delta_{X,Y}$ upper bounds $\delta_{Y,X}$ when covariances in Loewner order; Proof is in Section E.2). *Let $X \sim \mathcal{N}(0, \Sigma_1)$ and $Y \sim \mathcal{N}(0, \Sigma_2)$ on $\mathbb{R}^d$, where $\Sigma_1, \Sigma_2 \succ 0$ and $\Sigma_1 \succeq \Sigma_2$. Then, for all $\varepsilon \geq 0$ and $d \in \mathbb{N}_{\geq 0}$,*

$$\delta_{X,Y}(\varepsilon) \geq \delta_{Y,X}(\varepsilon).$$

To prove Lemma 3, we first whiten by Σ_2 to reduce the comparison to the pair (M, I_d) with $M = \Sigma_2^{-1/2} \Sigma_1 \Sigma_2^{-1/2} \succeq I_d$, so it suffices to show $\delta_{M,I_d}(\varepsilon) \geq \delta_{I_d,M}(\varepsilon)$. Using the NDIS lemma together with the radial-angular expression of the standard multi-normal random variable $Z = RU$, both divergences can be written as expectations over the sphere: conditional on a direction U , they reduce to one-dimensional “radial” expressions of the form $g_d(\cdot, \alpha(U))$ and $h_d(\cdot, \beta(U))$, where $\alpha(U) = U^\top (M - I)U$ and $\beta(U) = U^\top (I - M^{-1})U$. The eigenvalue dominance $M \succeq I$ implies a pointwise coupling $\beta(U) \leq \alpha(U)/(1 + \alpha(U))$. Combining this coupling with Proposition 2 below (and monotonicity properties of the resulting radial functions) gives a *direction-wise dominance*: for every fixed direction U , the contribution to $\delta_{I_d,M}(\varepsilon)$ is bounded by the corresponding contribution to $\delta_{M,I_d}(\varepsilon)$. Averaging over U then proves the claimed ordering.

Proposition 2 (Proof is in Section E.2). *Let $d \geq 2$, $t \geq 0$, and $a \in \mathbb{R}$. Define function, for $R^2 \sim \chi_d^2$,*

$$g_d(a, t) \stackrel{\text{def}}{=} \mathbb{E} \left[\left(1 - \exp \left(a - \frac{t}{2} R^2 \right) \right)_+ \right],$$

$$h_d(a, t) \stackrel{\text{def}}{=} \mathbb{E} \left[\left(1 - \exp \left(a + \frac{t}{2} R^2 \right) \right)_+ \right].$$

Then, for every $t \geq 0$ and every $a \in [-\frac{1}{2} \log(1+t), 0]$, we have

$$h_d \left(a, \frac{t}{1+t} \right) \leq g_d(a + \log(1+t), t).$$

Loewner-comparable mean and covariance shift. We now consider the more general setting where X and Y may differ in both mean and covariance. When the two covariances are comparable in Loewner order, Theorem 2 provides a

4. This asymmetry property is useful beyond saving a second computation: in the hard direction $\delta_{X,Y}(\varepsilon)$, the NDIS coefficients satisfy $1 - \tau_i \leq 0$, so the privacy-loss event reduces to a one-sided upper-tail constraint on a nonnegative weighted χ^2 sum. This sign structure is what enables the Markov-type product bound used in Lemma 2.

simple analytic upper bound on $\delta_{X,Y}(\varepsilon)$ by decomposing the privacy leakage into two parts: (i) a *mean-shift* component, corresponding to the case of equal covariance but different means, and (ii) a *covariance-shift* component, corresponding to the case of equal mean but different covariances. The theorem then gives an explicit analytic bound by combining these two special-cases, whose results were established above, via triangle inequality.

Theorem 2 (IS for mean and covariance shift; Proof is in Section E.2). *Let $\varepsilon \geq 0$ and $X \sim \mathcal{N}(\mu_1, \Sigma_1), Y \sim \mathcal{N}(\mu_2, \Sigma_2)$ on $\mathbb{R}^d$, where $\Sigma_1, \Sigma_2 \succ 0$. Assume either $\Sigma_1 \succeq \Sigma_2$ or $\Sigma_2 \succeq \Sigma_1$. Let $\Sigma_{\min}, \Sigma_{\max} \in \{\Sigma_1, \Sigma_2\}$ be such that $\Sigma_{\max} \succeq \Sigma_{\min}$. For $\varepsilon \geq 0$, define the two upper bounds*

$$\bar{\delta}^{(1)}(\varepsilon) \stackrel{\text{def}}{=} \inf_{\substack{\varepsilon_1, \varepsilon_2 \geq 0 \\ \varepsilon_1 + \varepsilon_2 = \varepsilon}} \left[\delta_{\mathcal{N}(\mu_1, \Sigma_{\min}), \mathcal{N}(\mu_2, \Sigma_{\min})}(\varepsilon_1) + e^{\varepsilon_1} \delta_{\mathcal{N}(\mu_2, \Sigma_{\max}), \mathcal{N}(\mu_2, \Sigma_{\min})}(\varepsilon_2) \right],$$

$$\bar{\delta}^{(2)}(\varepsilon) \stackrel{\text{def}}{=} \inf_{\substack{\varepsilon_1, \varepsilon_2 \geq 0 \\ \varepsilon_1 + \varepsilon_2 = \varepsilon}} \left[\delta_{\mathcal{N}(\mu_1, \Sigma_{\max}), \mathcal{N}(\mu_1, \Sigma_{\min})}(\varepsilon_1) + e^{\varepsilon_1} \delta_{\mathcal{N}(\mu_1, \Sigma_{\min}), \mathcal{N}(\mu_2, \Sigma_{\min})}(\varepsilon_2) \right].$$

Then,

$$\delta_{X,Y}(\varepsilon) \leq \min \{ \bar{\delta}^{(1)}(\varepsilon), \bar{\delta}^{(2)}(\varepsilon) \}.$$

Moreover, each component admits a simple analytic form: the mean-shift terms admits an analytic form in Proposition 1, and the covariance-shift terms admit an analytic upper bound in Lemma 2.

At a high level, Theorem 2 bounds the $\delta_{X,Y}$ by *separating* the change from X to Y into two simpler steps. We introduce an intermediate Gaussian W so that one step changes only the mean (holding the covariance fixed) and the other step changes only the covariance (holding the mean fixed). The triangle inequality for the DP divergence (Proposition 3) then upper bounds $\delta_{X,Y}(\varepsilon)$ as the sum of the divergences incurred on these two steps, with a privacy-budget split $\varepsilon = \varepsilon_1 + \varepsilon_2$ that allocates how much “privacy cost” we pay on each step.

There are two natural ways to choose the intermediate distribution, corresponding to whether we perform the mean change first or the covariance change first. These two choices lead to the bounds $\bar{\delta}^{(1)}(\varepsilon)$ and $\bar{\delta}^{(2)}(\varepsilon)$, respectively. Optimizing over the budget split and taking the better of the two paths yields

$$\delta_{X,Y}(\varepsilon) \leq \min \{ \bar{\delta}^{(1)}(\varepsilon), \bar{\delta}^{(2)}(\varepsilon) \}.$$

Proposition 3 (Triangle inequality for DP divergence; Proof is in Section E.2). *Let X, Y, Z be random variables, then for all $\varepsilon_1, \varepsilon_2 \geq 0$,*

$$\delta_{X,Z}(\varepsilon_1 + \varepsilon_2) \leq \delta_{X,Y}(\varepsilon_1) + e^{\varepsilon_1} \delta_{Y,Z}(\varepsilon_2).$$

5. Random Projection DP Mechanism

We present an application of NDIS (and associated tools) to construct an improved random projection (RP)-based DP mechanism. We treat Gaussian RP $\text{gRP}_r(D)$ as a *Gaussian-output algorithm* and use NDIS (Lemma 1 and theorem 1) to characterize the indistinguishability between neighboring outputs $\text{gRP}_r(D)$ and $\text{gRP}_r(D')$. This leads to an explicit privacy characterization, which we then use to “lift” RP into a mechanism that can meet desired privacy (ε, δ) . Building on this, we prove the privacy and utility guarantees of the resulting Random Projection mechanism $\mathcal{M}_{RP}$. We also directly show that using leverage to calibrate noise compares favourably with LSV-based and GaussMix-style differentially-private RP mechanisms, both at the calibration level and in downstream linear regression benchmarks (Figure 3 and Figure 2). In App. B, we show how $\mathcal{M}_{RP}$ can be optimized with tools in DP literature—subsampling and *Propose-Test-Release (PTR)* techniques.

5.1. Construction

Our RP mechanism $\mathcal{M}_{RP}$ is in Figure 1. Its high level design is guided by Proposition 4, which formalizes the principle that *larger leverage yields weaker privacy*, and builds on the pairwise indistinguishability analysis in Lemma 4. Concretely, the mechanism matches a desired $\delta(\varepsilon)$ to a leverage threshold $p_{(i)}$. Then, given any input database, applies regularization to achieve (at most) such leverage. We note that the ‘amount’ of regularization depends only on the leverage threshold and input space, and not on the input database itself.

We begin by Lemma 4 studying the *inherent* privacy of Gaussian RP under the row-deletion neighboring relation, i.e., one database is a neighbor of another through deleting a row. Concretely, fix the i -th record and let D' be obtained by removing row i from D . We analyze the indistinguishability $\delta_{X,Y}(\varepsilon)$ between the neighboring outputs $X = \text{gRP}_r(D)$ and $Y = \text{gRP}_r(D')$.⁵ Lemma 4 expresses $\delta_{X,Y}(\varepsilon)$ as an explicit analytic function of the leverage $p_{(i)}$, showing that the privacy impact of record i is entirely captured by its leverage score in D . Moreover, the target dimension r governs the concentration of the RP output, and hence the tail behavior that determines δ .

Lemma 4 (Proof is in Section F.1). *Let $n, r, d \in \mathcal{N}_{>0}$, and fix $i \in [n]$. Let $D \in \mathbb{R}^{n \times d}$ such that $D^\top D \succ 0$, and let D' be obtained from D by removing row i . Let $p_{(i)} < 1$ be the leverage score of row i in D (cf. Equation (1)), and let*

$$\rho = \frac{1}{1 - p_{(i)}} \geq 1, \quad t_0 = \frac{2(\varepsilon + \frac{r}{2} \log \rho)}{\rho - 1}.$$

5. Deletion lets the RP output covariance have $\Sigma(D') \preceq \Sigma(D)$, and by covariance-order asymmetry in Lemma 3, the deletion orientation is the worst case in the sense that $\delta_{X,Y} \preceq \delta_{Y,X}$. Hence it suffices to analyze the row-deletion direction; the add-one direction follows automatically (details in proof of Theorem 3).

Then, for random matrices $X \stackrel{\text{def}}{=} \text{gRP}_r(D), Y \stackrel{\text{def}}{=} \text{gRP}_r(D_{-i})$ ($\text{gRP}_r(\cdot)$ is defined in Definition 2), and for every $\varepsilon \geq 0$,

$$\delta_{X,Y}(\varepsilon) = \frac{1}{\Gamma(\frac{r}{2})} \left(\Gamma\left(\frac{r}{2}, \frac{t_0}{2}\right) - e^\varepsilon \Gamma\left(\frac{r}{2}, \frac{\rho t_0}{2}\right) \right),$$

where $\Gamma(\cdot, \cdot)$ denotes the upper incomplete gamma function.

To prove Lemma 4, we treat it as a direct application of the NDIS lemma specialized to the *zero-mean, covariance-shift* setting. Indeed, $\text{gRP}_r(D)$ produces r i.i.d. Gaussian columns with covariance $\Sigma_1 = D^\top D$, whereas for the row-deleted database D' the output has the same form with covariance $\Sigma_2 = D'^\top D' = \Sigma_1 - vv^\top$, where $v^\top$ is the removed row. After vectorizing the $d \times r$ output, the neighboring pair becomes two centered Gaussians in $\mathbb{R}^{dr}$ with covariances $I_r \otimes \Sigma_1$ and $I_r \otimes \Sigma_2$, placing us exactly in the zero-mean covariance-shift regime.

We then inspect the resulting NDIS quadratic form $g_\varepsilon(Z)$ and observe that all dependence on the database enters through two quantities: (i) the determinant ratio in the constant term $c(\varepsilon)$ and (ii) the spectrum of $\Sigma_1^{1/2} \Sigma_2^{-1} \Sigma_1^{1/2}$. Both are controlled by the leverage of the removed record $v^\top$. By the matrix determinant lemma,

$$\frac{\det(\Sigma_1)}{\det(\Sigma_2)} = (1 - v^\top \Sigma_1^{-1} v)^{-1} = (1 - p_{(i)})^{-1} = \rho,$$

and writing $u = \Sigma_1^{-1/2} v$ yields

$$\Sigma_1^{1/2} \Sigma_2^{-1} \Sigma_1^{1/2} = (I - uu^\top)^{-1},$$

whose eigenvalues are ρ in the direction u and 1 on $u^\perp$. Consequently, the privacy impact, $\delta_{X,Y}(\varepsilon)$, is fully captured by the single leverage-induced scalar $\rho = (1 - p_{(i)})^{-1}$.

Finally, the same rank-one structure explains why the resulting expression is *clean*. While NDIS in general reduces computing $\delta_{X,Y}(\varepsilon)$ to computing generalized- χ^2 CDF, here the rank-one update $\Sigma_2 = \Sigma_1 - vv^\top$ leaves only r nontrivial directions in the quadratic form (one per RP column), with all remaining directions canceling. Consequently, $g_\varepsilon(Z)$ depends only on the simple statistic $T = \sum_{k=1}^r Z_k^2 \sim \chi_r^2$, so the dr -dimensional expectation collapses to a one-dimensional χ^2 tail integral. Evaluating this tail yields upper incomplete gamma functions, resulting in our stated analytic formula.

Building on Lemma 4, Proposition 4 formalizes the intuition that *larger leverage implies weaker privacy* by showing that, for every fixed $\varepsilon \geq 0$, the resulting smallest δ value of the RP mechanism is nondecreasing in the leverage parameter. This monotone ordering is central to our subsequent DP analysis: once we enforce a uniform leverage bound for all records (e.g., via regularization), the worst-case neighboring pair is realized at the maximal leverage. Consequently, proving that a random-projection mechanism is (ε, δ) -DP reduces to analyzing the indistinguishability $\delta_{X,Y}(\varepsilon)$ for a *single* neighboring pair $(X, Y) = (\text{gRP}_r(D), \text{gRP}_r(D'))$ — precisely the type of Gaussian indistinguishability problem that the NDIS toolbox is designed to handle.

Proposition 4 (Proof is in Section F.1). *Let $r \in \mathcal{N}_{>0}$ and $\varepsilon \geq 0$. For $p \in [0, 1]$ define*

$$\rho(p) \stackrel{\text{def}}{=} \frac{1}{1-p} \geq 1, \quad t_0(p) \stackrel{\text{def}}{=} \frac{2(\varepsilon + \frac{r}{2} \log \rho(p))}{\rho(p) - 1}.$$

Define the RP δ -curve (as a function of leverage) as

$$\delta_r^{\text{gRP}}(\varepsilon; p) \stackrel{\text{def}}{=} \frac{1}{\Gamma(\frac{r}{2})} \left(\Gamma\left(\frac{r}{2}, \frac{t_0(p)}{2}\right) - e^\varepsilon \Gamma\left(\frac{r}{2}, \frac{\rho(p)t_0(p)}{2}\right) \right).$$

Then $p \mapsto \delta_r^{\text{gRP}}(\varepsilon; p)$ is nondecreasing on $[0, 1]$.

Input: A database $D \in \mathbb{R}^{n \times d}$; target dimension $r \geq 1$; privacy parameters $\varepsilon \in \mathbb{R}_{\geq 0}$, $\delta \in [0, 1]$; and a public parameter $l \in \mathbb{R}_{>0}$ that upper bounds the row norm in the record universe.

Output: A matrix $\widetilde{M} \in \mathbb{R}^{d \times r}$.

- 1) Use binary search to find $p^* \in (0, 1)$ such that

$$\delta_r^{\text{gRP}}(\varepsilon; p^*) \leq \delta,$$

where $\delta_r^{\text{gRP}}(\varepsilon; p)$ is defined in Prop. 4 ($p \mapsto \delta_r^{\text{gRP}}(\varepsilon; p)$ is monotone in p)

- 2) Set the ridge level $\lambda = \frac{l^2}{p^*}$ and define the augmented (regularized) database

$$\overline{D} \stackrel{\text{def}}{=} \begin{bmatrix} D \\ \sqrt{\lambda} I_d \end{bmatrix} \in \mathbb{R}^{(n+d) \times d}.$$

- 3) Return $\widetilde{M} = \text{gRP}_r(\overline{D})$.

Figure 1. Random Projection (RP) mechanism $\mathcal{M}_{RP}$.

Equipped with the above relation between leverage and privacy (Lemma 4 and proposition 4), our $\mathcal{M}_{RP}$ (Figure 1) ‘upgrades’/‘lifts’ Gaussian RP into an (ε, δ) -DP mechanism: it (1) converts the desired (ε, δ) into the maximum allowable leverage, and (2) regularizes the input domain to be below such leverage threshold. Recall that though Lemma 4 is stated for the row-deletion neighboring relation; the discussion above for the add-one-row direction follows automatically from the covariance-order asymmetry in Lemma 3.

5.2. Privacy and utility analysis of $\mathcal{M}_{RP}$

We present the concrete mechanism $\mathcal{M}_{RP}$ in Figure 1, and Theorem 3 states that $\mathcal{M}_{RP}$ is (ε, δ) -DP.

Theorem 3 (Privacy of $\mathcal{M}_{RP}$; Proof is in Section F.1). *Let $\varepsilon \geq 0, \delta \in (0, 1)$. The Random Projection (RP) mechanism $\mathcal{M}_{RP}$ in Figure 1 is (ε, δ) -DP under the standard add/remove (unbounded) notion of neighboring databases.*

In more detail, $\mathcal{M}_{RP}(D)$ first applies a *public* ridge regularization, producing the augmented matrix $\overline{D}$. Because this regularization is data-independent, the same transformation applies to the deletion neighbor D' , yielding the paired augmented matrices $\overline{D}$ and $\overline{D}'$. Distinguishing $X = \text{gRP}_r(\overline{D})$ and $Y = \text{gRP}_r(\overline{D}')$ then places us exactly in the rank-one covariance-shift regime of Lemma 4.

The role of the ridge is to enforce a uniform leverage bound. Writing $\Sigma_1 = \overline{D}^\top \overline{D} = D^\top D + \lambda I_d$, we have $\Sigma_1 \succeq$

λI_d , and hence the leverage of the deleted row $v^\top$ in the regularized design satisfies

$$\overline{p}_{(i)} = v^\top (D^\top D + \lambda I_d)^{-1} v \leq \frac{1}{\lambda} \|v\|_2^2 \leq \frac{l^2}{\lambda} = p^*.$$

With this leverage control in hand, Lemma 4 identifies the deletion-direction indistinguishability as $\delta_{X,Y}(\varepsilon) = \delta_r^{\text{gRP}}(\varepsilon; \overline{p}_{(i)})$, and Proposition 4 turns the leverage bound $\overline{p}_{(i)} \leq p^*$ into the worst-case bound $\delta_{X,Y}(\varepsilon) \leq \delta_r^{\text{gRP}}(\varepsilon; p^*) \leq \delta$ by the choice of p^* in the mechanism (cf. Figure 1). Finally, the add-one direction follows from the covariance-order asymmetry (Lemma 3), so analysing deletions is sufficient in obtaining (ε, δ) -DP under add/remove adjacency.

Theorem 4 studies the utility of $\mathcal{M}_{RP}$ by examining how accurately the released sketch preserves the Gram matrix $\Sigma = D^\top D$. This is a natural criterion as Gaussian RP depends on the data only through Σ : each released column of $\widetilde{M} = \text{gRP}_r(\overline{D})$ is an independent draw from $\mathcal{N}(0, \Sigma + \lambda I_d)$.⁶ Accordingly, the estimator $\widehat{\Sigma} = \frac{1}{r} \widetilde{M} \widetilde{M}^\top - \lambda I_d$ is *unbiased* for the original Gram matrix, i.e., $\mathbb{E}[\widehat{\Sigma}] = \Sigma$.

Theorem 4 also shows that $\widehat{\Sigma}$ concentrates to $D^\top D$ in *spectral norm* at the standard sample-covariance rate, with scale $\|D\|_2^2 + \lambda$, making the utility–privacy tradeoff explicit. In particular, accuracy improves with the sketch dimension r , while privacy is enforced through the ridge level $\lambda = \lambda(\varepsilon, \delta, r)$ chosen via the monotone calibration in Step 1 of Figure 1. It highlights that tighter privacy targets (ε, δ) translate into stronger regularization and hence larger estimation error.

Theorem 4 (Utility of $\mathcal{M}_{RP}$; Proof is in Section F.2). *Given a database $D \in \mathbb{R}^{n \times d}$, and let $\widetilde{M} = \mathcal{M}_{RP}(D)$, where $\mathcal{M}_{RP}$ is defined in Figure 1. Let*

$$\widehat{\Sigma} \stackrel{\text{def}}{=} \frac{1}{r} \widetilde{M} \widetilde{M}^\top - \lambda I_d, \quad \Sigma \stackrel{\text{def}}{=} D^\top D,$$

where $\lambda = l^2/p^$ and $p^* = p^*(\varepsilon, \delta, r) \in (0, 1)$ is the value selected in Step 1 of Figure 1. Then:*

- 1) **Unbiasedness.** $\mathbb{E}[\widehat{\Sigma}] = \Sigma$.
- 2) **High-probability spectral-norm error.** *There exists an absolute constant $C > 0$ such that for every $\beta \in (0, 1)$, with probability at least $1 - \beta$,*

$$\|\widehat{\Sigma} - \Sigma\|_2 \leq C \left(\sqrt{\frac{d + \log(2/\beta)}{r}} + \frac{d + \log(2/\beta)}{r} \right) (\|D\|_2^2 + \lambda).$$

Leverage Calibration: NDIS vs LSV-Based Bounds. Our results directly connect *leverage* to the privacy of Gaussian RP. Thus, to show the tightness of our results, we compare the leverage thresholds implied by our NDIS-based calibration versus that of a LSV-based calibration [11], [12]. We fix $\delta = 10^{-6}$ and a row-norm bound $l = 1$, and vary the target privacy parameter ε over a grid (here,

6. This “sketch-as-second-moment” viewpoint is standard in randomized numerical linear algebra and sketching [69], [70].

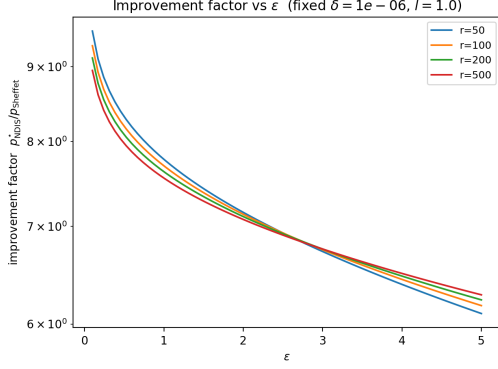


Figure 2. Improvement factor in allowable leverage: $p_{\text{NDIS}}^*/p_{\text{LSV}}$ versus ε (log scale), with $\delta = 10^{-6}$, $l = 1$, and $r \in \{50, 100, 200, 500\}$. Here p_{NDIS}^* is the maximal leverage threshold computed by our NDIS-based RP analysis, while p_{LSV} is the leverage bound implied by the LSV-based calibration of [12], [11]. Ratios above 1 indicate that LSV-based calibration is more conservative, forcing smaller allowable leverage and thus larger ridge regularization.

$\varepsilon \in [0.1, 5]$). We report results for several sketch dimensions $r \in \{50, 100, 200, 500\}$.

For each (ε, δ, r) , our method computes the maximal leverage threshold

$$p_{\text{NDIS}}^*(\varepsilon, \delta; r) = \sup\{p \in [0, 1) : \delta_r^{\text{RP}}(\varepsilon; p) \leq \delta\},$$

obtained using binary search. As a baseline, we compute the leverage threshold *implicitly* enforced by the LSV-based analysis: Theorem 2 of [12] calibrates a ridge parameter w (equivalently $w^2 I$ added to $D^\top D$), and the condition $\sigma_{\min}(A) \geq w$ together with $\|x_i\|_2 \leq l$ implies

$$p_{\text{LSV}}(\varepsilon, \delta; r) \stackrel{\text{def}}{=} \ell_{\max}(A) \leq \frac{\varepsilon}{4(\sqrt{2r \log(4/\delta)} + \log(4/\delta))}.$$

Figure 2 plots the improvement factor $\frac{p_{\text{NDIS}}^*(\varepsilon, \delta; r)}{p_{\text{LSV}}(\varepsilon, \delta; r)}$ as a function of ε , for several values of r . A ratio larger than 1 means that our analysis guarantees (ε, δ) -DP while permitting a strictly larger worst-case leverage, translating into less required regularization. Across the full range of ε shown, the ratios are consistently above 1, demonstrating that the LSV-based condition controls privacy only through a conservative upper bound on leverage and can substantially over-regularize relative to an analysis that characterizes privacy directly in terms of leverage.

Empirical comparison with prior RP mechanisms. We complement the leverage-threshold comparison with an end-to-end evaluation against prior Gaussian-RP mechanisms, including the LSV-based calibrations of Sheffet [11], [12] and the recent GaussMix mechanism [24]⁷. We use the same linear-regression release task and utility metric, the test MSE, as in GaussMix. In addition, we measure the

7. A preliminary version of this work appeared online in 2023, prior to the appearance of GaussMix. We nevertheless include GaussMix as a closely related mechanism for Gaussian sketching and compare against it empirically, since it provides a recent refined analysis and implementation baseline for private linear regression via Gaussian sketches.

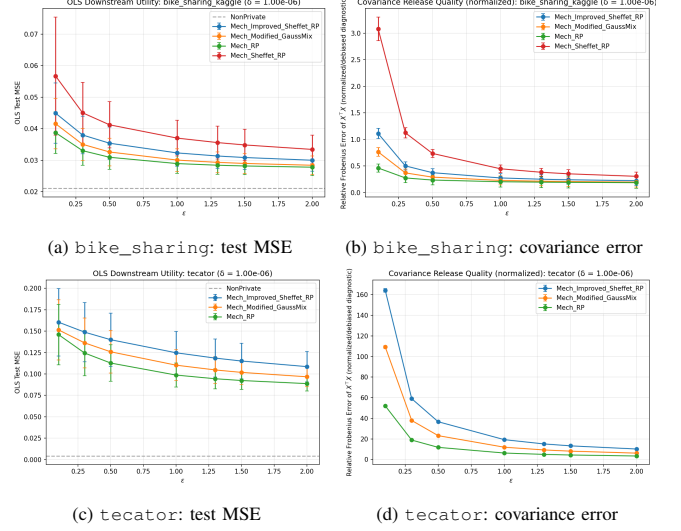


Figure 3. Empirical comparison of $\mathcal{M}_{\text{RP}}$ against prior RP-based mechanisms on linear-regression release tasks. Panels (a) and (c) report downstream test MSE; panels (b) and (d) report relative Frobenius error of the released covariance matrix.

utility by relative Frobenius error of the released covariance matrix, since the downstream linear-regression estimator is computed solely by post-processing this differentially private release. For each dataset and privacy level, we average over 250 random seeds with $\delta = 10^{-6}$ and $\varepsilon \in \{0.1, 0.3, 0.5, 1.0, 1.3, 1.5, 2.0\}$.

Figure 3 reports the comparison on resulting test MSE and covariance-release error. On *bike_sharing*, $\mathcal{M}_{\text{RP}}$ achieves 2.16%–6.82% lower test MSE than GaussMix, and on *tecator* it achieves 3.62%–10.50% lower test MSE. The direct covariance-release metric shows a larger separation: the relative Frobenius error improves over GaussMix by $1.06\times$ – $1.65\times$ on *bike_sharing* and by $1.82\times$ – $2.10\times$ on *tecator*. These gains are consistent with the sharper calibration provided by NDIS: the privacy guarantee is calibrated directly in (ε, δ) to the leverage parameter governing neighboring Gaussian-RP outputs, rather than passing through an RDP bound and conversion. The improvement over Sheffet-style LSV calibrations is larger still, since LSV controls leverage only indirectly, while NDIS calibrates to the privacy-relevant quantity itself.

6. Lifting any Gaussian-output algorithm to a DP mechanism

Many algorithms in modern statistics and ML naturally output a Gaussian distribution whose *mean and covariance both depend on the dataset*—for example, posterior or asymptotic normal approximations [5], [8], randomized estimators [6], and second-order procedures [7], [9]. However, it remains a question whether we can in general take advantage of this inherent (Gaussian) randomness for privacy. An answer to this question is not only of theoretical interest (i.e., feasibility), but can be a practical starting point for constructing/refining new DP mechanisms.

In this section, we present a ‘wrapper’ which calibrates a general Gaussian-output algorithm’s output distribution so that it is turned/‘lifted’ into a Gaussian-output DP mechanism. Importantly, this wrapper does not depend on how the algorithm is implemented, only on sensitivity bounds for how its output distribution changes on neighboring databases. Our notion of sensitivity generalizes that of the standard Gaussian-mechanism. Here, a Gaussian-output algorithm has NDIS sensitivity if one can bound *both* the mean shift and the covariance shift of neighboring outputs, and if the neighboring covariances satisfy a mild *Loewner-comparability* condition (required to apply our NDIS tools). Given this NDIS sensitivity exists (we show examples for useful Gaussian-output algorithms), our wrapper uses it to *calibrate/compute* the amount of additional noise needed to achieve (ε, δ) -DP.

Unlike the RP instantiation in Section 5, where NDIS yields an exact worst-case calibration in terms of leverage, the generic wrapper below is a conservative calibration framework for arbitrary Gaussian-output algorithms. The final mechanism uses easy-to-compute sensitivity bounds to enclose all neighboring Gaussian pairs, and then applies the NDIS-derived analytic upper bound on their indistinguishability to obtain an automatic worst-case DP guarantee.

We first introduce an abstraction of *Gaussian-output algorithms* (Definition 3) via an underlying deterministic map $q(D) = (\mu(D), \Sigma(D))$. We then describe a family of *additive-noise mechanisms* (cf. Definition 4) that increases/inflates the covariance of the output of these algorithms (which preserves Gaussianity). The core challenge is that privacy depends on how the *entire* Gaussian distribution changes across neighboring datasets; when the covariance is data-dependent, proving DP would in principle require controlling indistinguishability over *all* neighboring pairs, which is typically intractable.

NDIS makes this problem tractable via a *pairwise reduction*: as shown in Theorem 1, the IS for any fixed neighboring pair can be written as the scalar functional $\delta_\varepsilon(m, R)$ of an invariant pair (m, R) derived from two Gaussians (cf. Eq. 3). We then define a NDIS-based sensitivity model (Def. 5), and use it to upper-bound the collection of invariants (over pairs of neighboring databases) by an *envelope* that is tractable to work with. Finally, leveraging our NDIS tool for the bound on IS when covariances are Loewner-comparable (Theorem 2) allows us to simplify and upper bound $\delta_\varepsilon(m, R)$ via a small number of meaningful parameters. This gives us an efficient upper bound of the envelope’s worst-case privacy and an explicit (ε, δ) -DP guarantee. The resulting NDIS-Calibrated Gaussian Mechanism is shown in Figure 4.

6.1. Our NDIS-Calibrated Gaussian Mechanism

Definition 3 formalizes the class of algorithms studied in this section: randomized procedures whose output distribution is multivariate Gaussian, with both the mean and covariance could be data-dependent.

Definition 3 (Gaussian-output algorithm). *Let $\mathcal{D}$ be a dataset space with a remove/add adjacency relation. Let $m \in \mathcal{N}_{>0}$. A randomized algorithm $\mathcal{M}_q$ is called a Gaussian-output algorithm if there exists a deterministic map*

$$q : \mathcal{D} \rightarrow \mathbb{R}^m \times \mathbb{S}_{++}^m, \quad q(D) = (\mu(D), \Sigma(D)),$$

such that given any input $D \in \mathcal{D}$, $\mathcal{M}_q$ outputs $\mathcal{M}_q(D) \sim \mathcal{N}(\mu(D), \Sigma(D))$.

A natural way to privatize such outputs while preserving their Gaussianity is to add independent Gaussian noise. Definition 4 models this by a family of *additive-noise mechanisms* parameterized by a positive semi-definite (PSD) matrix $N \succeq 0$, which inflates the output covariance from $\Sigma(D)$ to $\Sigma(D) + N$. This family include the standard isotropic Gaussian noise, when $N = \sigma^2 I_m$.

Definition 4 (Family of additive-noise mechanisms for Gaussian-output algorithms). *Let $\mathcal{M}_q$ be a Gaussian-output algorithm (Definition 3). For any positive-semi-definite (PSD) matrix $N \succeq 0$, the additive-noise mechanism $\mathcal{M}_{N,q}$ is defined as:*

$$\mathcal{M}_{N,q}(D) \sim \mathcal{N}(\mu(D), \Sigma(D) + N).$$

Next, we connect the privacy of $\mathcal{M}_{N,q}$ to the NDIS toolbox. Concretely, Theorem 1 gives a representation of Gaussian indistinguishability, and Proposition 5 is an immediate specialization of that result to the two Gaussian distributions induced by a neighboring pair under additive Gaussian noise. Proposition 5 states that, for any ordered neighbors $D \sim D'$, the δ -curve of $\mathcal{M}_{N,q}$ is fully captured by a pair of invariants $(m_N(D, D'), R_N(D, D'))$ (Eq. 2) through the scalar functional $\delta_\varepsilon(m, R)$ (Eq. 3).

Proposition 5. *Let $q(D) = (\mu(D), \Sigma(D))$ be a Gaussian-output map and $N \succeq 0$, and let $\mathcal{M}_{N,q}$ (cf. Definition 4) be the additive-noise mechanism. For any ordered neighboring pair $D \sim D'$, define*

$$\begin{aligned} m_N(D, D') &\stackrel{\text{def}}{=} (\Sigma(D') + N)^{-1/2} (\mu(D) - \mu(D')), \\ R_N(D, D') &\stackrel{\text{def}}{=} (\Sigma(D') + N)^{-1/2} (\Sigma(D) + N) (\Sigma(D') + N)^{-1/2}. \end{aligned} \quad (2)$$

Define the NDIS function

$$\delta_\varepsilon(m, R) \stackrel{\text{def}}{=} \delta_{X,Y}(\varepsilon) \text{ where } X \sim \mathcal{N}(m, R), Y \sim \mathcal{N}(0, I_d). \quad (3)$$

Then, for every such (D, D') ,

$$\delta_{\mathcal{M}_{N,q}(D), \mathcal{M}_{N,q}(D')}(\varepsilon) = \delta_\varepsilon(m_N(D, D'), R_N(D, D')).$$

To turn the pairwise reduction into a DP guarantee, we must control the invariants $(m_N(D, D'), R_N(D, D'))$ uniformly over all neighboring pairs. Let

$$\mathcal{S}_{q,N} \stackrel{\text{def}}{=} \{(m_N(D, D'), R_N(D, D')) : D \sim D'\} \quad (4)$$

denote the set of feasible invariant pairs induced by (q, N) . By Proposition 5, calibrating N so that $\mathcal{M}_{N,q}$ is (ε, δ) -DP is equivalent to enforcing

$$\sup_{(m,R) \in \mathcal{S}_{q,N}} \delta_\varepsilon(m, R) \leq \delta.$$

which is, however, usually intractable because the feasible set $S_{q,N}$ can be highly complex for an arbitrary data-dependent Gaussian map q . We address this issue by replacing the exact worst-case condition with a single conservative check obtained from two relaxations: (i) we upper bound the unknown feasible set by an *envelope* $S_{q,N} \subseteq \mathcal{C}(q, N)$ via an NDIS-based sensitivity model (cf. Definition 5) which is tractable to work with, and (ii) we upper bound the NDIS functional itself by an analytic expression $\bar{\delta}_{LC}$ (cf. Definition 6) on that envelope. Combining the two reduces privacy calibration to verifying

$$\sup_{(m,R) \in \mathcal{C}(q,N)} \bar{\delta}_{LC}(\varepsilon; \cdot) \leq \delta,$$

which is efficiently computable and directly supports automatic noise selection.

Definition 5 introduces a model for the sensitivity of general Gaussian-output algorithms, which we call Loewner-comparable NDIS sensitivity, that we use to induce an envelope $\mathcal{C}(q, N)$ of $S_{q,N}$, which we can work with. Definition 5 is tailored to *Gaussian-output* queries in that it constrains exactly the NDIS invariants (m_N, R_N) appearing in the pairwise reduction. The *mean sensitivity* $\|m_N(D, D')\|_2 \leq \Delta$ bounds the largest standardized mean difference (i.e., a Mahalanobis distance under the noised covariance). The *Loewner-comparability* condition requires the two noised covariances to be ordered, which is precisely the structural regime in which our analytic NDIS bounds apply. Finally, the *covariance sensitivity* constraints $\|\ell_N\|_\infty \leq \rho_\infty$ and $|\sum_i \ell_{N,i}| \leq \nu$ bound, respectively, the worst per-direction multiplicative distortion and the aggregate log-determinant change.

We note that the Loewner-comparable NDIS sensitivity is “tight” as a modeling abstraction: since δ_ε depends on the output distribution only through (m_N, R_N) , the sensitivity conditions are stated directly in terms of these invariants. The conservatism enters only in two explicit steps: replacing the feasible set $S_{q,N}$ by a tractable envelope, and upper bounding the exact NDIS indistinguishability functional on this envelope by the analytic quantity δ_{LC} .

Definition 5 (Loewner-comparable NDIS sensitivity). Let $q(D) = (\mu(D), \Sigma(D))$ (corresponding to the output distribution of Gaussian-output algorithm) be a map with $\Sigma(D) \succ 0$ for all $D \in \mathcal{D}$, and fix a PSD matrix $N \succeq 0$. Let $(m_N(D, D'), R_N(D, D'))$ be as defined in Equation 2 for an ordered neighboring pair $D \sim D'$.

We say that q has $(\Delta, \rho_\infty, \nu, N)$ -Loewner-comparable NDIS sensitivity if for every ordered neighboring pair $D \sim D'$:

- 1) **Mean sensitivity:** $\|m_N(D, D')\|_2 \leq \Delta$.
- 2) **Loewner comparability:** $R_N(D, D') \succeq I$ or $R_N(D, D') \preceq I$.
- 3) **Covariance sensitivity:** Let $\tau_1, \dots, \tau_d$ be the eigenvalues of $R_N(D, D')$ and define $\ell_N(D, D') \stackrel{\text{def}}{=} (\log \tau_1, \dots, \log \tau_d)$. Then

$$\|\ell_N(D, D')\|_\infty \leq \rho_\infty, \quad \left| \sum_{i=1}^d \ell_N(D, D')_i \right| \leq \nu.$$

Definition 6 defines an analytic upper bound on $\delta_\varepsilon(m, R)$ over the envelope induced by Definition 5. Concretely, Theorem 2 shows that under Loewner comparability, the indistinguishability between two Gaussians can be upper bounded by splitting the privacy budget $\varepsilon = \varepsilon_1 + \varepsilon_2$ and separating the effect of (i) a *covariance shift* under a common mean and (ii) a *mean shift* under a common covariance. Definition 6 is a direct instantiation of this template in terms of the sensitivity parameters: the term $\bar{\delta}_{\text{cov}}(\varepsilon_1; \rho_\infty, \nu)$ upper bounds the covariance-shift contribution (cf. Lemma 2) using the hard covariance direction, when the log-eigenvalue statistics of the relative covariance satisfy $\|\ell\|_\infty \leq \rho_\infty$ and $|\sum_i \ell_i| \leq \nu$, while $\bar{\delta}_{\text{mean}}(\varepsilon_2; \Delta)$ is exactly the analytic form of the mean-shift bound (cf. Proposition 1) when the standardized mean difference is bounded by Δ .

As a result, once map q satisfies Loewner-comparable NDIS sensitivity with parameters $(\Delta, \rho_\infty, \nu)$, the worst-case privacy over the entire envelope $\mathcal{C}(q, \sigma I_m)$ can be controlled by the single scalar quantity $\bar{\delta}_{LC}(\varepsilon; \Delta, \rho_\infty, \nu)$, enabling an efficient calibration rule for σI_m in the mechanism of Figure 4.

Definition 6. Let $\varepsilon \geq 0$, $\Delta \geq 0$, $\rho_\infty \geq 0$ and $\nu \geq 0$. Define

$$\bar{\delta}_{LC}(\varepsilon; \Delta, \rho_\infty, \nu) \stackrel{\text{def}}{=} \inf_{\substack{\varepsilon_1, \varepsilon_2 \geq 0 \\ \varepsilon_1 + \varepsilon_2 = \varepsilon}} \left[\bar{\delta}_{\text{cov}}(\varepsilon_1; \rho_\infty, \nu) + e^{\varepsilon_1} \bar{\delta}_{\text{mean}}(\varepsilon_2; \Delta) \right],$$

where $\bar{\delta}_{\text{mean}}$ and $\bar{\delta}_{\text{cov}}$ are defined as follows.

Mean-shift upper bound. Define, for $\Delta \geq 0$,

$$\bar{\delta}_{\text{mean}}(\varepsilon; \Delta) \stackrel{\text{def}}{=} \begin{cases} \Phi\left(\frac{\Delta}{2} - \frac{\varepsilon}{\Delta}\right) - e^\varepsilon \Phi\left(-\frac{\Delta}{2} - \frac{\varepsilon}{\Delta}\right), & \Delta > 0, \\ 0, & \Delta = 0. \end{cases}$$

Covariance-shift upper bound. Define

$$c(\rho_\infty) \stackrel{\text{def}}{=} e^{\rho_\infty} - 1, \quad k(\rho_\infty) \stackrel{\text{def}}{=} \frac{1}{2c(\rho_\infty)} \in (0, \infty],$$

with the convention $k(\rho_\infty) = \infty$ when $\rho_\infty = 0$. For $s \in [0, k(\rho_\infty))$ and $u \geq 0$, define the scalar functions

$$\begin{aligned} \varphi_s(\ell) &\stackrel{\text{def}}{=} -s\ell - \frac{1}{2} \log(1 - 2s(e^\ell - 1)), \\ \psi_u(\ell) &\stackrel{\text{def}}{=} u\ell - \frac{1}{2} \log(1 + 2u(1 - e^{-\ell})). \end{aligned}$$

Next define the envelope functionals

$$\begin{aligned} A_\star(s; \rho_\infty, \nu) &\stackrel{\text{def}}{=} \sup \left\{ \sum_{i=1}^d \varphi_s(\ell_i) : 0 \leq \ell_i \leq \rho_\infty \ \forall i, \sum_{i=1}^d \ell_i \leq \nu \right\}, \\ B_\star(u; \rho_\infty, \nu) &\stackrel{\text{def}}{=} \sup \left\{ \sum_{i=1}^d \psi_u(\ell_i) : 0 \leq \ell_i \leq \rho_\infty \ \forall i, \sum_{i=1}^d \ell_i \leq \nu \right\}. \end{aligned}$$

Then define $\bar{\delta}_{\text{cov}}(\varepsilon; \rho_\infty, \nu)$ as

$$\begin{aligned} \bar{\delta}_{\text{cov}}(\varepsilon; \rho_\infty, \nu) &\stackrel{\text{def}}{=} \inf_{\substack{s \in [0, k(\rho_\infty)) \\ u \geq 0}} \left[\exp(-2\varepsilon s + A_\star(s; \rho_\infty, \nu)) \right. \\ &\quad \left. - e^\varepsilon \left(1 - \exp(2\varepsilon u + B_\star(u; \rho_\infty, \nu)) \right) \right]. \end{aligned}$$

We are now ready to state the NDIS-Calibrated Gaussian Mechanism (cf. Figure 4). The mechanism takes as input the

Input: A database $D \in \mathbb{R}^{n \times d}$; privacy parameters $\varepsilon \in \mathbb{R}_{\geq 0}$, $\delta \in [0, 1)$.
Output: A vector $\tilde{M} \in \mathbb{R}^m$
1) Use binary search to find σ^* such that

$$\bar{\delta}_{\text{LC}}(\varepsilon; \Delta(\sigma^*), \rho_\infty(\sigma^*), \nu(\sigma^*)) \leq \delta.$$

2) Return $\mathcal{M}_{\sigma^* I_m, q}(D)$

Figure 4. An NDIS-Calibrated Gaussian Mechanism for Loewner-comparable Gaussian-output algorithms $\mathcal{M}_q^{\text{NDIS}}$

TABLE 1. EVALUATION OF GAUSSIAN-OUTPUT WRAPPER ON BLR AND GPR AT $(\varepsilon, \delta) = (1, 10^{-5})$.

Alg.	Dataset	Metric	Non-private	Private (Wrapper)
BLR	Breast Cancer	Acc. $\uparrow$	0.9912	0.7649 ± 0.1229
GPR	Linnerud	Abs. err. $\downarrow$	0.4718	0.7563 ± 0.5290

Results average over five runs. The calibrated covariance inflation levels are $\sigma^* = 3.1002$ for BLR and $\sigma^* = 0.8213$ for GPR.

target privacy parameters (ε, δ) and adds *isotropic* Gaussian noise $N = \sigma I_m$ to the underlying Gaussian-output algorithm. The key step is to choose the smallest noise level σ^* that makes the worst-case privacy bound fall below δ . Concretely, for each candidate σ , the query-dependent sensitivity bounds produce parameters $(\Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma))$, and Definition 6 yields the upper bound

$$\bar{\delta}_{\text{LC}}(\varepsilon; \Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma))$$

on the worst-case indistinguishability over all neighboring pairs. For the sensitivity envelopes used, these quantities are monotone in σ , so the calibration predicate is monotone and σ^* can be located efficiently by binary search. Finally, the mechanism outputs $\mathcal{M}_{\sigma^* I_m, q}(D)$, i.e., a Gaussian draw with the same mean as the original algorithm and covariance inflated by $\sigma^* I_m$.

Theorem 5 (Proof is in Section G.1). *Let $\varepsilon \geq 0, \delta \in (0, 1)$. If q has $(\Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma), \sigma I_d)$ -Loewner-comparable NDIS sensitivity then the mechanism $\mathcal{M}_q^{\text{NDIS}}$ in Figure 4 is (ε, δ) -DP under the add/remove neighboring notion.*

6.2. Evaluations of Concrete Applications

We demonstrate that our general Gaussian wrapper retains nontrivial utility, through two specific Gaussian-output algorithms that have not received attention for their inherent privacy—Bayesian Logistic Regression and Gaussian Process Regression (though privacy-preserving *versions* of these functionalities exist). See App. C for the definitions of these algorithms and their NDIS sensitivities (Def. 5). While future work may optimize these generic wrapper mechanisms, our evaluations show a promising first step in leveraging existing ML algorithms for their inherent privacy. Table 1 evaluates the generic wrapper on BLR and GPR at $(\varepsilon, \delta) = (1, 10^{-5})$. For test accuracy (BLR), higher is better; for absolute prediction error at a fixed public query point (GPR), lower is better.

TABLE 2. NDIS AUDIT OF THE GAUSSIAN MECHANISM IN PARAMETRIC BLACK-BOX SETTING AT $\varepsilon = 1, \delta = 10^{-5}$.

Samples n	plug-in estimate	95% bootstrap upper
10^3	$2.46 \cdot 10^{-6}$	$1.97 \cdot 10^{-4}$
10^4	$6.69 \cdot 10^{-6}$	$3.20 \cdot 10^{-5}$
10^5	$8.64 \cdot 10^{-6}$	$1.37 \cdot 10^{-5}$

The mechanism is calibrated so that the exact pairwise δ value at $\varepsilon = 1$ is 10^{-5} . For all three sample sizes, the pointwise 95% bootstrap interval contains this exact value.

7. Whitebox Auditing Tool Via NDIS

We construct a tool for *white-box auditing* of mechanisms with Gaussian outputs. The tool does not require understanding of NDIS—only that we know the mean/covariance of the mechanism’s output on the tested inputs, and have access to a CDF oracle (e.g., Davies method)—which makes it a simple addition to any white-box auditor’s toolbox. Intuitively, our new tool accurately and deterministically estimates whether a mechanism whose output has such a mean/covariance could satisfy (ε, δ) -DP, through our Tool 1 (Thm. 1)—see App. D for more detail.

Auditing can also be performed when the output is Gaussian but its parameters are unknown. In this parametric/semi-black-box regime, the auditor samples the mechanism on the two neighboring inputs, fits a Gaussian distribution to each sample set, and plugs the fitted parameters into the NDIS formula to estimate $\delta_{\mathcal{M}(D), \mathcal{M}(D')}(\varepsilon)$. Since the Gaussian parameters are estimated from finitely many samples, we instead report a conservative interval for the pairwise δ -curve with confidence intervals, rather than a certificate.

We illustrate the tool on the Gaussian mechanism calibrated to $(\varepsilon, \delta) = (1, 10^{-5})$. In the white-box regime, NDIS recovers the exact pairwise privacy curve from the known parameters. In the parametric black-box regime, Table 2 shows that the plug-in estimate approaches the exact curve as the sample size increases. At $\varepsilon = 1$, the conservative pointwise 95% bootstrap upper estimate of δ decreases from $1.97 \cdot 10^{-4}$ at (sample size $n = 10^3$), to $3.20 \cdot 10^{-5}$ at ($n = 10^4$), and to $1.37 \cdot 10^{-5}$ at $n = 10^5$.

Auditing beyond Gaussian-output mechanisms. More broadly, the same auditing principle applies whenever an auditor can extract a Gaussian, or approximately Gaussian, statistic from a mechanism transcript: the statistic’s pairwise indistinguishability lower-bounds that of the full transcript. For example, in one-run DP-SGD auditing, canary-aligned scores have been used to obtain privacy lower bounds from a single training run [71], [72], [73]. Recent work further formalizes that, in the white-box DP-SGD setting, these canary-aligned signals naturally yield normalized sums that are asymptotically Gaussian [74]. This gives a concrete setting in which NDIS can be useful beyond mechanisms whose full output is itself Gaussian.

References

- [1] C. Dwork, F. McSherry, K. Nissim, and A. Smith, “Calibrating noise to sensitivity in private data analysis,” in *TCC 2006*, ser. LNCS,

- S. Halevi and T. Rabin, Eds., vol. 3876. Springer, Berlin, Heidelberg, Mar. 2006, pp. 265–284.
- [2] B. Balle, G. Barthe, and M. Gaboardi, “Privacy profiles and amplification by subsampling,” *Journal of Privacy and Confidentiality*, vol. 10, no. 1, 2020.
 - [3] —, “Privacy amplification by subsampling: Tight analyses via couplings and divergences,” in *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018*, 2018, pp. 6280–6290.
 - [4] C. Dwork and A. Roth, “The algorithmic foundations of differential privacy,” *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014.
 - [5] L. Tierney and J. B. Kadane, “Accurate approximations for posterior moments and marginal densities,” *Journal of the American Statistical Association*, vol. 81, no. 393, pp. 82–86, 1986.
 - [6] P. McCullagh and J. A. Nelder, *Generalized Linear Models*, 2nd ed. Chapman and Hall, 1989.
 - [7] A. W. van der Vaart, *Asymptotic Statistics*. Cambridge University Press, 1998.
 - [8] H. Rue, S. Martino, and N. Chopin, “Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations,” *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 71, no. 2, pp. 319–392, 2009.
 - [9] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.
 - [10] J. Blocki, A. Blum, A. Datta, and O. Sheffet, “The johnson-lindenstrauss transform itself preserves differential privacy,” in *Proceedings of the 53rd Annual IEEE Symposium on Foundations of Computer Science*. IEEE Computer Society, 2012, pp. 410–419.
 - [11] O. Sheffet, “Differentially private ordinary least squares,” in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 70. PMLR, 2017, pp. 3105–3114.
 - [12] —, “Old techniques in differentially private linear regression,” in *Algorithmic Learning Theory*, ser. Proceedings of Machine Learning Research, vol. 98. PMLR, 2019, pp. 788–826.
 - [13] R. Arora and J. Upadhyay, “On differentially private graph sparsification and applications,” in *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019*, 2019, pp. 13 378–13 389.
 - [14] J. Upadhyay and S. Upadhyay, “A framework for private matrix analysis in sliding window model,” in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 10 465–10 475.
 - [15] D. Kharkovskii, Z. Dai, and B. K. H. Low, “Private outsourced bayesian optimization,” in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119. PMLR, 2020, pp. 5231–5242.
 - [16] J. Upadhyay, “Differentially private linear algebra in the streaming model,” *CoRR*, vol. abs/1409.5414, 2014.
 - [17] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. The MIT Press, 2006.
 - [18] J. P. Imhof, “Computing the distribution of quadratic forms in normal variables,” *Biometrika*, vol. 48, no. 3–4, pp. 419–426, 1961.
 - [19] R. B. Davies, “Numerical inversion of a characteristic function,” *Biometrika*, vol. 60, no. 2, pp. 415–417, 1973.
 - [20] P. Duchesne and P. Lafaye de Micheaux, “Computing the distribution of quadratic forms: Further comparisons between the liu–tang–zhang approximation and exact methods,” *Computational Statistics & Data Analysis*, vol. 54, no. 4, pp. 858–862, 2010.
 - [21] P. Lafaye de Micheaux, P. Duchesne, R. B. Davies, R. W. Farebrother, J. Sheil, and I. O’Muircheartaigh, *CompQuadForm: Distribution Function of Quadratic Forms in Normal Variables*, 2025, r package, version 1.4.4. [Online]. Available: <https://CRAN.R-project.org/package=CompQuadForm>
 - [22] A. Das, “New methods to compute the generalized chi-square distribution,” *Journal of Statistical Computation and Simulation*, vol. 95, no. 12, pp. 2608–2642, 2025.
 - [23] C. Li, G. Miklau, M. Hay, A. McGregor, and V. Rastogi, “The matrix mechanism: Optimizing linear counting queries under differential privacy,” *The VLDB Journal*, vol. 24, no. 6, pp. 757–781, 2015.
 - [24] O. Lev, V. Srinivasan, M. Shenfeld, K. Ligett, A. Sekhari, and A. C. Wilson, “The gaussian mixing mechanism: Renyi differential privacy via gaussian sketches,” *CoRR*, vol. abs/2505.24603, 2025.
 - [25] S. Lan, Y. Ma, W. Huang, W. Wang, H. Yang, and P. Li, “DSTAGNN: Dynamic spatial-temporal aware graph neural network for traffic flow forecasting,” in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 2022, pp. 11 906–11 917.
 - [26] N. S. Sohoni, J. Dunnmon, G. Angus, A. Gu, and C. Ré, “No subclass left behind: Fine-grained robustness in coarse-grained classification problems,” in *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020*, 2020.
 - [27] D. Alabi, P. K. Kothari, P. Tankala, P. Venkat, and F. Zhang, “Privately estimating a gaussian: Efficient, robust, and optimal,” in *Proceedings of the 55th Annual ACM Symposium on Theory of Computing*. ACM, 2023, pp. 483–496.
 - [28] X. Zhang, H. Gu, L. Fan, K. Chen, and Q. Yang, “No free lunch theorem for security and utility in federated learning,” *ACM Transactions on Intelligent Systems and Technology*, vol. 14, no. 1, pp. 1:1–1:35, 2023.
 - [29] L. Devroye, A. Mehrabian, and T. Reddad, “The total variation distance between high-dimensional gaussians with the same mean,” *arXiv preprint arXiv:1810.08693*, 2018.
 - [30] J. Arbas, H. Ashtiani, and C. Liaw, “Polynomial time and private learning of unbounded gaussian mixture models,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 1018–1040.
 - [31] B. Bhusal, R. Chadha, A. P. Sistla, and M. Viswanathan, “Approximate algorithms for verifying differential privacy with gaussian distributions,” in *ACM CCS 2025*, C.-Y. Huang, J.-C. Chen, S.-P. Shieh, D. Lie, and V. Cortier, Eds. ACM Press, Oct. 2025, pp. 1439–1453.
 - [32] C. Dwork, F. McSherry, K. Nissim, and A. Smith, “Calibrating noise to sensitivity in private data analysis,” in *Theory of cryptography conference*. Springer, 2006, pp. 265–284.
 - [33] C. Dwork and A. Roth, “The algorithmic foundations of differential privacy,” *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014.
 - [34] B. Balle and Y.-X. Wang, “Improving the gaussian mechanism for differential privacy: Analytical calibration and optimal denoising,” in *Proceedings of the 35th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 80. PMLR, 2018, pp. 403–412.
 - [35] T. Chanyaswad, A. Dytso, H. V. Poor, and P. Mittal, “MVG mechanism: Differential privacy under matrix-valued query,” in *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security*. ACM, 2018, pp. 230–246.
 - [36] F. Liu, “Generalized gaussian mechanism for differential privacy,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 31, no. 4, pp. 747–756, 2019.

- [37] A. Blum and A. Roth, “Fast private data release algorithms for sparse queries,” in *Approximation, Randomization, and Combinatorial Optimization: Algorithms and Techniques*, ser. Lecture Notes in Computer Science, vol. 8096. Springer, 2013, pp. 395–410.
- [38] J. Upadhyay, “Circulant matrices and differential privacy,” *CoRR*, vol. abs/1410.2470, 2014.
- [39] —, “Randomness efficient fast-johnson-lindenstrauss transform with applications in differential privacy and compressed sensing,” *arXiv preprint arXiv:1410.2470*, 2014.
- [40] B. Bartan and M. Pilanci, “Distributed sketching methods for privacy preserving regression,” *CoRR*, vol. abs/2002.06538, 2020.
- [41] B. Balle, G. Barthe, and M. Gaboardi, “Privacy amplification by subsampling: Tight analyses via couplings and divergences,” in *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, NeurIPS 2018, December 3-8, 2018, Montréal, Canada*, S. Bengio, H. M. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, Eds., 2018, pp. 6280–6290. [Online]. Available: <https://proceedings.neurips.cc/paper/2018/hash/3b5020bb891119b9f5130f1fea9bd773-Abstract.html>
- [42] M. Park, J. R. Foulds, K. Chaudhuri, and M. Welling, “Variational bayes in private settings (VIPS),” *Journal of Artificial Intelligence Research*, vol. 68, pp. 109–157, 2020.
- [43] T. Steinke, “Composition of differential privacy & privacy amplification by subsampling,” *CoRR*, vol. abs/2210.00597, 2022.
- [44] M. Abadi, A. Chu, I. J. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, “Deep learning with differential privacy,” in *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*. ACM, 2016, pp. 308–318.
- [45] C. Hu, J. Li, Z. Liu, X. Guo, Y. Wei, X. Guang, G. Loukides, and C. Dong, “How to make private distributed cardinality estimation practical, and get differential privacy for free,” in *30th USENIX Security Symposium, USENIX Security 2021, August 11-13, 2021, M. D. Bailey and R. Greenstadt, Eds.* USENIX Association, 2021, pp. 965–982. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity21/presentation/hu-changhui>
- [46] A. D. Smith, S. Song, and A. Thakurta, “The flajolet-martin sketch itself preserves differential privacy: Private counting with minimal space,” in *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., 2020. [Online]. Available: <https://proceedings.neurips.cc/paper/2020/hash/e3019767b1b23f82883c9850356b71d6-Abstract.html>
- [47] T. Ou, R. Cummings, and M. A. Medina, “Thompson sampling itself is differentially private,” in *International Conference on Artificial Intelligence and Statistics 2024*, ser. Proceedings of Machine Learning Research, vol. 238. PMLR, 2024, pp. 1576–1584.
- [48] G. Barthe and F. Olmedo, “Beyond differential privacy: Composition theorems and relational logic for f-divergences between probabilistic programs,” in *Automata, Languages, and Programming: 40th International Colloquium*, ser. Lecture Notes in Computer Science, vol. 7966. Springer, 2013, pp. 49–60.
- [49] C. Dwork and G. N. Rothblum, “Concentrated differential privacy,” *CoRR*, vol. abs/1603.01887, 2016.
- [50] C. J. Lebeda, M. Regehr, G. Kamath, and T. Steinke, “Avoiding pitfalls for privacy accounting of subsampled mechanisms under composition,” *CoRR*, vol. abs/2405.20769, 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2405.20769>
- [51] D. M. Sommer, S. Meiser, and E. Mohammadi, “Privacy loss classes: The central limit theorem in differential privacy,” *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 2, pp. 245–269, 2019.
- [52] J. T. Wang, S. Mahloujifar, T. Wu, R. Jia, and P. Mittal, “A randomized approach to tight privacy accounting,” in *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023*, 2023.
- [53] Y. Zhu, J. Dong, and Y. Wang, “Optimal accounting of differential privacy via characteristic function,” in *International Conference on Artificial Intelligence and Statistics, AISTATS 2022, 28-30 March 2022, Virtual Event*, ser. Proceedings of Machine Learning Research, G. Camps-Valls, F. J. R. Ruiz, and I. Valera, Eds., vol. 151. PMLR, 2022, pp. 4782–4817. [Online]. Available: <https://proceedings.mlr.press/v151/zhu22c.html>
- [54] M. Bun and T. Steinke, “Concentrated differential privacy: Simplifications, extensions, and lower bounds,” in *Theory of Cryptography: 14th International Conference*, ser. Lecture Notes in Computer Science, vol. 9985, 2016, pp. 635–658.
- [55] I. Mironov, “Rényi differential privacy,” in *Proceedings of the 30th IEEE Computer Security Foundations Symposium*. IEEE Computer Society, 2017, pp. 263–275.
- [56] I. Mironov, K. Talwar, and L. Zhang, “Rényi differential privacy of the sampled gaussian mechanism,” *CoRR*, vol. abs/1908.10530, 2019. [Online]. Available: <http://arxiv.org/abs/1908.10530>
- [57] J. Dong, A. Roth, and W. J. Su, “Gaussian differential privacy,” *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 84, 2022.
- [58] D. Zhang and D. Kifer, “Lightdp: Towards automating differential privacy proofs,” in *Proceedings of the 44th ACM SIGPLAN Symposium on Principles of Programming Languages*, 2017, pp. 888–901.
- [59] Y. Wang, Z. Ding, G. Wang, D. Kifer, and D. Zhang, “Proving differential privacy with shadow execution,” in *Proceedings of the 40th ACM SIGPLAN Conference on Programming Language Design and Implementation*, 2019, pp. 655–669.
- [60] Y. Wang, Z. Ding, D. Kifer, and D. Zhang, “CheckDP: An automated and integrated approach for proving differential privacy or finding precise counterexamples,” in *ACM CCS 2020*, J. Ligatti, X. Ou, J. Katz, and G. Vigna, Eds. ACM Press, Nov. 2020, pp. 919–938.
- [61] H. Zhang, E. Roth, A. Haebleren, B. C. Pierce, and A. Roth, “Testing differential privacy with dual interpreters,” *Proc. ACM Program. Lang.*, vol. 4, no. OOPSLA, nov 2020.
- [62] S. Wang, L. Xiang, B. Cheng, Z. Ji, T. Sun, and X. Wang, “Curator attack: When blackbox differential privacy auditing loses its power,” in *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security*, 2024, pp. 3540–3554.
- [63] B. Bichsel, T. Gehr, D. Drachler-Cohen, P. Tsankov, and M. T. Vechev, “DP-finder: Finding differential privacy violations by sampling and optimization,” in *ACM CCS 2018*, D. Lie, M. Mannan, M. Backes, and X. Wang, Eds. ACM Press, Oct. 2018, pp. 508–524.
- [64] Y. Lu, M. Magdon-Ismail, Y. Wei, and V. Zikas, “Eureka: A general framework for black-box differential privacy estimators,” in *2024 IEEE Symposium on Security and Privacy*, 2024, pp. 169–169.
- [65] X. Liu and S. Oh, “Minimax optimal estimation of approximate differential privacy on neighboring databases,” *Advances in neural information processing systems*, vol. 32, 2019.
- [66] Ö. Askin, H. Dette, M. Dunsche, T. Kutta, Y. Lu, Y. Wei, and V. Zikas, “General-purpose f-dp estimation and auditing in a black-box setting,” in *USENIX Security Symposium*, 2025.
- [67] P. Kairouz, B. McMahan, S. Song, O. Thakkar, A. Thakurta, and Z. Xu, “Practical and private (deep) learning without sampling or shuffling,” in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 5213–5225.
- [68] S. Denisov, H. B. McMahan, J. Rush, A. D. Smith, and A. G. Thakurta, “Improved differential privacy for SGD via optimal private linear operators on adaptive streams,” in *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022*, 2022.
- [69] N. Halko, P. Martinsson, and J. A. Tropp, “Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions,” *SIAM Review*, vol. 53, no. 2, pp. 217–288, 2011.

- [70] D. P. Woodruff, “Sketching as a tool for numerical linear algebra,” *Foundations and Trends in Theoretical Computer Science*, vol. 10, no. 1–2, pp. 1–157, 2014.
- [71] M. Nasr, J. Hayes, T. Steinke, B. Balle, F. Tramèr, M. Jagielski, N. Carlini, and A. Terzis, “Tight auditing of differentially private machine learning,” in *USENIX Security 2023*, J. A. Calandrino and C. Troncoso, Eds. USENIX Association, Aug. 2023, pp. 1631–1648. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity23/presentation/nasr>
- [72] T. Steinke, M. Nasr, and M. Jagielski, “Privacy auditing with one (1) training run,” *CoRR*, vol. abs/2305.08846, 2023.
- [73] S. Mahloujifar, L. Melis, and K. Chaudhuri, “Auditing f -differential privacy in one run,” *CoRR*, vol. abs/2410.22235, 2024.
- [74] A. Agrawal, Y. Wei, J. Singh, M. Magdon-Ismail, and V. Zikas, “Let’s ask gauss: Improved one-run privacy auditing,” 2026.
- [75] R. Vershynin, *High-Dimensional Probability: An Introduction with Applications in Data Science*, ser. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2018, vol. 47.

Appendix A. Proof for Lemma 1

Proof of Lemma 1. Fix a set $\mathcal{S} \subseteq \mathbb{R}^d$. Let p_X and p_Y denote the densities of X and Y . We start from the density formulas.

$$\Pr[X \in \mathcal{S}] = \int_{\mathcal{S}} \frac{\exp\left(-\frac{1}{2}(x - \mu_1)^T \Sigma_1^{-1}(x - \mu_1)\right)}{(2\pi)^{d/2} \det(\Sigma_1)^{1/2}} dx,$$

$$\Pr[Y \in \mathcal{S}] = \int_{\mathcal{S}} \frac{\exp\left(-\frac{1}{2}(x - \mu_2)^T \Sigma_2^{-1}(x - \mu_2)\right)}{(2\pi)^{d/2} \det(\Sigma_2)^{1/2}} dx.$$

Plugging the above into $\Pr[X \in \mathcal{S}] - e^\varepsilon \Pr[Y \in \mathcal{S}]$, we have that

$$\begin{aligned} & \Pr[X \in \mathcal{S}] - e^\varepsilon \Pr[Y \in \mathcal{S}] \\ &= \int_{\mathcal{S}} \left[\frac{1}{(2\pi)^{d/2} \det(\Sigma_1)^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_1)^T \Sigma_1^{-1}(x - \mu_1)\right) \right. \\ & \quad \left. - \frac{e^\varepsilon}{(2\pi)^{d/2} \det(\Sigma_2)^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_2)^T \Sigma_2^{-1}(x - \mu_2)\right) \right] dx. \end{aligned}$$

Performing a change of variable,

$$x = \mu_1 + \Sigma_1^{1/2} z, \quad z = \Sigma_1^{-1/2}(x - \mu_1), \quad dx = \det(\Sigma_1)^{1/2} dz.$$

and let

$$\mathcal{S}_z \stackrel{\text{def}}{=} \Sigma_1^{-1/2}(\mathcal{S} - \mu_1) = \{z \in \mathbb{R}^d : \mu_1 + \Sigma_1^{1/2} z \in \mathcal{S}\}.$$

gives

$$\begin{aligned} & \Pr[X \in \mathcal{S}] - e^\varepsilon \Pr[Y \in \mathcal{S}] \\ &= \int_{\mathcal{S}_z} \left[\frac{1}{(2\pi)^{d/2} \det(\Sigma_1)^{1/2}} \exp\left(-\frac{1}{2}z^T z\right) - \frac{e^\varepsilon}{(2\pi)^{d/2} \det(\Sigma_2)^{1/2}} \right. \\ & \quad \cdot \exp\left(-\frac{1}{2}(\Sigma_1^{1/2} z + \Delta\mu)^T \Sigma_2^{-1}(\Sigma_1^{1/2} z + \Delta\mu)\right) \Big] \det(\Sigma_1)^{1/2} dz \\ &= \int_{\mathcal{S}_z} \frac{\exp(-\frac{1}{2}z^T z)}{(2\pi)^{d/2}} \left[1 - \frac{e^\varepsilon \det(\Sigma_1)^{1/2}}{\det(\Sigma_2)^{1/2}} \right. \\ & \quad \cdot \exp\left(-\frac{1}{2}(\Sigma_1^{1/2} z + \Delta\mu)^T \Sigma_2^{-1}(\Sigma_1^{1/2} z + \Delta\mu) + \frac{1}{2}z^T z\right) \Big] dz. \end{aligned}$$

Define $\psi_\varepsilon(z)$ as the log of the exponential factor inside the brackets:

$$\begin{aligned} \psi_\varepsilon(z) &\stackrel{\text{def}}{=} \varepsilon + \frac{1}{2} \log \frac{\det \Sigma_1}{\det \Sigma_2} - \frac{1}{2}(\Sigma_1^{1/2} z + \Delta\mu)^T \Sigma_2^{-1}(\Sigma_1^{1/2} z + \Delta\mu) + \frac{1}{2}z^T z \\ &= c(\varepsilon) - \Delta\mu^T \Sigma_2^{-1} \Sigma_1^{1/2} z + \frac{1}{2}z^T \left(I_d - \Sigma_1^{1/2} \Sigma_2^{-1} \Sigma_1^{1/2} \right) z \\ &= c(\varepsilon) - \Delta\mu^T \Sigma_2^{-1} \Sigma_1^{1/2} z + \frac{1}{2}z^T (I_d - M) z. \end{aligned}$$

Then the bracket becomes $1 - \exp(\psi_\varepsilon(z))$, and we obtain

$$\Pr[X \in \mathcal{S}] - e^\varepsilon \Pr[Y \in \mathcal{S}] = \int_{\mathcal{S}_z} \frac{\exp(-\frac{1}{2}z^T z)}{(2\pi)^{d/2}} (1 - \exp(\psi_\varepsilon(z))) dz.$$

Now diagonalize $M = U \text{diag}(\tau_1, \dots, \tau_d) U^T$ and let $v = U^T z$. Since U is orthogonal, $dz = dv$ and $z^T z = v^T v$, and moreover

$$z^T (I_d - M) z = v^T \left(I_d - \text{diag}(\tau_1, \dots, \tau_d) \right) v = \sum_{i=1}^d a_i v_i^2.$$

Also,

$$\begin{aligned} -\Delta\mu^T \Sigma_2^{-1} \Sigma_1^{1/2} z &= -\Delta\mu^T \Sigma_2^{-1} \Sigma_1^{1/2} U v \\ &= (-U^T \Sigma_1^{1/2} \Sigma_2^{-1} \Delta\mu)^T v = b^T v. \end{aligned}$$

Therefore $\psi_\varepsilon(z) = g_\varepsilon(v)$, where

$$g_\varepsilon(v) = c(\varepsilon) + b^T v + \frac{1}{2} \sum_{i=1}^d a_i v_i^2.$$

Let $\mathcal{T} \stackrel{\text{def}}{=} U^T \mathcal{S}_z$. Then

$$\begin{aligned} & \Pr[X \in \mathcal{S}] - e^\varepsilon \Pr[Y \in \mathcal{S}] \\ &= \int_{\mathcal{T}} \frac{\exp(-\frac{1}{2}v^T v)}{(2\pi)^{d/2}} (1 - \exp(g_\varepsilon(v))) dv. \end{aligned}$$

To maximize this expression over $\mathcal{S}$ (equivalently, over $\mathcal{T}$), observe that the Gaussian density factor is nonnegative, hence the integral is maximized by taking $\mathcal{T}$ to include exactly the points where the integrand $(1 - \exp(g_\varepsilon(v)))$ is positive, i.e., where $g_\varepsilon(v) \leq 0$. Thus the optimal value is

$$\begin{aligned} \delta_{X,Y}(\varepsilon) &= \int_{\mathbb{R}^d} \frac{\exp(-\frac{1}{2}v^T v)}{(2\pi)^{d/2}} \left(1 - \exp(g_\varepsilon(v)) \right)_+ dv \\ &= \mathbb{E} \left[\left(1 - \exp(g_\varepsilon(Z)) \right)_+ \right], \end{aligned}$$

where $Z \sim \mathcal{N}(0, I_d)$. This concludes the proof. $\square$

Appendix B. Optimizing $\mathcal{M}_{RP}$ via Subsampling and Propose-Test-Release (PTR)

We have shown that the NDIS lemma gives an *exact* characterization of the IS for random projection given neighboring databases, which in turn yields a method to calibrate $\mathcal{M}_{RP}$ to satisfy (ε, δ) -DP. Such *global* calibration, however, is inherently conservative: it injects noise uniformly over *all* databases in the universe, and thus can over-regularize

well-conditioned instances. In this section, we show that the NDIS-based analysis composes cleanly with today’s standard DP toolkit, enabling *database-agnostic* privacy guarantees while delivering *database-dependent* utility improvements.

Input: A database $D \in \mathbb{R}^{n \times d}$, target dimension $r \geq 1$; privacy parameters $\varepsilon \in \mathbb{R}_{\geq 0}$, $\delta \in [0, 1]$; a Poisson subsampling rate $1 \geq q > \delta$; and a public parameter $l \in \mathbb{R}_{>0}$ that upper bounds the row norm in the record universe.

Output: A matrix $\tilde{M} \in \mathbb{R}^{d \times r}$.

- 1) Construct a sub-database D_S by including each row of D independently with probability q .
- 2) Set $\varepsilon_0 = \ln\left(1 + \frac{e^\varepsilon - 1}{q}\right)$, and $\delta_0 = \frac{\delta}{q}$.
- 3) Return $\tilde{M} \leftarrow \mathcal{M}_{RP}(D_S; r, \varepsilon_0, \delta_0, l)$, where $\mathcal{M}_{RP}$ is defined in Figure 1.

Figure 5. Poisson-subsampled wrapper $\mathcal{M}_{RP}^{\text{Pois}}$ for $\mathcal{M}_{RP}$.

Figure 5 presents a Poisson-subsampled wrapper, denoted $\mathcal{M}_{RP}^{\text{Pois}}$. On input D , the $\mathcal{M}_{RP}^{\text{Pois}}$ first forms a random sub-database D_S by including each row independently with probability q , and then runs the baseline mechanism $\mathcal{M}_{RP}$ on D_S with *weaker* privacy parameters $(\varepsilon_0, \delta_0)$ chosen so that the overall output remains (ε, δ) -DP. Theorem 6 states that $\mathcal{M}_{RP}^{\text{Pois}}$ is (ε, δ) -DP.

Subsampling improves utility in two complementary ways. From a privacy perspective, subsampling making the privacy “event” that distinguishes neighboring databases occurs less often, so that we can afford a weaker level of regularization inside $\mathcal{M}_{RP}$. From a utility perspective, when the dataset is sufficiently redundant so that each record is, in aggregate, representable by the others, dropping a random $(1 - q)$ fraction of rows has limited effect on the same geometric signal $(D^\top D)$ that drives the random projection.

Theorem 6 (Proof is in Section F.3). *Let $\varepsilon \geq 0, \delta \in (0, 1)$. The mechanism $\mathcal{M}_{RP}^{\text{Pois}}$ in Figure 5 is (ε, δ) -DP under the standard add/remove neighboring notion.*

Figure 6 presents a PTR-based wrapper, denoted as $\mathcal{M}_{RP}^{\text{PTR}}$. It follows classical *Propose–Test–Release* (PTR) paradigm, and the idea is: i) *proposing* a data-dependent condition under which random projection is inherently more stable when removing a record; here a larger minimum eigenvalue $\lambda_{\min}(D^\top D)$ implies smaller leverage scores ii) *testing* this condition privately by releasing a conservative lower bound λ_{lb} on $\lambda_{\min}(D^\top D)$ with failure probability δ_{ptr} ; the lower bound is derived from a differentially private query on $\lambda_{\min}$ iii) *releasing* a projection computed on an augmented database $\bar{D} = [D; \sqrt{\lambda} I_d]$ with the ridge level λ chosen using the lower bound λ_{lb} . In particular, the ridge is set to satisfy the same NDIS-based indistinguishability constraint as used in Figure 1, while the privacy budget is split between the test and release steps. Theorem 7 states that $\mathcal{M}_{RP}^{\text{PTR}}$ is (ε, δ) -DP.

Theorem 7 (Proof is in Section F.3). *Let $\varepsilon \geq 0, \delta \in (0, 1)$. The mechanism $\mathcal{M}_{RP}^{\text{PTR}}$ in Figure 6 is (ε, δ) -DP under the*

Input: A database $D \in \mathbb{R}^{n \times d}$, target dimension $r \geq 1$; privacy parameters $\varepsilon \in \mathbb{R}_{\geq 0}$, $\delta \in [0, 1]$; and a public parameter $l \in \mathbb{R}_{>0}$ that upper bounds the row norm in the record universe.

Parameters: A privacy split $\delta = \delta_R + \delta_T + \delta_{\text{ptr}}$ and an eigenvalue-noise scale $\tau \in \mathbb{R}_{>0}$.

Output: A matrix $\tilde{M} \in \mathbb{R}^{d \times r}$.

- 1) Use binary search to find $\varepsilon_T \geq 0$ such that

$$\Phi\left(\frac{l^2}{2\tau} - \frac{\varepsilon_T \cdot \tau}{l^2}\right) - e^{\varepsilon_T} \Phi\left(-\frac{l^2}{2\tau} - \frac{\varepsilon_T \cdot \tau}{l^2}\right) \leq \delta_T$$

$$\text{Set } \varepsilon_R = \max\{\varepsilon - \varepsilon_T, 0\}.$$

- 2) Use binary search to find $p^* \in (0, 1)$ such that

$$\delta_r^{\text{gRP}}(\varepsilon_R; p^*) \leq \delta_R,$$

where $\delta_r^{\text{gRP}}(\cdot; \cdot)$ is defined in Proposition 4.

- 3) Let $\lambda = \lambda_{\min}(D^\top D)$, and $\alpha = \tau \cdot \Phi^{-1}(1 - \delta_{\text{ptr}})$. Set

$$\lambda_{\text{lb}} = \max\{\lambda + \eta - \alpha, 0\}, \quad \text{where } \eta \sim \mathcal{N}(0, \tau^2).$$

- 4) Set the ridge level $\lambda = \max\left\{\frac{l^2}{p^*} - \lambda_{\text{lb}}, 0\right\}$, and define the augmented (regularized) database

$$\bar{D} \stackrel{\text{def}}{=} \begin{bmatrix} D \\ \sqrt{\lambda} I_d \end{bmatrix} \in \mathbb{R}^{(n+d) \times d}.$$

- 5) Return $\tilde{M} = \text{gRP}_r(\bar{D})$.

Figure 6. PTR-based wrapper $\mathcal{M}_{RP}^{\text{PTR}}$ for $\mathcal{M}_{RP}$.

standard add/remove neighboring notion.

Proposition 6 states that, $\mathcal{M}_{RP}^{\text{PTR}}$ improves utility on databases that has a large enough $\lambda_{\min}(D^\top D)$. Informally, whenever the data geometry is sufficiently favorable, the (small) privacy cost of privately estimating a lower bound on $\lambda_{\min}(D^\top D)$ is repaid by a strictly less conservative release, i.e., the strictly less regularization in random projection. More precisely, if $\lambda_{\min}(D^\top D)$ exceeds a concrete threshold (capturing the “slack” needed to pay for the noisy lower-bound test, via α , and the NDIS-calibrated noise scale, via l^2/p^*), then the PTR-selected ridge level λ_{ptr} is strictly smaller than the baseline ridge λ_{rp} used by $\mathcal{M}_{RP}$.

Proposition 6. *Let λ_{rp} denote the ridge level used by $\mathcal{M}_{RP}$ (cf. Step 2 of Figure 1). Let $(\varepsilon_R, p^*, \alpha, \lambda_{\text{lb}})$ be the variables defined in $\mathcal{M}_{RP}^{\text{PTR}}$ (cf. Steps 1–3 of Figure 6). If a database D satisfies*

$$\lambda_{\min}(D^\top D) > \alpha + \frac{l^2}{p^*} - \lambda_{\text{rp}},$$

then the PTR ridge level $\lambda_{\text{ptr}} = \max\left\{\frac{l^2}{p^} - \lambda_{\text{lb}}, 0\right\}$ is strictly smaller than λ_{rp} .*

Appendix C.

Applications of our General Wrapper, Sec. 6.2

Throughout this appendix, n denotes a public upper bound on the dataset size used for calibration. Under

add/remove adjacency, all sensitivity bounds below should be read with $|D|, |D'| \leq n$ for the ordered neighboring pair.

Definition 7 (Bayesian Logistic Regression surrogate as a Gaussian-output algorithm (BLR)). *Let $\lambda > 0$ be a regularization parameter and $\mathcal{D} = (\mathbb{R}^d \times \{-1, 1\})^n$ be the dataset space. The Böhning-style Bayesian Logistic Regression surrogate, denoted $\mathcal{M}_{q_{\text{BLR}}}$, is a Gaussian-output algorithm (cf. Definition 3) determined by the map q_{BLR} . Given a dataset $D = \{(x_i, y_i)\}_{i=1}^n \in \mathcal{D}$, define the regularized negative log-likelihood*

$$\mathcal{L}(\theta; D) \stackrel{\text{def}}{=} \sum_{i=1}^n \ln(1 + \exp(-y_i x_i^\top \theta)) + \frac{\lambda}{2} \|\theta\|_2^2.$$

The map $q_{\text{BLR}} : \mathcal{D} \rightarrow \mathbb{R}^d \times \mathbb{S}_{++}^d$ is defined as $q_{\text{BLR}}(D) = (\hat{\theta}, \Sigma)$, where:

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^d} \mathcal{L}(\theta; D), \quad \Sigma = \left(\lambda I_d + \frac{1}{4} \sum_{i=1}^n x_i x_i^\top \right)^{-1}.$$

Proposition 7 (Proof is in Section G.2). *Let $l \in \mathbb{R}_{>0}$ be the row norm upper bound, i.e., for every $(x, y) \in \mathbb{R}^d \times \{-1, 1\}$, $\|x\|_2 \leq l$. Then q_{BLR} has $(\Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma), \sigma I_d)$ -Loewner-comparable NDIS sensitivity, where:*

$$\Delta(\sigma) = \frac{l/\lambda}{\sqrt{\sigma + \frac{1}{\lambda + \frac{n l^2}{4}}}}, \quad \rho_\infty(\sigma) = \log\left(\frac{\sigma + 1/\lambda}{\sigma + 1/(\lambda + \frac{n l^2}{4})}\right)$$

$$\nu(\sigma) = d \rho_\infty(\sigma).$$

Definition 8 (Gaussian Process Regression [17] as a Gaussian-output algorithm (GPR)). *Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the input domain. Fix a positive definite kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ and a noise variance $\sigma_n^2 > 0$. Fix a public query point $x_* \in \mathcal{X}$. The Gaussian Process Regression (GPR) mechanism, denoted $\mathcal{M}_{q_{\text{GPR}}}$, is a Gaussian-output algorithm (cf. Definition 3) determined by the map q_{GPR} .*

Given a dataset $D = \{(x_i, y_i)\}_{i=1}^n \in (\mathcal{X} \times \mathbb{R})^n$, define the Gram matrix $K \in \mathbb{R}^{n \times n}$ and the kernel vector $k_* \in \mathbb{R}^n$ by

$$K_{ij} \stackrel{\text{def}}{=} k(x_i, x_j), \quad (k_*)_i \stackrel{\text{def}}{=} k(x_*, x_i), \quad k_{**} \stackrel{\text{def}}{=} k(x_*, x_*).$$

Let $A \stackrel{\text{def}}{=} K + \sigma_n^2 I_n$. The map $q_{\text{GPR}} : (\mathcal{X} \times \mathbb{R})^n \rightarrow \mathbb{R} \times \mathbb{R}$ is defined as $q_{\text{GPR}}(D) = (\mu_*, \Sigma_*)$, where

$$\mu_* \stackrel{\text{def}}{=} k_*^\top A^{-1} y, \quad \Sigma_* \stackrel{\text{def}}{=} k_{**} - k_*^\top A^{-1} k_* \in \mathbb{R}.$$

Proposition 8 (Proof is in Section G.2). *Let $B \in \mathbb{R}_{>0}$ be a public upper bound on the label magnitude, i.e., for every $(x, y) \in \mathcal{X} \times \mathbb{R}$, we have $|y| \leq B$. Let $l \in \mathbb{R}_{>0}$ be a public upper bound on the kernel diagonal, i.e., for every $x \in \mathcal{X}$, we have $k(x, x) \leq l^2$. Then q_{GPR} has $(\Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma), \sigma)$ -Loewner-comparable NDIS sensitivity, where*

$$\Delta(\sigma) = \frac{lB\sqrt{n}}{\sigma_n \sqrt{\sigma}}, \quad \rho_\infty(\sigma) = \log\left(\frac{\sigma + l^2}{\sigma}\right), \quad \nu(\sigma) = \rho_\infty(\sigma).$$

Appendix D. White-box Auditing, Section 7

Below, we formalize how to (empirically) estimate $\delta_{X,Y}(\varepsilon)$ through the connection between NDIS and generalized- χ^2 distribution, and thus enable white-box auditing of a pair of Gaussians. We define a CDF oracle (with some error η_0) as the following.

Definition 9 (CDF oracle for the generalized- χ^2 distribution). *A CDF oracle for the generalized- χ^2 distribution with error η_0 is one which, on any generalized- χ^2 distribution X , outputs $\hat{p}$ such that*

$$|\hat{p} - \Pr(X \leq 0)| \leq \eta_0$$

Given a CDF oracle described above, we can construct an IS estimator, which can be purposed into a white-box auditor by treating the estimator's output as a lower-bound on the tested mechanism's DP.

Theorem 8. *Suppose there exists a CDF oracle for the generalized- χ^2 distribution with error $\eta_0 = \frac{\eta}{2(1+e^\varepsilon)}$. Then we can construct a deterministic algorithm which, for any X, Y, ε , outputs $\underline{\delta}$ such that*

$$\underline{\delta} \leq \delta_{X,Y}(\varepsilon) \quad \text{and} \quad \delta_{X,Y}(\varepsilon) - \underline{\delta} \leq \eta.$$

In particular, $\underline{\delta}$ is an η -accurate certified lower bound on $\delta_{X,Y}(\varepsilon)$.

Proof of Theorem 8. We define a deterministic algorithm that compute an η -accurate lower bound using the CDF oracle given in Definition 9. For simplicity, let $p_0 = \Pr[g_\varepsilon(Z) \leq 0]$ and $p_1 = \Pr[g_\varepsilon(\tilde{Z}) \leq 0]$. Since we have shown that both $g_\varepsilon(Z)$ and $g_\varepsilon(\tilde{Z})$ are generalized- χ^2 , then each probability p_0, p_1 can be evaluated up to additive error by one CDF oracle call.

Let the per-call accuracy $\eta_0 = \frac{\eta}{2(1+e^\varepsilon)}$. Then, the deterministic algorithm is defined as follows:

- 1) Invoke the η_0 -generalized- χ^2 CDF oracle on the instance corresponding to $\Pr[g_\varepsilon(Z) \leq 0]$, and obtain an estimate $\hat{p}_0$ such that $|\hat{p}_0 - p_0| \leq \eta_0$.
- 2) Invoke the η_0 -generalized- χ^2 CDF oracle on the instance corresponding to $\Pr[g_\varepsilon(\tilde{Z}) \leq 0]$, and obtain an estimate $\hat{p}_1$ such that $|\hat{p}_1 - p_1| \leq \eta_0$.
- 3) Set $\hat{\delta} = \hat{p}_0 - e^\varepsilon \hat{p}_1$ and output

$$\underline{\delta} = \max\left\{0, \hat{\delta} - \eta/2\right\}.$$

Correctness of the Algorithm.. By the oracle guarantee, $|\hat{p}_0 - p_0| \leq \eta_0, |\hat{p}_1 - p_1| \leq \eta_0$. Therefore,

$$\begin{aligned} |\hat{\delta} - \delta_{X,Y}(\varepsilon)| &= |(\hat{p}_0 - p_0) - e^\varepsilon(\hat{p}_1 - p_1)| \\ &\leq |\hat{p}_0 - p_0| + e^\varepsilon |\hat{p}_1 - p_1| \\ &\leq (1 + e^\varepsilon) \eta_0 = \frac{\eta}{2}. \end{aligned}$$

It follows that $\hat{\delta} - \eta/2 \leq \delta_{X,Y}(\varepsilon) \leq \hat{\delta} + \eta/2$. Combining all together, we then complete the proof of Theorem 8. $\square$

Appendix E.

Missing Proofs for NDIS toolbox

E.1. Proof for Theorem 1

Proof of Theorem 1. Let $a \in \mathbb{R}^d$, $b \in \mathbb{R}^d$, and $c(\varepsilon) \in \mathbb{R}$ be as in Lemma 1, and recall that for $Z \sim \mathcal{N}(0, I_d)$,

$$\delta_{X,Y}(\varepsilon) = \mathbb{E} \left[(1 - \exp(g_\varepsilon(Z)))_+ \right], \text{ where}$$

$$g_\varepsilon(z) = c(\varepsilon) + b^\top z + \frac{1}{2} \sum_{i=1}^d a_i z_i^2.$$

Then, we have that

$$\begin{aligned} \delta_{X,Y}(\varepsilon) &= \mathbb{E} \left[(1 - \exp(g_\varepsilon(Z)))_+ \right] \\ &= \mathbb{E} \left[(1 - \exp(g_\varepsilon(Z))) \mathbf{1}_{\{g_\varepsilon(Z) \leq 0\}} \right] \\ &= \Pr[g_\varepsilon(Z) \leq 0] - \mathbb{E} \left[e^{g_\varepsilon(Z)} \mathbf{1}_{\{g_\varepsilon(Z) \leq 0\}} \right]. \end{aligned}$$

Our first step is to rewrite $\delta_{X,Y}(\varepsilon)$ in terms of two quadratic-form CDF values. Denote $D = \text{diag}(\tau_1, \dots, \tau_d)$, where $a_i = 1 - \tau_i$, and define

$$m \stackrel{\text{def}}{=} D^{-1}b, \quad \tilde{Z} \sim \mathcal{N}(m, D^{-1}).$$

Let $\varphi(z) = (2\pi)^{-d/2} \exp(-\|z\|_2^2/2)$ be the standard Gaussian density on $\mathbb{R}^d$. Express $g_\varepsilon(z)$ in quadratic form:

$$g_\varepsilon(z) = c(\varepsilon) + b^\top z + \frac{1}{2} z^\top (I - D) z.$$

Indeed, performing change of variable gives

$$\begin{aligned} &\mathbb{E} \left[e^{g_\varepsilon(Z)} \mathbf{1}_{\{g_\varepsilon(Z) \leq 0\}} \right] \\ &= \int_{g_\varepsilon(z) \leq 0} e^{g_\varepsilon(z)} \varphi(z) dz \\ &= \int_{g_\varepsilon(z) \leq 0} (2\pi)^{-d/2} \exp(c(\varepsilon) + b^\top z - \frac{1}{2} z^\top D z) dz \\ &= \int_{g_\varepsilon(z) \leq 0} (2\pi)^{-d/2} \\ &\quad \cdot \exp(c(\varepsilon) + \frac{1}{2} m^\top D m) \exp(-\frac{1}{2} (z - m)^\top D (z - m)) dz \\ &= \int_{g_\varepsilon(z) \leq 0} e^\varepsilon \tilde{\varphi}(z) dz \\ &= e^\varepsilon \Pr[g_\varepsilon(\tilde{Z}) \leq 0], \end{aligned}$$

where $\tilde{\varphi}(z)$ is the density of $\tilde{Z} \sim \mathcal{N}(m, D^{-1})$. Plugging this back yields

$$\delta_{X,Y}(\varepsilon) = \Pr[g_\varepsilon(Z) \leq 0] - e^\varepsilon \Pr[g_\varepsilon(\tilde{Z}) \leq 0],$$

with $\tilde{Z} \sim \mathcal{N}(m, D^{-1})$ as defined above.

The important observation is that both $g_\varepsilon(Z)$ and $g_\varepsilon(\tilde{Z})$ are (possibly indefinite) quadratic polynomials in independent Gaussian coordinates, and hence are generalized- χ^2

random variables. To fit it in the characterization given in Definition 1, we rewrite both $g_\varepsilon(Z)$ and $g_\varepsilon(\tilde{Z})$ as a weighted sum of independent (noncentral) χ^2 variables with an Gaussian linear term if needed.

Term 1: $g_\varepsilon(Z)$. Recall that

$$g_\varepsilon(Z) = c(\varepsilon) + b^\top Z + \frac{1}{2} \sum_{i=1}^d a_i Z_i^2, \quad Z \sim \mathcal{N}(0, I_d).$$

Define the set $\mathcal{I} = \{i \in [d] : a_i \neq 0\}$, and for each $i \in \mathcal{I}$, define

$$\lambda_i = \frac{a_i}{2}, \quad \mu_i = \frac{b_i}{a_i}, \quad X_i \sim \chi_1^2(\mu_i^2).$$

Then we can rewrite $g_\varepsilon(Z)$ as

$$g_\varepsilon(Z) = c(\varepsilon) - \sum_{i \in \mathcal{I}} \frac{b_i^2}{2a_i} + \sum_{i \in \mathcal{I}} \lambda_i X_i + \sum_{i \notin \mathcal{I}} b_i Z_i.$$

In particular, if $|\mathcal{I}| \neq d$, then $\sum_{i \notin \mathcal{I}} b_i Z_i$ is Gaussian. Therefore, $g_\varepsilon(Z)$ is a generalized- χ^2 random variable (cf. Definition 1) with parameters

$$r = |\mathcal{I}|, \quad (\lambda_i, \nu_i, \rho_i) = \left(\frac{a_i}{2}, 1, \left(\frac{b_i}{a_i} \right)^2 \right) \text{ for } i \in \mathcal{I}, \quad \sigma = \left(\sum_{i \in [d] \setminus \mathcal{I}} b_i^2 \right)^{1/2},$$

and threshold

$$c \stackrel{\text{def}}{=} - \left(c(\varepsilon) - \sum_{i \in \mathcal{I}} \frac{b_i^2}{2a_i} \right),$$

since $\Pr[g_\varepsilon(Z) \leq 0] = \Pr[\sum_{i \in \mathcal{I}} \lambda_i X_i + \sigma G \leq c]$.

Term 2: $g_\varepsilon(\tilde{Z})$. For $i \in [d]$, write

$$\tilde{Z}_i = m_i + \frac{1}{\sqrt{\tau_i}} G_i,$$

where $G_1, \dots, G_d \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ are independent. Recall that

$$g_\varepsilon(\tilde{Z}) = c(\varepsilon) + b^\top \tilde{Z} + \frac{1}{2} \sum_{i=1}^d a_i \tilde{Z}_i^2.$$

Recall that $\mathcal{I} = \{i \in [d] : a_i \neq 0\}$. For each $i \in \mathcal{I}$, complete the square in $\tilde{Z}_i$:

$$b_i \tilde{Z}_i + \frac{a_i}{2} \tilde{Z}_i^2 = \frac{a_i}{2} \left(\tilde{Z}_i + \frac{b_i}{a_i} \right)^2 - \frac{b_i^2}{2a_i}.$$

Moreover,

$$\tilde{Z}_i + \frac{b_i}{a_i} = \frac{1}{\sqrt{\tau_i}} \left(G_i + \tilde{\mu}_i \right), \quad \tilde{\mu}_i = \sqrt{\tau_i} \left(m_i + \frac{b_i}{a_i} \right) = \frac{b_i}{a_i \sqrt{\tau_i}},$$

where the last equality uses $m_i = b_i/\tau_i$ and $a_i + \tau_i = 1$.

$$\tilde{\lambda}_i = \frac{a_i}{2\tau_i}, \quad \tilde{X}_i \sim \chi_1^2(\tilde{\mu}_i^2) \quad \text{for } i \in \mathcal{I}.$$

Then we can rewrite

$$g_\varepsilon(\tilde{Z}) = c(\varepsilon) - \sum_{i \in \mathcal{I}} \frac{b_i^2}{2a_i} + \sum_{i \in \mathcal{I}} \tilde{\lambda}_i \tilde{X}_i + \sum_{i \in [d] \setminus \mathcal{I}} b_i \tilde{Z}_i.$$

In particular, for $i \notin \mathcal{I}$ we have $a_i = 0$ and hence $\tau_i = 1$, so $\tilde{Z}_i = m_i + G_i = b_i + G_i$, and thus

$$\sum_{i \in [d] \setminus \mathcal{I}} b_i \tilde{Z}_i = \sum_{i \in [d] \setminus \mathcal{I}} b_i^2 + \sum_{i \in [d] \setminus \mathcal{I}} b_i G_i \stackrel{d}{=} \sum_{i \in [d] \setminus \mathcal{I}} b_i^2 + \sigma \tilde{G},$$

where $\sigma = (\sum_{i \in [d] \setminus \mathcal{I}} b_i^2)^{1/2}$ and $\tilde{G} \sim \mathcal{N}(0, 1)$. Therefore, $g_\varepsilon(\tilde{Z})$ is a generalized- χ^2 random variable (cf. Definition 1) with parameters

$$r = |\mathcal{I}|, (\lambda_i, \nu_i, \rho_i) = \left(\frac{a_i}{2\tau_i}, 1, \left(\frac{b_i}{a_i \sqrt{\tau_i}} \right)^2 \right) \text{ for } i \in \mathcal{I}$$

$$\sigma = \left(\sum_{i \in [d] \setminus \mathcal{I}} b_i^2 \right)^{1/2}.$$

Let $\tilde{c}(\varepsilon) \stackrel{\text{def}}{=} c(\varepsilon) - \sum_{i \in \mathcal{I}} \frac{b_i^2}{2a_i} + \sum_{i \in [d] \setminus \mathcal{I}} b_i^2$, and threshold $c \stackrel{\text{def}}{=} -\tilde{c}(\varepsilon)$ since $\Pr[g_\varepsilon(Z) \leq 0] = \Pr\left[\sum_{i \in \mathcal{I}} \tilde{\lambda}_i X_i + \sigma \tilde{G} \leq c\right]$. $\square$

E.2. Proof for Theorem 2

Proof for Proposition 1

Proof of Proposition 1. Let $\Sigma \stackrel{\text{def}}{=} \Sigma_1 = \Sigma_2 \succ 0$ and $\Delta\mu \stackrel{\text{def}}{=} \mu_1 - \mu_2$. If $\Delta\mu = 0$ (equivalently $\tau = 0$), then X and Y are identical and hence $\delta_{X,Y}(\varepsilon) = 0$ for all $\varepsilon \geq 0$. Assume henceforth $\tau > 0$.

We specialize Lemma 1 and Theorem 1 to the equal-covariance case. Since

$$M = \Sigma^{1/2} \Sigma^{-1} \Sigma^{1/2} = I_d,$$

we may take $U = I_d$ and $\tau_i = 1$ for all $i \in [d]$, hence $a_i = 1 - \tau_i = 0$. Moreover,

$$b = -U^\top \Sigma^{1/2} \Sigma^{-1} \Delta\mu = -\Sigma^{-1/2} \Delta\mu$$

$$c(\varepsilon) = \varepsilon - \frac{1}{2} \Delta\mu^\top \Sigma^{-1} \Delta\mu = \varepsilon - \frac{\tau^2}{2}.$$

Therefore, for $Z \sim \mathcal{N}(0, I_d)$,

$$g_\varepsilon(Z) = c(\varepsilon) + b^\top Z = \varepsilon - \frac{\tau^2}{2} + b^\top Z.$$

Since $\|b\|_2^2 = \Delta\mu^\top \Sigma^{-1} \Delta\mu = \tau^2$, let the scalar $W = \frac{b^\top Z}{\tau} \sim \mathcal{N}(0, 1)$, and thus

$$\Pr[g_\varepsilon(Z) \leq 0] = \Pr\left[\varepsilon - \frac{\tau^2}{2} + \tau W \leq 0\right]$$

$$= \Pr\left[W \leq -\frac{\varepsilon}{\tau} + \frac{\tau}{2}\right] = \Phi\left(-\frac{\varepsilon}{\tau} + \frac{\tau}{2}\right).$$

Next, in Theorem 1 we have $D = I_d$ and $m = D^{-1}b = b$, so $\tilde{Z} \sim \mathcal{N}(b, I_d)$; write $\tilde{Z} = b + Z'$ with $Z' \sim \mathcal{N}(0, I_d)$ independent. Then

$$g_\varepsilon(\tilde{Z}) = \varepsilon - \frac{\tau^2}{2} + b^\top (b + Z') = \varepsilon + \frac{\tau^2}{2} + b^\top Z'.$$

Let $W' = \frac{b^\top Z'}{\tau} \sim \mathcal{N}(0, 1)$. Hence

$$\Pr[g_\varepsilon(\tilde{Z}) \leq 0] = \Pr\left[\varepsilon + \frac{\tau^2}{2} + \tau W' \leq 0\right]$$

$$= \Pr\left[W' \leq -\frac{\varepsilon}{\tau} - \frac{\tau}{2}\right] = \Phi\left(-\frac{\varepsilon}{\tau} - \frac{\tau}{2}\right).$$

Finally, applying Theorem 1 gives

$$\delta_{X,Y}(\varepsilon) = \Pr[g_\varepsilon(Z) \leq 0] - e^\varepsilon \Pr[g_\varepsilon(\tilde{Z}) \leq 0]$$

$$= \Phi\left(-\frac{\varepsilon}{\tau} + \frac{\tau}{2}\right) - e^\varepsilon \Phi\left(-\frac{\varepsilon}{\tau} - \frac{\tau}{2}\right),$$

as claimed. $\square$

Proof for Lemma 2

Proof of Lemma 2. In this lemma, we compute the indistinguishability $\delta_{X,Y}$, where the leakage is only from the covariance, i.e., $\mu_1 = \mu_2$ and hence $\Delta\mu = 0$. In this case, $b = 0$ in Lemma 1, and therefore

$$c(\varepsilon) = \varepsilon + \frac{1}{2} \log \frac{\det \Sigma_1}{\det \Sigma_2} = \frac{1}{2} f(\varepsilon), \quad g_\varepsilon(Z) = c(\varepsilon) + \frac{1}{2} \sum_{i=1}^d (1 - \tau_i) Z_i^2.$$

Moreover, by Theorem 1 with $b = 0$ and $m = 0$,

$$\delta_{X,Y}(\varepsilon) = \Pr[g_\varepsilon(Z) \leq 0] - e^\varepsilon \Pr[g_\varepsilon(\tilde{Z}) \leq 0], \quad (5)$$

where $Z \sim \mathcal{N}(0, I_d)$ and $\tilde{Z} \sim \mathcal{N}(0, D^{-1})$, with $D = \text{diag}(\tau_1, \dots, \tau_d)$.

By the condition $\Sigma_1 \succeq \Sigma_2$, we have that $\tau_i \geq 1$ for all $i \in [d]$. We then define

$$w_i \stackrel{\text{def}}{=} \tau_i - 1 \geq 0 \quad \text{and} \quad w'_i \stackrel{\text{def}}{=} \frac{\tau_i - 1}{\tau_i} \geq 0.$$

Using $c(\varepsilon) = \frac{1}{2} f(\varepsilon)$ and $1 - \tau_i = -w_i$, we can rewrite the NDIS polynomial as

$$g_\varepsilon(Z) = \frac{1}{2} f(\varepsilon) - \frac{1}{2} \sum_{i=1}^d w_i Z_i^2.$$

Therefore, $g_\varepsilon(Z) \leq 0 \iff \sum_{i=1}^d w_i Z_i^2 \geq f(\varepsilon)$, and hence

$$\Pr[g_\varepsilon(Z) \leq 0] = \Pr\left[\sum_{i=1}^d w_i Z_i^2 \geq f(\varepsilon)\right].$$

We next upper bound the first probability in Equation (5) and lower bound the second probability.

Upper bound $\Pr[g_\varepsilon(Z) \leq 0]$. Let $\chi_i^2 \stackrel{\text{def}}{=} Z_i^2 \sim \chi_1^2$ be i.i.d. For any $s \in [0, k]$, where

$$k = \left(2 \max_{i \in [d]} w_i\right)^{-1} = \left(2 \max_{i \in [d]} (\tau_i - 1)\right)^{-1},$$

we have $2sw_i < 1$ for all i . By Markov's inequality,

$$\begin{aligned} \Pr \left[\sum_{i=1}^d w_i \chi_i^2 \geq f(\varepsilon) \right] &= \Pr \left[\exp \left(s \sum_{i=1}^d w_i \chi_i^2 \right) \geq e^{sf(\varepsilon)} \right] \\ &\leq e^{-sf(\varepsilon)} \mathbb{E} \left[\exp \left(s \sum_{i=1}^d w_i \chi_i^2 \right) \right] \\ &= e^{-sf(\varepsilon)} \prod_{i=1}^d \mathbb{E} \left[e^{sw_i \chi_i^2} \right] \\ &= e^{-sf(\varepsilon)} \prod_{i=1}^d (1 - 2sw_i)^{-1/2} \\ &= \exp \left(-sf(\varepsilon) - \frac{1}{2} \sum_{i=1}^d \log(1 - 2sw_i) \right), \end{aligned}$$

where we used $\mathbb{E} [e^{t\chi_1^2}] = (1 - 2t)^{-1/2}$ for $t < 1/2$.

Lower bound $\Pr [g_\varepsilon(\tilde{Z}) \leq 0]$. Since $\tilde{Z} \sim \mathcal{N}(0, D^{-1})$, we can write $\tilde{Z}_i = \tau_i^{-1/2} Z'_i$ with $Z' \sim \mathcal{N}(0, I_d)$. Hence

$$\begin{aligned} g_\varepsilon(\tilde{Z}) &= \frac{1}{2} f(\varepsilon) - \frac{1}{2} \sum_{i=1}^d \frac{\tau_i - 1}{\tau_i} (Z'_i)^2 \\ &= \frac{1}{2} f(\varepsilon) - \frac{1}{2} \sum_{i=1}^d w'_i (Z'_i)^2. \end{aligned}$$

Thus $\Pr [g_\varepsilon(\tilde{Z}) \leq 0] = \Pr \left[\sum_{i=1}^d w'_i (Z'_i)^2 \geq f(\varepsilon) \right]$. Let $\chi_i'^2 \stackrel{\text{def}}{=} (Z'_i)^2 \sim \chi_1^2$ be i.i.d. Then

$$\Pr \left[\sum_{i=1}^d w'_i \chi_i'^2 \geq f(\varepsilon) \right] = 1 - \Pr \left[\sum_{i=1}^d w'_i \chi_i'^2 < f(\varepsilon) \right].$$

For any $u \geq 0$, by Markov's inequality,

$$\begin{aligned} \Pr \left[\sum_{i=1}^d w'_i \chi_i'^2 < f(\varepsilon) \right] &\leq \Pr \left[\sum_{i=1}^d w'_i \chi_i'^2 \leq f(\varepsilon) \right] \\ &= \Pr \left[\exp \left(-u \sum_{i=1}^d w'_i \chi_i'^2 \right) \geq e^{-uf(\varepsilon)} \right] \\ &\leq e^{uf(\varepsilon)} \mathbb{E} \left[\exp \left(-u \sum_{i=1}^d w'_i \chi_i'^2 \right) \right] \\ &= e^{uf(\varepsilon)} \prod_{i=1}^d \mathbb{E} \left[e^{-uw'_i \chi_i'^2} \right] \\ &= e^{uf(\varepsilon)} \prod_{i=1}^d (1 + 2uw'_i)^{-1/2} \\ &= \exp \left(uf(\varepsilon) - \frac{1}{2} \sum_{i=1}^d \log(1 + 2uw'_i) \right), \end{aligned}$$

where we used $\mathbb{E} [e^{-t\chi_1^2}] = (1 + 2t)^{-1/2}$ for $t \geq 0$. Therefore,

$$\Pr [g_\varepsilon(\tilde{Z}) \leq 0] \geq 1 - \exp \left(uf(\varepsilon) - \frac{1}{2} \sum_{i=1}^d \log(1 + 2uw'_i) \right).$$

Plugging the bounds above into Equation (5) gives, for all $s \in [0, k]$ and $u \geq 0$,

$$\begin{aligned} \delta_{X,Y}(\varepsilon) &\leq \exp \left(-sf(\varepsilon) - \frac{1}{2} \sum_{i=1}^d \log(1 - 2s(\tau_i - 1)) \right) \\ &\quad - e^\varepsilon \left(1 - \exp \left(uf(\varepsilon) - \frac{1}{2} \sum_{i=1}^d \log \left(1 + 2u \frac{\tau_i - 1}{\tau_i} \right) \right) \right). \end{aligned}$$

Taking the infimum over $s \in [0, k]$ and $u \geq 0$ completes the proof. $\square$

Proof for Proposition 2

Proof of Proposition 2. If $t = 0$, then both sides equal $(1 - e^a)_+$ and the claim holds with equality. Assume henceforth $t > 0$.

Let $Y \sim \chi_d^2$ and set $k \stackrel{\text{def}}{=} d/2 \geq 1$. Equivalently, $Y/2 \sim \Gamma(k, 1)$. Let P_k and Q_k denote the CDF and survival function of $\Gamma(k, 1)$, i.e., $P_k(x) = \Pr[G \leq x]$ and $Q_k(x) = \Pr[G \geq x]$ for $G \sim \Gamma(k, 1)$. A direct change-of-variables calculation gives the following truncated exponential moments: for every $y_0 \geq 0$,

$$\begin{aligned} \mathbb{E} \left[\exp \left(-\frac{u}{2} Y \right) \mathbf{1}\{Y \geq y_0\} \right] &= (1 + u)^{-k} Q_k \left(\frac{1+u}{2} y_0 \right), \quad u \geq 0, \\ \mathbb{E} \left[\exp \left(\frac{u}{2} Y \right) \mathbf{1}\{Y \leq y_0\} \right] &= (1 - u)^{-k} P_k \left(\frac{1-u}{2} y_0 \right), \quad u \in [0, 1). \end{aligned}$$

Our first step is to get closed forms for g_d and h_d . Since $a \in [-\frac{1}{2} \log(1 + t), 0]$, we have $a + \log(1 + t) \geq 0$, hence the positivity region $\{1 - \exp(a + \log(1 + t) - \frac{t}{2} Y) > 0\}$ is $\{Y \geq y_g\}$ with

$$y_g \stackrel{\text{def}}{=} \frac{2(a + \log(1 + t))}{t} \geq 0, \quad x_g \stackrel{\text{def}}{=} \frac{y_g}{2} = \frac{a + \log(1 + t)}{t}.$$

Therefore,

$$\begin{aligned} &g_d(a + \log(1 + t), t) \\ &= \mathbb{E} \left[\left(1 - \exp \left(a + \log(1 + t) - \frac{t}{2} Y \right) \right) \mathbf{1}\{Y \geq y_g\} \right] \\ &= \Pr[Y \geq y_g] - \exp(a + \log(1 + t)) \mathbb{E} \left[\exp \left(-\frac{t}{2} Y \right) \mathbf{1}\{Y \geq y_g\} \right] \\ &= Q_k(x_g) - \exp(a) (1 + t)^{1-k} Q_k((1 + t)x_g), \end{aligned}$$

where in the last line we used the first truncated-moment identity with $u = t$.

Similarly, since $a \leq 0$, the positivity region $\{1 - \exp(a + \frac{t}{2(1+t)} Y) > 0\}$ is $\{Y \leq y_h\}$ with

$$y_h \stackrel{\text{def}}{=} \frac{-2a(1 + t)}{t} \geq 0, \quad x_h \stackrel{\text{def}}{=} \frac{y_h}{2} = \frac{-a(1 + t)}{t}.$$

Using the second truncated-moment identity with $u = t/(1 + t) \in [0, 1)$ yields

$$\begin{aligned} &h_d \left(a, \frac{t}{1 + t} \right) \\ &= \mathbb{E} \left[\left(1 - \exp \left(a + \frac{t}{2(1+t)} Y \right) \right) \mathbf{1}\{Y \leq y_h\} \right] \\ &= \Pr[Y \leq y_h] - \exp(a) \mathbb{E} \left[\exp \left(\frac{t}{2(1+t)} Y \right) \mathbf{1}\{Y \leq y_h\} \right] \\ &= P_k(x_h) - \exp(a) (1 + t)^k P_k \left(\frac{x_h}{1 + t} \right). \end{aligned}$$

Subtracting and regrouping, we have that

$$\begin{aligned} & g_d(a + \log(1+t), t) - h_d\left(a, \frac{t}{1+t}\right) \\ &= \left[Q_k(x_g) - \exp(a) (1+t)^{1-k} Q_k((1+t)x_g) \right] \\ & \quad + \left[\exp(a) (1+t)^k P_k\left(\frac{x_h}{1+t}\right) - P_k(x_h) \right]. \end{aligned}$$

Using that Q_k is nonincreasing and P_k is nondecreasing, we have $Q_k((1+t)x_g) \leq Q_k(x_g)$ and $P_k(x_h) \geq P_k(x_h/(1+t))$, hence

$$\begin{aligned} & g_d(a + \log(1+t), t) - h_d\left(a, \frac{t}{1+t}\right) \\ & \geq Q_k(x_g) \left[1 - \exp(a) (1+t)^{1-k} \right] + P_k\left(\frac{x_h}{1+t}\right) \left[\exp(a) (1+t)^k - 1 \right]. \end{aligned}$$

It remains to check that both bracketed coefficients are nonnegative under $a \in [-\frac{1}{2} \log(1+t), 0]$ and $k \geq 1$. Indeed, since $\exp(a) \leq 1$ and $(1+t)^{1-k} \leq 1$, we get $\exp(a) (1+t)^{1-k} \leq 1$, so the first bracket is ≥ 0 . Also, $\exp(a) \geq (1+t)^{-1/2}$, hence

$$\exp(a) (1+t)^k \geq (1+t)^{k-\frac{1}{2}} \geq 1,$$

so the second bracket is nonnegative as well. Since $Q_k(\cdot) \geq 0$ and $P_k(\cdot) \geq 0$, the RHS is nonnegative, proving the desired inequality. $\square$

Proof for Lemma 3

Proof of Lemma 3. Recall the definition of the matrix $M = \Sigma_2^{-1/2} \Sigma_1 \Sigma_2^{-1/2}$. Since $\Sigma_1 \succeq \Sigma_2$, we know that $M \succeq I_d$, hence all eigenvalues $\tau_1, \dots, \tau_d$ of M satisfy $\tau_i \geq 1$. Write $L = \log \det M = \sum_{i=1}^d \log \tau_i \geq 0$.

Reduction to (M, I_d) . Applying the NDIS lemma (Lemma 1) to the pair $(\mathcal{N}(0, \Sigma_1), \mathcal{N}(0, \Sigma_2))$ and using the change of variables $z = \Sigma_2^{-1/2} x$, we obtain the invariance

$$\begin{aligned} \delta_{\mathcal{N}(0, \Sigma_1), \mathcal{N}(0, \Sigma_2)}(\varepsilon) &= \delta_{\mathcal{N}(0, M), \mathcal{N}(0, I_d)}(\varepsilon) \\ \delta_{\mathcal{N}(0, \Sigma_2), \mathcal{N}(0, \Sigma_1)}(\varepsilon) &= \delta_{\mathcal{N}(0, I_d), \mathcal{N}(0, M)}(\varepsilon). \end{aligned}$$

Therefore it suffices to prove that, for all $\varepsilon \geq 0$,

$$\delta_{M, I_d}(\varepsilon) \geq \delta_{I_d, M}(\varepsilon).$$

Radial-Angular decomposition. Let $Z \sim \mathcal{N}(0, I_d)$ and write $Z = RU$, where $R \geq 0$ and $U \in \mathbb{S}^{d-1}$ are independent, $R^2 \sim \chi_d^2$, and U is uniform on the unit sphere. Define the Rayleigh quotients

$$\alpha(U) \stackrel{\text{def}}{=} U^\top (M - I_d) U \geq 0, \quad \beta(U) \stackrel{\text{def}}{=} U^\top (I_d - M^{-1}) U \geq 0.$$

Applying Lemma 1 to the zero-mean case gives

$$\begin{aligned} \delta_{M, I_d}(\varepsilon) &= \mathbb{E} \left[1 - \exp\left(\varepsilon + \frac{1}{2} \log \det M + \frac{1}{2} Z^\top (I_d - M) Z\right) \right]_+ \\ &= \mathbf{E} \mathbf{x}_U \left[\mathbf{E} \mathbf{x}_{R^2} \left[\left(1 - \exp\left(\varepsilon + \frac{L}{2} - \frac{\alpha(U)}{2} R^2\right) \right)_+ \right] \right] \\ &= \mathbf{E} \mathbf{x}_U \left[g_d\left(\varepsilon + \frac{L}{2}, \alpha(U)\right) \right], \end{aligned}$$

and similarly

$$\begin{aligned} & \delta_{I_d, M}(\varepsilon) \\ &= \mathbb{E} \left[1 - \exp\left(\varepsilon - \frac{1}{2} \log \det M + \frac{1}{2} Z^\top (I_d - M^{-1}) Z\right) \right]_+ \\ &= \mathbf{E} \mathbf{x}_U \left[\mathbf{E} \mathbf{x}_{R^2} \left[\left(1 - \exp\left(\varepsilon - \frac{L}{2} + \frac{\beta(U)}{2} R^2\right) \right)_+ \right] \right] \\ &= \mathbf{E} \mathbf{x}_U \left[h_d\left(\varepsilon - \frac{L}{2}, \beta(U)\right) \right], \end{aligned}$$

where g_d, h_d are exactly the functions in Proposition 2.

Coupling $\beta(U)$ and $\alpha(U)$. Recall that $M = Q^\top \text{diag}(\tau_1, \dots, \tau_d) Q$ with $\tau_i \geq 1$. Let $V = QU$, note that V is also uniform on the sphere and $\|V\|_2 = 1$. Then, we have that

$$\alpha(U) = \sum_{i=1}^d (\tau_i - 1) V_i^2, \quad \beta(U) = \sum_{i=1}^d \left(1 - \frac{1}{\tau_i}\right) V_i^2 = \sum_{i=1}^d \frac{\tau_i - 1}{\tau_i} V_i^2.$$

With $f(x) \stackrel{\text{def}}{=} \frac{x}{1+x}$ (concave on $\mathbb{R}_{\geq 0}$), Jensen's inequality gives

$$\beta(U) = \sum_i f(\lambda_i - 1) V_i^2 \leq f\left(\sum_i (\lambda_i - 1) V_i^2\right) = \frac{\alpha(U)}{1 + \alpha(U)}.$$

Fix U and set $t = \alpha(U) \geq 0$. The above inequality and the fact that $h_d(a, \cdot)$ is nondecreasing in its second argument give

$$h_d\left(\varepsilon - \frac{L}{2}, \beta(U)\right) \leq h_d\left(\varepsilon - \frac{L}{2}, \frac{t}{1+t}\right).$$

We now first consider the case that $0 \leq \varepsilon \leq L/2$, and let $a = \varepsilon - \frac{L}{2} \in [-L/2, 0]$. Since

$$\log(1+t) = \log(U^\top M U) \leq \log \tau_{\max} \leq \sum_{i=1}^d \log \tau_i = L,$$

we have $a \geq -\frac{1}{2} \log(1+t)$, i.e., $a \in [-\frac{1}{2} \log(1+t), 0]$, and thus Proposition 2 applies:

$$h_d\left(a, \frac{t}{1+t}\right) \leq g_d(a + \log(1+t), t).$$

Since $g_d(\cdot, t)$ is nondecreasing in its first argument and $a + \log(1+t) \leq a + L = \varepsilon + \frac{L}{2}$, we get

$$g_d(a + \log(1+t), t) \leq g_d\left(\varepsilon + \frac{L}{2}, t\right) = g_d\left(\varepsilon + \frac{L}{2}, \alpha(U)\right).$$

Combining the last three, we have that for all U and all $\varepsilon \in [0, L/2]$

$$h_d\left(\varepsilon - \frac{L}{2}, \beta(U)\right) \leq g_d\left(\varepsilon + \frac{L}{2}, \alpha(U)\right).$$

Taking expectation over U gives $\delta_{I_d, M}(\varepsilon) \leq \delta_{M, I_d}(\varepsilon)$ for $\varepsilon \in [0, L/2]$.

We last discuss the case that $\varepsilon \geq L/2$. It is easy to see that $\delta_{I_d, M}(\varepsilon) = 0$, hence trivially $\delta_{M, I_d}(\varepsilon) \geq \delta_{I_d, M}(\varepsilon)$.

Putting all cases together establishes that, for all $\varepsilon \geq 0$,

$$\delta_{M, I_d}(\varepsilon) \geq \delta_{I_d, M}(\varepsilon)$$

and therefore

$$\delta_{X, Y}(\varepsilon) \geq \delta_{Y, X}(\varepsilon).$$

□ $[X^{(1)}, \dots, X^{(r)}]$ and $Y = [Y^{(1)}, \dots, Y^{(r)}]$ by columns, we have i.i.d. columns

$$X^{(j)} \sim \mathcal{N}(0, \Sigma_1), \quad Y^{(j)} \sim \mathcal{N}(0, \Sigma_2)$$

$$\Sigma_1 \stackrel{\text{def}}{=} D^\top D, \quad \Sigma_2 \stackrel{\text{def}}{=} D_{-i}^\top D_{-i} = \Sigma_1 - vv^\top.$$

Identifying $\mathbb{R}^{d \times r}$ with $\mathbb{R}^{dr}$ via vectorization (a bijection), the hockey-stick divergence is invariant under this relabeling, hence we may work with

$$\hat{X} \stackrel{\text{def}}{=} \text{vec}(X) \sim \mathcal{N}(0, I_r \otimes \Sigma_1), \quad \hat{Y} \stackrel{\text{def}}{=} \text{vec}(Y) \sim \mathcal{N}(0, I_r \otimes \Sigma_2).$$

Reduction to the NDIS quadratic form. Applying Lemma 1 with $\mu_1 = \mu_2 = 0$ and $\Sigma'_1 \stackrel{\text{def}}{=} I_r \otimes \Sigma_1$, $\Sigma'_2 \stackrel{\text{def}}{=} I_r \otimes \Sigma_2$ gives

$$\Delta\mu = 0, \quad b = 0, \quad c(\varepsilon) = \varepsilon + \frac{r}{2} \log \frac{\det \Sigma_1}{\det \Sigma_2}.$$

To compute $\det \Sigma_2$, use the matrix determinant lemma:

$$\det(\Sigma_2) = \det(\Sigma_1 - vv^\top) = \det(\Sigma_1)(1 - v^\top \Sigma_1^{-1} v)$$

$$= \det(\Sigma_1)(1 - p_{(i)}),$$

recalling $p_{(i)}$ is the leverage score of the record v in D , as defined in Equation (1). Thus $\det \Sigma_1 / \det \Sigma_2 = 1/(1 - p_{(i)}) = \rho$, and hence

$$c(\varepsilon) = \varepsilon + \frac{r}{2} \log \rho.$$

Next, we compute the eigenvalues of

$$M \stackrel{\text{def}}{=} (\Sigma'_1)^{1/2} (\Sigma'_2)^{-1} (\Sigma'_1)^{1/2} = I_r \otimes \left(\Sigma_1^{1/2} \Sigma_2^{-1} \Sigma_1^{1/2} \right).$$

Let $u \stackrel{\text{def}}{=} \Sigma_1^{-1/2} v$, so that $u^\top u = v^\top \Sigma_1^{-1} v = p_{(i)}$ and

$$\Sigma_2 = \Sigma_1 - vv^\top = \Sigma_1^{1/2} (I_d - uu^\top) \Sigma_1^{1/2}.$$

Therefore

$$\Sigma_1^{1/2} \Sigma_2^{-1} \Sigma_1^{1/2} = (I_d - uu^\top)^{-1}.$$

The rank-one matrix $uu^\top$ has eigenvalue $p_{(i)}$ in direction u and 0 on $u^\perp$; hence $(I_d - uu^\top)^{-1}$ has eigenvalues $\rho = (1 - p_{(i)})^{-1}$ in direction u and 1 on $u^\perp$. Consequently, M has exactly r eigenvalues equal to ρ and the remaining $r(d-1)$ eigenvalues equal to 1.

Plugging in Lemma 1, this means the coefficients $a_k = 1 - \tau_k$ satisfy

$$a_k = 1 - \rho = -(\rho - 1) \quad \text{for } k \in [r]$$

$$a_k = 0 \quad \text{for } k \in \{r+1, \dots, dr\}.$$

Let $Z \sim \mathcal{N}(0, I_{dr})$ and write $T = \sum_{k=1}^r Z_k^2 \sim \chi_r^2$. Lemma 1 yields

$$\delta_{X,Y}(\varepsilon) = \mathbb{E} \left[1 - \exp(g_\varepsilon(Z)) \right]_+$$

where $g_\varepsilon(Z) = \varepsilon + \frac{r}{2} \log \rho - \frac{\rho - 1}{2} T$.

Proof for Proposition 3

Proof of Proposition 3. Fix any event A , we have

$$\begin{aligned} & \Pr[X \in A] - e^{\varepsilon_1 + \varepsilon_2} \Pr[Z \in A] \\ &= (\Pr[X \in A] - e^{\varepsilon_1} \Pr[Y \in A]) + e^{\varepsilon_1} (\Pr[Y \in A] - e^{\varepsilon_2} \Pr[Z \in A]) \\ &\leq (\Pr[X \in A] - e^{\varepsilon_1} \Pr[Y \in A])_+ + e^{\varepsilon_1} (\Pr[Y \in A] - e^{\varepsilon_2} \Pr[Z \in A])_+ \\ &= \delta_{X,Y}(\varepsilon_1) + e^{\varepsilon_1} \delta_{Y,Z}(\varepsilon_2), \end{aligned}$$

which yields the claim. □

Proof for Theorem 2

Proof of Theorem 2. Fix $\varepsilon \geq 0$ and any split $\varepsilon = \varepsilon_1 + \varepsilon_2$ with $\varepsilon_1, \varepsilon_2 \geq 0$. We introduce an intermediate Gaussian $W_1 \sim \mathcal{N}(\mu_2, \Sigma_{\min})$, which shares the *target mean* μ_2 and the *smaller covariance* $\Sigma_{\min}$. This induces the two-step path $X = \mathcal{N}(\mu_1, \Sigma_{\min}) \rightarrow W_1 = \mathcal{N}(\mu_2, \Sigma_{\min}) \rightarrow Y = \mathcal{N}(\mu_2, \Sigma_{\max})$.

By the triangle inequality for DP divergence (Proposition 3),

$$\begin{aligned} & \delta_{X,Y}(\varepsilon_1 + \varepsilon_2) \\ &\leq \delta_{X,W_1}(\varepsilon_1) + e^{\varepsilon_1} \delta_{W_1,Y}(\varepsilon_2) \\ &= \delta_{\mathcal{N}(\mu_1, \Sigma_{\min}), \mathcal{N}(\mu_2, \Sigma_{\min})}(\varepsilon_1) + e^{\varepsilon_1} \delta_{\mathcal{N}(\mu_2, \Sigma_{\min}), \mathcal{N}(\mu_2, \Sigma_{\max})}(\varepsilon_2) \\ &\leq \delta_{\mathcal{N}(\mu_1, \Sigma_{\min}), \mathcal{N}(\mu_2, \Sigma_{\min})}(\varepsilon_1) + e^{\varepsilon_1} \delta_{\mathcal{N}(\mu_2, \Sigma_{\max}), \mathcal{N}(\mu_2, \Sigma_{\min})}(\varepsilon_2). \end{aligned}$$

(by Lemma 3)

Taking the infimum over all splits $\varepsilon_1 + \varepsilon_2 = \varepsilon$ yields

$$\delta_{X,Y}(\varepsilon) \leq \bar{\delta}^{(1)}(\varepsilon).$$

For the second quantity $\bar{\delta}^{(2)}(\varepsilon)$, we could similarly create an intermediate Gaussian $W_2 \sim \mathcal{N}(\mu_1, \Sigma_{\min})$. We then take the path that first change the covariance under mean μ_1 , then change the mean under $\Sigma_{\min}$. The rest part of proof just follow the above.

Combining the two bounds gives

$$\delta_{X,Y}(\varepsilon) \leq \min\{\bar{\delta}^{(1)}(\varepsilon), \bar{\delta}^{(2)}(\varepsilon)\}.$$

Finally, the stated analytic form claims follow immediately from Proposition 1 (mean-only terms) and Lemma 2 (covariance-only terms). □

Appendix F.

Missing Proofs for Random-Projection DP Mechanisms

F.1. Privacy analysis of $\mathcal{M}_{RP}$

Proof for Lemma 4

Proof of Lemma 4. Let $v^\top \in \mathbb{R}^{1 \times d}$ be the i -th row of D , so that $D_{-i}^\top D_{-i} = D^\top D - vv^\top$. By Definition 2, writing $X =$

Expression of NDIS quadratic form in closed form. The integrand is positive iff $g_\varepsilon(Z) \leq 0$, i.e.,

$$T \geq t_0 \quad \text{where} \quad t_0 = \frac{2(\varepsilon + \frac{r}{2} \log \rho)}{\rho - 1}.$$

Hence, with $s = r/2$ and $T \sim \chi_r^2$,

$$\begin{aligned} \delta_{X,Y}(\varepsilon) &= \mathbf{E} \mathbf{x}_T \left[\left(1 - \exp\left(\varepsilon + \frac{r}{2} \log \rho - \frac{\rho-1}{2} T\right) \right) \mathbf{1}\{T \geq t_0\} \right] \\ &= \Pr[T \geq t_0] - e^\varepsilon \rho^s \mathbf{E} \mathbf{x}_T \left[e^{-(\rho-1)T/2} \mathbf{1}\{T \geq t_0\} \right]. \end{aligned}$$

Using the χ_r^2 density $f(t) = \frac{1}{2^s \Gamma(s)} t^{s-1} e^{-t/2}$, the second term equals

$$\begin{aligned} &e^\varepsilon \rho^s \int_{t_0}^{\infty} e^{-(\rho-1)t/2} f(t) dt \\ &= e^\varepsilon \rho^s \cdot \frac{1}{2^s \Gamma(s)} \int_{t_0}^{\infty} t^{s-1} e^{-\rho t/2} dt \\ &= e^\varepsilon \rho^s \cdot \frac{1}{2^s \Gamma(s)} \cdot \rho^{-s} \int_{\rho t_0}^{\infty} u^{s-1} e^{-u/2} du \quad (u = \rho t) \\ &= \frac{e^\varepsilon}{\Gamma(s)} \Gamma\left(s, \frac{\rho t_0}{2}\right). \end{aligned}$$

Moreover,

$$\Pr[T \geq t_0] = \frac{1}{\Gamma(s)} \Gamma\left(s, \frac{t_0}{2}\right).$$

Combining the two and recalling $s = r/2$ gives the claimed expression

$$\delta_{X,Y}(\varepsilon) = \frac{1}{\Gamma(\frac{r}{2})} \left(\Gamma\left(\frac{r}{2}, \frac{t_0}{2}\right) - e^\varepsilon \Gamma\left(\frac{r}{2}, \frac{\rho t_0}{2}\right) \right).$$

□

Proof for Proposition 4

Proof is in Proposition 4. Let $s = r/2$ and recall that $\bar{F}(x) = \Gamma(s, x/2)/\Gamma(s)$ is the upper tail probability of χ_r^2 , evaluating at point x . From Lemma 4 we may rewrite, for $\rho > 1$,

$$\begin{aligned} \delta_r^{\text{gRP}}(\varepsilon; \rho) &= \bar{F}(t_0(\rho)) - e^\varepsilon \bar{F}(\rho t_0(\rho)), \\ t_0(\rho) &\stackrel{\text{def}}{=} \frac{2(\varepsilon + s \log \rho)}{\rho - 1}. \end{aligned}$$

Differentiating and using $\bar{F}'(x) = -f_{\chi_r^2}(x)$ gives

$$\frac{d}{d\rho} \delta_r^{\text{gRP}}(\varepsilon; \rho) = -f_{\chi_r^2}(t_0) t'_0 + e^\varepsilon f_{\chi_r^2}(\rho t_0) (t_0 + \rho t'_0).$$

By the definition $t_0(\rho) \stackrel{\text{def}}{=} 2(\varepsilon + s \log \rho)/(\rho - 1)$, we have

$$\varepsilon + s \log \rho = \frac{\rho - 1}{2} t_0(\rho) \iff e^\varepsilon e^{-(\rho-1)t_0(\rho)/2} = \rho^{-s}.$$

Recall that the χ_r^2 density (with $s = r/2$) is

$$f_{\chi_r^2}(x) = \frac{1}{2^s \Gamma(s)} x^{s-1} e^{-x/2}.$$

Using the identity, and a substitution into the χ_r^2 density gives that $e^\varepsilon f_{\chi_r^2}(\rho t_0) = \frac{1}{\rho} f_{\chi_r^2}(t_0)$. Concretely,

$$\begin{aligned} e^\varepsilon f_{\chi_r^2}(\rho t_0(\rho)) &= \frac{1}{2^s \Gamma(s)} e^\varepsilon (\rho t_0)^{s-1} e^{-\rho t_0/2} \\ &= \frac{1}{2^s \Gamma(s)} (\rho t_0)^{s-1} e^{-t_0/2} \left(e^\varepsilon e^{-(\rho-1)t_0/2} \right) \\ &= \frac{1}{2^s \Gamma(s)} (\rho t_0)^{s-1} e^{-t_0/2} \rho^{-s} \\ &= \frac{1}{\rho} \cdot \frac{1}{2^s \Gamma(s)} t_0^{s-1} e^{-t_0/2} = \frac{1}{\rho} f_{\chi_r^2}(t_0(\rho)), \end{aligned}$$

where in the second line we used $e^{-\rho t_0/2} = e^{-t_0/2} e^{-(\rho-1)t_0/2}$ and in the third line we substituted $e^\varepsilon e^{-(\rho-1)t_0/2} = \rho^{-s}$.

Plugging this into the derivative above yields the cancellation

$$\frac{d}{d\rho} \delta_r^{\text{gRP}}(\varepsilon; \rho) = \frac{t_0(\rho)}{\rho} f_{\chi_r^2}(t_0(\rho)) \geq 0.$$

Finally, since $\rho(p) = 1/(1-p)$ and $\rho'(p) = \rho(p)^2$, the chain rule gives

$$\begin{aligned} &\frac{d}{dp} \delta_r^{\text{gRP}}(\varepsilon; p) \\ &= \frac{d}{d\rho} \delta_r^{\text{gRP}}(\varepsilon; \rho) \Big|_{\rho=\rho(p)} \cdot \rho'(p) \\ &= \rho(p) t_0(p) f_{\chi_r^2}(t_0(p)) \geq 0. \end{aligned}$$

□

Proof for Theorem 3

proof of Theorem 3. Let D and D' be neighboring databases under the standard add/remove notion. Without loss of generality, assume D is the larger database and D' is obtained from D by removing one row, i.e., $D' = D_{-i}$ for some $i \in [n]$. Let $v^\top$ be the removed row. Define the augmented databases

$$\bar{D} = \begin{bmatrix} D \\ \sqrt{\lambda} I_d \end{bmatrix}, \quad \bar{D}' = \begin{bmatrix} D' \\ \sqrt{\lambda} I_d \end{bmatrix}.$$

Let the mechanism outputs be

$$X \stackrel{\text{def}}{=} \mathcal{M}_{RP}(D) = \text{gRP}_r(\bar{D}), \quad Y \stackrel{\text{def}}{=} \mathcal{M}_{RP}(D') = \text{gRP}_r(\bar{D}').$$

By Definition 2, each column of X is $\mathcal{N}(0, \Sigma_1)$ and each column of Y is $\mathcal{N}(0, \Sigma_2)$, where

$$\begin{aligned} \Sigma_1 &= \bar{D}^\top \bar{D} = D^\top D + \lambda I_d, \\ \Sigma_2 &= \bar{D}'^\top \bar{D}' = D'^\top D' + \lambda I_d = \Sigma_1 - v v^\top, \end{aligned}$$

so $\Sigma_1 \succeq \Sigma_2 \succ 0$.

Leverage regularization under ridge. Let $\overline{p(i)}$ denote the leverage score of row i in the augmented database $\bar{D}$. Then

$$\overline{p(i)} = v^\top (\bar{D}^\top \bar{D})^{-1} v = v^\top (D^\top D + \lambda I_d)^{-1} v.$$

Since $D^\top D + \lambda I_d \succeq \lambda I_d$, we have $(D^\top D + \lambda I_d)^{-1} \preceq \frac{1}{\lambda} I_d$, and hence

$$\overline{p(i)} \leq \frac{1}{\lambda} \|v\|_2^2 \leq \frac{l^2}{\lambda} = p^*.$$

Applying Lemma 4 to the pair $(\overline{D}, \overline{D}')$ yields

$$\delta_{X,Y}(\varepsilon) = \delta_r^{\text{gRP}}(\varepsilon; \overline{p(i)}),$$

where $\delta_r^{\text{gRP}}(\varepsilon; p)$ is defined in Proposition 4. Using Proposition 4 and $\overline{p(i)} \leq p^*$ gives

$$\delta_{X,Y}(\varepsilon) \leq \delta_r^{\text{gRP}}(\varepsilon; p^*) \leq \delta.$$

By the definition of $\delta_{X,Y}(\varepsilon)$, this implies that for every measurable set $\mathcal{S}$,

$$\Pr[X \in \mathcal{S}] \leq e^\varepsilon \Pr[Y \in \mathcal{S}] + \delta.$$

Reverse DP inequality. Since $\Sigma_1 \succeq \Sigma_2 \succ 0$, Lemma 3 implies

$$\delta_{Y,X}(\varepsilon) \leq \delta_{X,Y}(\varepsilon) \leq \delta,$$

equivalently, for every measurable set $\mathcal{S}$,

$$\Pr[Y \in \mathcal{S}] \leq e^\varepsilon \Pr[X \in \mathcal{S}] + \delta.$$

Combining the two inequalities, by definition, $\mathcal{M}_{RP}$ is (ε, δ) -DP for add/remove adjacency. $\square$

F.2. Utility analysis of $\mathcal{M}_{RP}$

Proof of Theorem 4. Recall the notations $\lambda = l^2/p^*$ and $\overline{D} = \left[\frac{D}{\sqrt{\lambda} I_d} \right] \in \mathbb{R}^{(n+d) \times d}$ that are defined in Figure 1. Write $\overline{M} = \text{gRP}_r(\overline{D}) = [X^{(1)}, \dots, X^{(r)}]$ by columns. Then $X^{(1)}, \dots, X^{(r)}$ are i.i.d. and, by Definition 2, for each $j \in [r]$,

$$X^{(j)} \sim \mathcal{N}(0, \overline{D}^\top \overline{D}).$$

Since $\overline{D}^\top \overline{D} = D^\top D + \lambda I_d = \Sigma + \lambda I_d$, we have $X^{(j)} \sim \mathcal{N}(0, \Sigma + \lambda I_d)$.

Unbiasedness. By the above, we have that

$$\begin{aligned} \mathbb{E} \left[\frac{1}{r} \widetilde{M} \widetilde{M}^\top \right] &= \mathbb{E} \left[\frac{1}{r} \sum_{j=1}^r X^{(j)} (X^{(j)})^\top \right] \\ &= \mathbb{E} \left[X^{(1)} (X^{(1)})^\top \right] = \Sigma + \lambda I_d. \end{aligned}$$

Hence

$$\mathbb{E} [\widehat{\Sigma}] = \mathbb{E} \left[\frac{1}{r} \widetilde{M} \widetilde{M}^\top - \lambda I_d \right] = \Sigma.$$

High-probability spectral-norm error. Define $\Sigma_{\text{reg}} \stackrel{\text{def}}{=} \Sigma + \lambda I_d$ and $\widehat{\Sigma}_{\text{reg}} \stackrel{\text{def}}{=} \frac{1}{r} \widetilde{M} \widetilde{M}^\top$. Then $\widehat{\Sigma} = \widehat{\Sigma}_{\text{reg}} - \lambda I_d$, hence

$$\|\widehat{\Sigma} - \Sigma\|_2 = \|\widehat{\Sigma}_{\text{reg}} - \Sigma_{\text{reg}}\|_2.$$

Since $X^{(1)}, \dots, X^{(r)}$ are i.i.d. Gaussian with covariance Σ_{reg} , $\widehat{\Sigma}_{\text{reg}}$ then is the sample covariance of r i.i.d. mean-zero Gaussian samples in $\mathbb{R}^d$. We then can apply standard covariance concentration bounds for subgaussian vectors. By [75][Remark 4.7.3], there exists an absolute constant $C > 0$ such that for every $\beta \in (0, 1)$, with probability at least $1 - \beta$,

$$\|\widehat{\Sigma}_{\text{reg}} - \Sigma_{\text{reg}}\| \leq C \left(\sqrt{\frac{d + \log(2/\beta)}{r}} + \frac{d + \log(2/\beta)}{r} \right) \|\Sigma_{\text{reg}}\|.$$

Finally,

$$\|\Sigma_{\text{reg}}\|_2 = \|\Sigma + \lambda I_d\|_2 \leq \|\Sigma\|_2 + \lambda = \|D\|_2^2 + \lambda,$$

which yields the claimed bound. $\square$

F.3. Privacy analysis of $\mathcal{M}_{RP}$'s optimizations

Proof of Theorem 6. By Theorem 3, $\mathcal{M}_{RP}$ is $(\varepsilon_0, \delta_0)$ -DP under the unbounded (add/remove) neighboring relation.

The wrapper $\mathcal{M}_{RP}^{\text{Pois}}$ is exactly: Poisson subsample each row with rate q to form D_S , then output $\mathcal{M}_{RP}(D_S)$. By the standard privacy amplification theorem for Poisson subsampling (see, e.g., [41][Thm.8]), $\mathcal{M}_{RP}^{\text{Pois}}$ is (ε', δ') -DP with

$$\varepsilon' = \ln(1 + q(e^{\varepsilon_0} - 1)), \quad \delta' = q\delta_0.$$

With the choices in Figure 5, $\varepsilon_0 = \ln\left(1 + \frac{e^\varepsilon - 1}{q}\right)$ gives that $e^{\varepsilon_0} = 1 + \frac{e^\varepsilon - 1}{q}$. Therefore, $\varepsilon' = \varepsilon$, and $\delta' = q \cdot (\delta/q) = \delta$. This yields the desired claim. $\square$

Proof of Theorem 7. Fix neighboring databases under the add/remove notion: let $D' \stackrel{\text{def}}{=} D_{-i}$ be obtained from D by removing the i -th row $x_i^\top$.

Step 3 is $(\varepsilon_T, \delta_T)$ -DP. Define the query $f(D) \stackrel{\text{def}}{=} \lambda_{\min}(D^\top D)$. We know that f has global sensitivity at most l^2 . This is because, by Weyl's inequality,

$$\begin{aligned} |f(D) - f(D')| &= \left| \lambda_{\min}(D^\top D) - \lambda_{\min}(D'^\top D') \right| \leq \|x_i x_i^\top\|_2 \leq l^2. \end{aligned}$$

In Step 3, the noised query $f(D) + \eta$ is essentially $(\varepsilon_T, \delta_T)$ -DP by the Gaussian mechanism analysis (equivalently, Proposition 1 specialized to a scalar shift). Moreover, $\lambda_{\text{lb}} = \max\{\bar{\lambda} - \alpha, 0\}$ is also $(\varepsilon_T, \delta_T)$ -DP by post-processing.

We then argue that Step 4 certifies a leverage cap except with probability δ_{ptr} . Define the event $E \stackrel{\text{def}}{=} \{\eta \leq \alpha\}$, where $\alpha = \tau \cdot \Phi^{-1}(1 - \delta_{\text{ptr}})$. Since $\eta \sim \mathcal{N}(0, \tau^2)$, we have that

$$\Pr[E^c] = \Pr[\eta > \alpha] = \delta_{\text{ptr}}.$$

On event E , we have $\lambda_{\text{lb}} = \max\{f(D) + \eta - \alpha, 0\} \leq f(D)$. Let $\lambda_{\text{ridge}} \stackrel{\text{def}}{=} \max\{\frac{l^2}{p^*} - \lambda_{\text{lb}}, 0\}$ be the ridge chosen in Step 4. Then on E ,

$$\lambda_{\min}(D^\top D + \lambda_{\text{ridge}} I_d) \geq \frac{l^2}{p^*}.$$

Consequently, for any row x of D , its leverage score in the augmented database $\bar{D}$ is upper bounded by p^* on event E . An identical argument applies to D' when running the mechanism on D' .

Step 5 is $(\varepsilon_R, \delta_R + \delta_{\text{ptr}})$ -DP. Recall $\varepsilon_R = \max\{\varepsilon - \varepsilon_T, 0\}$ and p^* is chosen in Step 2 so that $\delta_r^{\text{gRP}}(\varepsilon_R; p^*) \leq \delta_R$. On event E , the leverage cap discussed above ensures that the same privacy argument as in Theorem 3 (via Lemma 4 and monotonicity in Proposition 4) on $\text{gRP}_r(\bar{D})$ holds and the release $\widetilde{M} = \text{gRP}_r(\bar{D})$ is $(\varepsilon_R, \delta_R)$ -DP with respect to removing row i .

Finally, combining the $(\varepsilon_T, \delta_T)$ -DP test step (Step 3) with the $(\varepsilon_R, \delta_R)$ -DP release step (Step 5) via sequential composition yields $(\varepsilon_T + \varepsilon_R, \delta_T + \delta_R)$ -DP on event E . Finally, E^c occurs with probability δ_{ptr} (independent of the data), so we add δ_{ptr} to δ . Since $\varepsilon_T + \varepsilon_R \leq \varepsilon$ and $\delta_T + \delta_R + \delta_{\text{ptr}} = \delta$, the mechanism is (ε, δ) -DP. $\square$

Appendix G.

Missing Proofs for NDIS-Calibrated Gaussian Mechanism

G.1. Proof for Theorem 5

Proof of Theorem 5. Fix an arbitrary ordered neighboring pair $D \sim D'$. For $\sigma \geq 0$, let

$$X_\sigma = \mathcal{M}_{\sigma I_m, q}(D) \sim \mathcal{N}(\mu(D), \Sigma(D) + \sigma I_m)$$

$$Y_\sigma = \mathcal{M}_{\sigma I_m, q}(D') \sim \mathcal{N}(\mu(D'), \Sigma(D') + \sigma I_m).$$

Binary search can find σ^* . Define the calibration objective

$$F(\sigma) \stackrel{\text{def}}{=} \bar{\delta}_{\text{LC}}(\varepsilon; \Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma)).$$

We first argue that we can use standard binary search on the monotone predicate $F(\sigma) \leq \delta$ to find σ^* (to any desired numerical tolerance). Towards that, we claim that $F(\sigma)$ is (i) nonincreasing and continuous in σ , and (ii) $F(\sigma) \rightarrow 0$ as $\sigma \rightarrow \infty$. We justify these properties from the σI_m regularization and the sensitivity definition.

Let $\Delta\mu = \mu(D) - \mu(D')$, and for each $\sigma \geq 0$ define

$$m_\sigma(D, D') \stackrel{\text{def}}{=} (\Sigma(D') + \sigma I_m)^{-1/2} \Delta\mu,$$

$$R_\sigma(D, D') \stackrel{\text{def}}{=} (\Sigma(D') + \sigma I_m)^{-1/2} (\Sigma(D) + \sigma I_m) (\Sigma(D') + \sigma I_m)^{-1/2}.$$

Note that these are exactly the (m_N, R_N) from Equation 2 with $N = \sigma I_m$. Let $\tau_1(\sigma), \dots, \tau_m(\sigma)$ be the eigenvalues of $R_\sigma(D, D')$.

Monotonicity/continuity of $\|m_\sigma(D, D')\|_2, \|\log \tau(\sigma)\|_\infty, |\sum_i \log \tau_i(\sigma)|$. The main task is to show that $\|m_\sigma(D, D')\|_2, \|\log \tau(\sigma)\|_\infty, |\sum_i \log \tau_i(\sigma)|$

(i) **Mean difference decreases in σ .** Since $\sigma \mapsto (\Sigma(D') + \sigma I_m)^{-1}$ is operator monotone decreasing, then

$$\|m_\sigma(D, D')\|_2^2 = \Delta\mu^\top (\Sigma(D') + \sigma I_m)^{-1} \Delta\mu$$

is nonincreasing and continuous in σ .

(ii) **Loewner comparability direction is σ -independent.** Note that

$$R_\sigma(D, D') \succeq I_m \iff \Sigma(D) + \sigma I_m \succeq \Sigma(D') + \sigma I_m$$

$$\iff \Sigma(D) \succeq \Sigma(D'),$$

and similarly for $\preceq I_m$. Hence the condition “ $R_\sigma \succeq I_m$ or $R_\sigma \preceq I_m$ ” does not depend on σ and keeps for all $\sigma \geq 0$.

(iii) **Covariance difference decreases in σ .** If $\Sigma(D) \succeq \Sigma(D')$, then $\tau_i(\sigma) \geq 1$ and $\tau_{\max}(\sigma) \stackrel{\text{def}}{=} \max_i \tau_i(\sigma)$ is nonincreasing and continuous in σ . Thus

$$\|\log \tau(\sigma)\|_\infty = \log \tau_{\max}(\sigma)$$

is nonincreasing and continuous in σ . If instead $\Sigma(D) \preceq \Sigma(D')$, then $\tau_i(\sigma) \leq 1$ and $\tau_{\min}(\sigma) \stackrel{\text{def}}{=} \min_i \tau_i(\sigma)$ is nondecreasing and continuous, so

$$\|\log \tau(\sigma)\|_\infty = -\log \tau_{\min}(\sigma)$$

is nonincreasing and continuous in σ . Moreover,

$$\sum_{i=1}^m \log \tau_i(\sigma) = \log \det R_\sigma(D, D')$$

$$= \log \frac{\det(\Sigma(D) + \sigma I_m)}{\det(\Sigma(D') + \sigma I_m)}$$

is continuous in σ , and its absolute value is nonincreasing in σ . This is because its derivative equals $\text{tr}((\Sigma(D) + \sigma I_m)^{-1} - (\Sigma(D') + \sigma I_m)^{-1})$, whose sign is fixed by $\Sigma(D) \succeq \Sigma(D')$ or $\preceq$.

Recall the Definition 5,

$$\Delta(\sigma) \geq \sup_{D \sim D'} \|m_\sigma(D, D')\|_2 \quad \rho_\infty(\sigma) \sup_{D \sim D'} \geq \|\log \tau(\sigma)\|_\infty$$

$$\nu(\sigma) \geq \sup_{D \sim D'} \left| \sum_i \log \tau_i(\sigma) \right|,$$

we have that $\Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma)$ is also nonincreasing and continuous in σ , indeed, taking the supremum over all neighboring pairs preserves these properties.

From Definition 6, it is easy to see that $\bar{\delta}_{\text{mean}}(\varepsilon; \Delta)$ is nondecreasing in Δ , and $\bar{\delta}_{\text{cov}}(\varepsilon; \rho_\infty, \nu)$ is nondecreasing in ρ_∞ and ν . Therefore $F(\sigma) = \bar{\delta}_{\text{LC}}(\varepsilon; \Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma))$ is nonincreasing and continuous in σ .

Existence of a feasible σ and the limit $F(\sigma) \rightarrow 0$. As $\sigma \rightarrow \infty$, we have

$$(\Sigma + \sigma I_m)^{-1/2} = \sigma^{-1/2} (I_m + o(1)),$$

so $\|m_N(D, D')\|_2 = O(\sigma^{-1/2}) \rightarrow 0$ and $R_\sigma(D, D') \rightarrow I_m$. Therefore $\|\log \tau(\sigma)\|_\infty \rightarrow 0$, and $|\sum_i \log \tau_i(\sigma)| \rightarrow 0$ for every neighbor pair. Taking the supremum over all neighboring pairs gives, as $\sigma \rightarrow \infty$,

$$\Delta(\sigma) \rightarrow 0, \quad \rho_\infty(\sigma) \rightarrow 0, \quad \nu(\sigma) \rightarrow 0.$$

Further, by Definition 6, $F(\sigma) \rightarrow \bar{\delta}_{\text{LC}}(\varepsilon; 0, 0, 0) = 0$.

$\bar{\delta}_{\text{LC}}(\varepsilon; \Delta(\sigma^*), \rho_\infty(\sigma^*), \nu(\sigma^*)) \leq \delta$ implies that $\mathcal{M}_{\sigma I_m, q}$ is ε, δ . We are left to show that if $\bar{\delta}_{\text{LC}}(\varepsilon; \Delta(\sigma^*), \rho_\infty(\sigma^*), \nu(\sigma^*)) \leq \delta$ then $\mathcal{M}_{\sigma I_m, q}$ is ε, δ for $\varepsilon \geq 0, \delta \in (0, 1)$.

By Proposition 5, we have that

$$\delta_{X_{\sigma^*}, Y_{\sigma^*}}(\varepsilon) = \delta_\varepsilon(m_{\sigma^*}(D, D'), R_{\sigma^*}(D, D')),$$

where $\delta_\varepsilon(m, R)$ is defined in Equation (3). As q has $(\Delta(\sigma^*), \rho_\infty(\sigma^*), \nu(\sigma^*), \sigma^* I_m)$ -Loewner-comparable NDIS sensitivity (Definition 5), applying the Loewner-comparable decomposition bound (by Theorem 2 together with Proposition 1), we obtain that

$$\delta_\varepsilon(m, R) \leq \bar{\delta}_{\text{LC}}(\varepsilon; \Delta(\sigma^*), \rho_\infty(\sigma^*), \nu(\sigma^*)).$$

Combining all together, for any ordered neighboring pair $D \sim D'$, we have that

$$\begin{aligned} \delta_{\mathcal{M}_{\sigma I_m, q}(D), \mathcal{M}_{\sigma I_m, q}(D')} &= \delta_\varepsilon(m_{\sigma^*}(D, D'), R_{\sigma^*}(D, D')) \\ &\leq \bar{\delta}_{\text{LC}}(\varepsilon; \Delta(\sigma^*), \rho_\infty(\sigma^*), \nu(\sigma^*)) \leq \delta \end{aligned}$$

We then conclude the proof. $\square$

G.2. Missing Proofs for Applications

Proof of Proposition 7. We verify the three function $(\Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma))$ defined in Proposition 7.

Mean sensitivity. By λ -strong convexity of $\mathcal{L}(\cdot; D')$ and $\|\nabla f_{(x, y)}(\theta)\|_2 \leq l$, the standard stability bound gives

$$\|\hat{\theta}(D) - \hat{\theta}(D')\|_2 \leq l/\lambda.$$

Next, we derive a uniform lower bound on $\Sigma(\tilde{D}) + \sigma I_d$ for $\tilde{D} \in \{D, D'\}$. By the definition of the Böhning-style covariance,

$$\Sigma(\tilde{D}) = \left(\lambda I_d + \frac{1}{4} \sum_{i=1}^{|\tilde{D}|} x_i x_i^\top \right)^{-1}.$$

Since $\|x_i\|_2 \leq l$, we have $x_i x_i^\top \preceq l^2 I_d$. Therefore,

$$\frac{1}{4} \sum_{i=1}^{|\tilde{D}|} x_i x_i^\top \preceq \frac{|\tilde{D}| l^2}{4} I_d \preceq \frac{nl^2}{4} I_d,$$

which implies $\frac{1}{\lambda + \frac{nl^2}{4}} I_d \preceq \Sigma(\tilde{D}) \preceq \frac{1}{\lambda} I_d$. Therefore,

$$\Sigma(\tilde{D}) + \sigma I_d \succeq \left(\sigma + \frac{1}{\lambda + \frac{nl^2}{4}} \right) I_d.$$

Applying this bound with $\tilde{D} = D'$, and using $N = \sigma I_d$, we obtain

$$\begin{aligned} \|m_N(D, D')\|_2 &= \|(\Sigma(D') + N)^{-1/2} (\hat{\theta}(D) - \hat{\theta}(D'))\|_2 \\ &\leq \|(\Sigma(D') + \sigma I_d)^{-1/2}\|_2 \cdot \|\hat{\theta}(D) - \hat{\theta}(D')\|_2 \\ &\leq \frac{1}{\sqrt{\sigma + \frac{1}{\lambda + \frac{nl^2}{4}}}} \cdot \frac{l}{\lambda} = \Delta(\sigma). \end{aligned}$$

Loewner comparability. Let $D' = D \setminus \{(x, y)\}$ be obtained from D by removing one record $x \in \mathbb{R}^d$. By the definition of the Böhning-style covariance,

$$\begin{aligned} \Sigma(D)^{-1} &= \lambda I_d + \frac{1}{4} \sum_{z \in D} z z^\top \\ &= \lambda I_d + \frac{1}{4} \sum_{z \in D'} z z^\top + \frac{1}{4} x x^\top \\ &= \Sigma(D')^{-1} + \frac{1}{4} x x^\top. \end{aligned}$$

Since $\frac{1}{4} x x^\top \succeq 0$, it follows that

$$\Sigma(D)^{-1} \succeq \Sigma(D')^{-1}.$$

Inverting both sides gives

$$\Sigma(D) \preceq \Sigma(D').$$

Finally, since $N = \sigma I_d \succeq 0$, adding N preserves the order:

$$\Sigma(D) + N \preceq \Sigma(D') + N.$$

Covariance sensitivity. We start by recalling an useful spectral sandwich

$$m(\sigma) I_d \preceq \Sigma(D) + \sigma I_d \leq \Sigma(D') + \sigma I_d \preceq M(\sigma) I_d,$$

where $m(\sigma) = \sigma + \frac{1}{\lambda + \frac{nl^2}{4}}$ and $M(\sigma) = \sigma + \frac{1}{\lambda}$. This is from the one we derived above

$$\frac{1}{\lambda + \frac{nl^2}{4}} I_d \preceq \Sigma(D) \preceq \frac{1}{\lambda} I_d.$$

Recalling the definition of $R_{\sigma I_d}(D, D')$, and using $m(\sigma) I_d \preceq \Sigma(D) + \sigma I_d \leq \Sigma(D') + \sigma I_d \preceq M(\sigma) I_d$, we have that for every $i \in [d]$,

$$|\log \tau_i| \leq \log \frac{M(\sigma)}{m(\sigma)} = \log \left(\frac{\sigma + 1/\lambda}{\sigma + \frac{1}{\lambda + \frac{nl^2}{4}}} \right) = \rho_\infty(\sigma).$$

Consequently,

$$|\log \det R_{\sigma I_d}(D, D')| = \left| \sum_{i=1}^d \log \tau_i \right| \leq d \rho_\infty(\sigma) = \nu(\sigma).$$

The other neighboring direction gives the inverse relative covariance and hence satisfies the same absolute log-eigenvalue and log-determinant bounds. $\square$

Proof for Proposition 8

Proof of Proposition 8. We verify the three function $(\Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma))$ stated in Proposition 8. Fix an ordered neighboring pair $D \sim D'$ and write $(\mu, \Sigma) \stackrel{\text{def}}{=} q_{\text{GPR}}(D)$ and $(\mu', \Sigma') \stackrel{\text{def}}{=} q_{\text{GPR}}(D')$. Since the GPR predictive distribution at a fixed x_* is one-dimensional, we must just write the covariance as variance (the scalar $\sigma > 0$), and the induced invariants are

$$m_\sigma(D, D') = (\Sigma' + \sigma)^{-1/2} (\mu - \mu'), \quad R_\sigma(D, D') = \frac{\Sigma + \sigma}{\Sigma' + \sigma}.$$

Mean sensitivity. Without lose of generality, let $D' = \{(x_i, y_i)\}_{i=1}^{n-1}$, and $D = D' \cup \{(x_n, y_n)\}$. Let

$$\begin{aligned} A' &\stackrel{\text{def}}{=} K' + \sigma_n^2 I_{n-1}, \\ u &\stackrel{\text{def}}{=} (k(x_i, x_n))_{i=1}^{n-1}, \\ c &\stackrel{\text{def}}{=} k(x_n, x_n) + \sigma_n^2, \\ k'_\star &\stackrel{\text{def}}{=} (k(x_\star, x_i))_{i=1}^{n-1}, \\ \kappa &\stackrel{\text{def}}{=} k(x_\star, x_n), \end{aligned}$$

so that

$$A = \begin{pmatrix} A' & u \\ u^\top & c \end{pmatrix}, \quad k_\star = \begin{pmatrix} k'_\star \\ \kappa \end{pmatrix}.$$

Let $\mathbf{y}' \in \mathbb{R}^{n-1}$ denote the label vector of D' , and let

$$\mathbf{y} = \begin{pmatrix} \mathbf{y}' \\ y_n \end{pmatrix} \in \mathbb{R}^n$$

be the label vector of D . Define

$$s \stackrel{\text{def}}{=} c - u^\top (A')^{-1} u.$$

Since $A \succ 0$, we have $s > 0$. By the block inverse formula,

$$A^{-1} = \begin{pmatrix} (A')^{-1} + (A')^{-1} u s^{-1} u^\top (A')^{-1} & -(A')^{-1} u s^{-1} \\ -s^{-1} u^\top (A')^{-1} & s^{-1} \end{pmatrix}.$$

Therefore,

$$\mu - \mu' = k_\star^\top A^{-1} \mathbf{y} - (k'_\star)^\top (A')^{-1} \mathbf{y}' = \frac{\gamma r}{s},$$

where

$$\gamma \stackrel{\text{def}}{=} \kappa - (k'_\star)^\top (A')^{-1} u, \quad r \stackrel{\text{def}}{=} y_n - u^\top (A')^{-1} \mathbf{y}'.$$

We next bound γ and r . Consider the block matrix

$$M \stackrel{\text{def}}{=} \begin{pmatrix} A' & u & k'_\star \\ u^\top & c & \kappa \\ (k'_\star)^\top & \kappa & k(x_\star, x_\star) \end{pmatrix}.$$

This matrix is positive semidefinite, since it can be written as the sum of the kernel Gram matrix on $(x_1, \dots, x_{n-1}, x_n, x_\star)$ and the positive semidefinite diagonal matrix

$$\text{diag}(\sigma_n^2 I_{n-1}, \sigma_n^2, 0).$$

Taking the Schur complement with respect to A' gives

$$\begin{pmatrix} s & \gamma \\ \gamma & \Sigma' \end{pmatrix} \succeq 0, \quad \Sigma' = k(x_\star, x_\star) - (k'_\star)^\top (A')^{-1} k'_\star.$$

Hence,

$$\gamma^2 \leq s \Sigma' \leq s l^2,$$

where the last inequality uses $\Sigma' \leq k(x_\star, x_\star) \leq l^2$.

Next, a direct expansion using the block inverse formula yields

$$\mathbf{y}^\top A^{-1} \mathbf{y} = (\mathbf{y}')^\top (A')^{-1} \mathbf{y}' + \frac{r^2}{s}.$$

Therefore,

$$\frac{r^2}{s} \leq \mathbf{y}^\top A^{-1} \mathbf{y}.$$

Since $A = K + \sigma_n^2 I_n \succeq \sigma_n^2 I_n$, we have $\|A^{-1}\|_2 \leq 1/\sigma_n^2$, and thus

$$\mathbf{y}^\top A^{-1} \mathbf{y} \leq \|\mathbf{y}\|_2^2 \|A^{-1}\|_2 \leq \frac{nB^2}{\sigma_n^2},$$

where we used $\|\mathbf{y}\|_2 \leq \sqrt{n}B$. Combining the above bounds,

$$|\mu - \mu'|^2 = \frac{\gamma^2 r^2}{s^2} = \frac{\gamma^2}{s} \cdot \frac{r^2}{s} \leq l^2 \cdot \frac{nB^2}{\sigma_n^2}.$$

Hence,

$$|\mu - \mu'| \leq \frac{lB\sqrt{n}}{\sigma_n}.$$

Since $\Sigma' + \sigma \geq \sigma$, we obtain

$$\|m_\sigma(D, D')\|_2 = |m_\sigma(D, D')| = \frac{|\mu - \mu'|}{\sqrt{\Sigma' + \sigma}} \leq \frac{lB\sqrt{n}}{\sigma_n \sqrt{\sigma}} = \Delta(\sigma).$$

Loewner comparability. Because $m = 1$, the comparability between the real number trivially has.

Covariance sensitivity. We first bound Σ and Σ' . Consider the block matrix

$$\begin{pmatrix} A & k_\star \\ k_\star^\top & k_{\star\star} \end{pmatrix} = \begin{pmatrix} K & k_\star \\ k_\star^\top & k_{\star\star} \end{pmatrix} + \begin{pmatrix} \sigma_n^2 I_n & 0 \\ 0 & 0 \end{pmatrix} \succeq 0,$$

where the first term is a Gram matrix and hence PSD, and the second term is PSD. Since $A \succ 0$, the Schur complement implies

$$\Sigma = k_{\star\star} - k_\star^\top A^{-1} k_\star \geq 0.$$

Moreover, by the diagonal bound $k_{\star\star} = k(x_\star, x_\star) \leq l^2$ and $k_\star^\top A^{-1} k_\star \geq 0$, we also have $\Sigma \leq l^2$. The same argument gives $0 \leq \Sigma' \leq l^2$. Hence

$$\Sigma + \sigma, \Sigma' + \sigma \in [\sigma, \sigma + l^2],$$

and therefore

$$\frac{\sigma}{\sigma + l^2} \leq \frac{\Sigma + \sigma}{\Sigma' + \sigma} \leq \frac{\sigma + l^2}{\sigma}.$$

Let τ be the (unique) eigenvalue of $R_\sigma(D, D')$, so $\tau = (\Sigma + \sigma)/(\Sigma' + \sigma)$. Then

$$|\log \tau| \leq \log\left(\frac{\sigma + l^2}{\sigma}\right) = \rho_\infty(\sigma).$$

Since $m = 1$, combining the above proves that q_{GPR} has $(\Delta(\sigma), \rho_\infty(\sigma), \nu(\sigma), \sigma)$ -Loewner-comparable NDIS sensitivity as stated. $\square$